

Innovation-Safe Compression of Financial Strategy Libraries: Semantics, Complexity, and Algorithms

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Abstract

Financial strategy libraries impose maintenance, validation, monitoring, governance, computing, and attention burdens. An entry that is redundant for current operation may nevertheless carry a capability used by a later strategy-generation process. We formulate source-relative minimum-burden retention that preserves both a regime-indexed operating frontier and a declared generative closure. Under explicit library-invariance assumptions, these objects form the library component of a controlled finite-state representation; this is a conditional semantic result, not a general minimality claim. Under identity closure, safe retention is exactly weighted set cover on disjoint frontier and module tags. This established combinatorial class yields a restricted NP-completeness boundary and supports enumeration, requirement-mask dynamic programming, exact preprocessing, mixed-integer optimization, weighted greedy construction, and certified deletion. A parameterized family proves that heaviest-safe-first deletion has no constant approximation guarantee. A registered and audited synthetic benchmark verifies exact-method agreement where applicable, reports certified burden gaps, and shows that preprocessing frequently resolves the generated instances; its cold-start concurrent timings are not used to rank algorithms. A retrospective point-in-time portfolio-library study finds large burden-index reductions and many additional innovation options, but its protected common-equity policy never exercises those options and its single ETF replacement loses held-out value. A separately design-locked synthetic mechanism experiment shows why option preservation and option exercise differ: the learner does not recover a small positive bridge value but does recover a prospectively powered one. These are bounded structural, synthetic, and retrospective findings, not evidence of alpha, forecasting ability, causal effects, or deployable performance.

Keywords: financial strategy libraries, innovation-safe compression, weighted set cover, exact algorithms, model governance

1 Introduction

Financial research is cumulative. Candidate signals, transformations, execution rules, and risk controls enter maintained catalogs after development and validation, where they remain available for operation, comparison, and reuse. As a catalog grows, so do its computing footprint, validation and monitoring queue, documentation burden, and the attention needed to understand it. The decision studied here is removal from an *active maintenance and validation set*. It is not destruction of code, data, model history, or records whose archival retention is required by law, contract, audit policy, or disaster-recovery practice.

The apparent redundancy of a strategy depends on which role is being assessed. For current operation, a strategy is redundant if removing it leaves every regime-dependent operating opportunity unchanged. For future research, the same strategy may be the only declared carrier of a signal, transformation, execution component, or risk-control tag used by a later project. A strategy can consequently be dominated at every current belief and still be generatively indispensable. Pruning only the current payoff envelope can preserve what the library can operate now while silently changing what it can help create later. This carrier interpretation is a modeling declaration: in the financial cases the tags describe a fixed tuple grammar, not evidence that the underlying code, data, tests, or organizational knowledge exist only in a strategy row.

We represent these roles by two source-library objects. The operating frontier F_L records the best current operating value attainable from a source library L at each declared belief. The generative closure C_L records the modules available to the modeled strategy-generation process after its closure rules have been applied. Conditional on the paper's finite library-invariance assumptions, the pair (F_L, C_L) is the library component of a controlled representation: replacing L by a sublibrary with the same frontier and closure leaves the declared operating opportunities and generative menus unchanged. The theorem is a factorization result for a specified information architecture, not a generic coarsest-state or minimality theorem. The filtered belief and any active-project variables remain part of the full controlled state.

This semantic reduction defines *innovation-safe compression*. Given a source library, additive retention burdens, and a mandatory inactive strategy, we seek a minimum-burden source sublibrary whose frontier and closure equal those of the source exactly. Equality is source-relative and exact for the declared finite tables: it is not a prediction criterion, statistical certainty, or a claim of equivalence in unobserved markets. The source itself is feasible, but choosing the least-burden feasible representation is a global combinatorial optimization problem. Sequential deletion can certify that every accepted step is safe, yet an inclusion-wise irreducible endpoint need not have minimum burden.

The identity-closure subclass exposes the combinatorial structure sharply. In that subclass, the requirements are tagged to keep frontier-attainment rows distinct from module-carrier rows, including frontier ties and requirements already met by the inactive strategy. A sublibrary is safe if and only if its strategies cover every remaining tagged requirement. This equivalence reduces the semantic retention problem to a weighted covering formulation; it does not claim set cover itself as a new problem. The restricted decision problem is NP-complete, and the broader compression model is NP-hard because it contains that subclass. The reduction therefore supplies both a

modeling bridge to the classical covering literature (Karp 1972; Chvátal 1979) and a reason not to expect local redundancy checks alone to solve global resource allocation.

We address this boundary with a portfolio of algorithms rather than a single method. Complete enumeration provides a small-library oracle. Requirement-mask dynamic programming is exact when the tagged requirement dimension is manageable. Proven preprocessing rules force unique carriers, remove safely dominated or empty variables, merge duplicate coverage and requirement patterns with explicit reconstruction maps, and can sometimes resolve an instance completely. An auditable mixed-integer formulation scales the covering model while retaining solver diagnostics and an independent exact postcheck of the returned library. Weighted greedy construction inherits its classical covering guarantee under identity closure, and reverse deletion can reduce its burden without weakening that guarantee. Deterministic and multi-start deletion heuristics provide certified feasible endpoints, but their feasibility certificates are not certificates of optimality.

The computational evidence asks a narrower question: how do certified burdens differ within one registered synthetic design? Small instances support exact cross-method comparisons, and the adversarial family connects finite calculation to the worst-case theorem. The structured generator frequently collapses under forced preprocessing and contains infeasible requested factor cells. Moreover, recorded wall times include first-call Julia compilation under concurrent worker contention. We therefore use the study for exact agreement, burden quality, failure accounting, and preprocessing prevalence, not for steady-state runtime rankings, complete factorial inference, or a population claim about financial libraries.

The financial evidence is a retrospective point-in-time portfolio-library study over annual common-equity origins with a separately registered ETF replication. It fixes point-in-time membership, delisting treatment, portfolio grammar, compression libraries, and proposal-year policies before opening each held-out evaluation block. The safe libraries retain the registered frontier and all capability artifacts at substantially lower burden indices and expose many more proposal options than budget-matched frontier-only libraries. Yet the protected policy never exercises an added common-equity option, and its single ETF replacement loses held-out value. A separate calibrated synthetic experiment isolates the bridge mechanism under equal current frontier and burden: small positive option value remains hard to detect, whereas a prospectively powered value is recovered. This pairing keeps option creation, statistical exercise, and realized financial value distinct. Neither study establishes population savings, causality, forecasting skill, alpha, or deployable performance.

1.1 Research questions

The analysis is organized around four questions.

- RQ1.** Under which declared library-invariance conditions does frontier-closure equality preserve both current operating opportunities and the modeled strategy-generation menu?

- RQ2.** How does exact minimum-burden preservation become a covering problem under identity closure, what computational barrier follows, and how far can a certified local-deletion rule be from the global optimum?
- RQ3.** Which exact, approximation-guaranteed, mixed-integer, and certified heuristic methods can exploit this structure while preserving exact feasibility and an auditable reconstruction to original strategy identities?
- RQ4.** Within the registered synthetic designs and the retrospective point-in-time financial panel, when does preserving closure create additional options, when does a protected policy exercise them, and what held-out value is realized when it does?

1.2 Contributions

The paper makes four principal contributions.

1. *Conditional semantic representation.* We identify operationally checkable library-invariance conditions under which the operating frontier and generative closure form a sufficient library summary, and we formulate innovation-safe compression as source-relative preservation of both objects. This separates operational from generative redundancy without claiming a generic minimal state.
2. *Complexity and worst-case theory.* For identity closure, we prove the tagged-cover equivalence and NP-completeness of a restricted safe-compression decision problem. A parameterized family gives the exact burden ratio of the heaviest-safe-first endpoint to the global optimum and shows that this named deletion rule has no constant approximation guarantee. The result does not apply to every deletion order.
3. *Certified computational reduction.* We map the semantic constraint to established weighted-cover machinery under identity closure and provide a common exact representation, optimum-preserving preprocessing with reconstruction, enumeration, requirement-mask dynamic programming, an auditable mixed-integer workflow, weighted greedy, and certified deletion. These are implementation and certification contributions, not claims that the underlying generic cover algorithms are new.
4. *Bounded computational and financial evidence.* The locked benchmark supplies exact-agreement, certified burden-quality, preprocessing, and adversarial records within its realized design. The point-in-time portfolio-library study reports exact structural preservation, burden-index compression, option coverage, protected proposal decisions, and held-out contrasts while preserving the licensed-data boundary. The calibrated closure-option experiment then isolates the theory’s bridge mechanism under prespecified null, adverse, low-dose, and powered regimes. Neither layer is generalized beyond its registered instances.

The theoretical, computational, solver, and financial evidence levels are not interchangeable. Universal claims rest on their stated human proofs and, where the theorem ledger says so, narrower kernel-checked declarations. Exact finite computation establishes facts about specified fixtures or instances. Solver diagnostics document a numerical global-search process, whereas independent exact rechecks certify the feasibility and burden of the returned sublibrary. Registered synthetic results describe the locked generators, and retrospective financial results describe the audited source instances. The algorithmic guarantees in this article apply to identity closure unless a

result explicitly states otherwise; we do not claim a covering algorithm for arbitrary nonidentity closure.

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1.3 Article organization

Section 2 positions the problem relative to covering optimization, policy compression, partial-information control, and financial research governance. Section 3 defines the financial strategy-library model and establishes the productive-state projection. Section 4 develops exact compression, the tagged covering representation, and the complexity result. Section 5 presents preprocessing, exact and approximate algorithms, mixed-integer optimization, and certified deletion. Section 6 interprets unsafe retention, reports the calibrated closure-option mechanism experiment, and summarizes capacity constraints, building on the local/global gap established in Section 5. Section 7 reports the registered computational study, and Section 8 reports the retrospective point-in-time financial study. Section 9 concludes. Proof details, certificates, extended outputs, registered-study documentation, and reproducibility records are provided in Online Resource 1.

2 Literature and problem positioning

The relevant literatures use similar terms—reduction, pruning, compression, coverage, and libraries—for decisions that preserve different objects. We therefore organize the comparison by research problem and preservation target. Established work in operations research, stochastic control, and combinatorial optimization comes first; recent agent-library and automated-strategy systems are considered afterward as application-level comparators. This ordering is important because the paper’s covering algorithms are inherited from a classical problem class, whereas its modeling contribution begins with the dynamic semantics of a retained financial strategy library.

2.1 Partial-information state reduction and semi-Markov control

Belief-state reduction is foundational in partially observed control. [Smallwood and Sondik \(1973\)](#) formulate finite-horizon control on the belief simplex and establish the familiar piecewise-linear convex value representation; [Sondik \(1978\)](#) develops the discounted infinite-horizon counterpart. Random decision durations do not remove the sufficient-statistic logic: [White \(1976\)](#) gives a finite-horizon, partially observed semi-Markov dynamic program on a sufficient statistic. Financial switching and regime-switching models likewise use filtered information to choose among a fixed menu of operating modes or positions ([Lakner 1998](#); [Li et al. 2015](#); [Dai et al. 2010](#)).

Those results supply the control architecture used here, not a novelty claim. The present reduction concerns a different component of state. Conditional on the filtered belief and any active-project variables, it asks which summary of an endogenous strategy library preserves both current operating choices and the modeled law of future strategy generation. The answer under the declared finite factorization is the operating frontier together with the generative closure. Thus the frontier–closure pair summarizes the *library* component of the productive state; it neither replaces the belief nor proposes a new general sufficient statistic for partially observed or semi-Markov control.

Two standard reductions sharpen the distinction. Incremental pruning removes alpha vectors that are redundant in a fixed partially observable dynamic-programming backup (Cassandra et al. 1997). Its preserved object is the pointwise value representation generated by that backup. A pruned vector does not also act as a parent carrying modules for later policy generation. State aggregation and model minimization instead identify states that are behaviorally equivalent for a specified comparison class (Givan et al. 2003). Their decision unit is a state or model component, not a retained strategy entry. Innovation-safe compression therefore differs from both alpha-vector pruning and state aggregation even though all three use exact equality to justify a smaller representation.

2.2 Policy, action, and control-library compression

The closest established library-selection problem is static control-library design. Dey et al. (2012) select and order compact maneuver or trajectory libraries using submodular-sequence structure, trading library size and search effort against expected task performance. The preserved or optimized object is static task usefulness and search efficiency. In the present problem, a retained entry can additionally alter the set of candidates that can be generated later, so static control coverage does not certify generative safety.

Mutti et al. (2022) provide the closest policy-compression comparison. They construct reward-free representatives of a policy space using coverage of induced state–action distributions and formulate the selection as set cover. Their preserved object is reward-independent distributional representativeness for downstream reinforcement-learning tasks. Here the feasible choices are sublibraries of a named source, and feasibility requires exact equality of a regime- or belief-indexed operating frontier and a parent-conditioned generative closure. Reward-free policy compression can therefore retain a representative for broad downstream usefulness without preserving the same objects as innovation-safe compression.

A directionally different problem expands rather than compresses the available action set. In such models the performance object may be learning or regret in a growing menu. Here generation is conditioned on modules carried by retained parents, and the decision is which existing entries can be deleted without changing either current opportunities or attainable descendants. The present problem is therefore distinct from generative action-set expansion: it concerns source-relative closure-preserving deletion rather than costly action creation.

2.3 Weighted covering and combinatorial representation

Set cover is the classical combinatorial core once a finite requirement universe has been specified. Karp’s set-covering decision problem is among the foundational NP-complete problems (Karp 1972). For positive column costs, Chvátal (1979) gives the standard binary weighted-cover formulation and the minimum-cost-per-newly-covered-requirement greedy rule with its harmonic approximation bound. These are established results. The paper neither introduces weighted set cover nor claims a new generic greedy algorithm.

The contribution is the direction of the reduction. The dynamic semantic model first determines what must be preserved: one tagged class records exact source frontier attainment and another records source modules. Under identity closure, a retained source sublibrary is innovation-safe if and only if it covers this tagged requirement universe, after fixing the mandatory inactive strategy. The tagged-cover equivalence is therefore a computational subclass *induced by* the dynamic frontier-closure model. It is not a semantic interpretation imposed on an arbitrary set-cover instance, and it does not extend automatically to a nonidentity closure with complementarities among raw modules.

Computational benchmarking has a separate methodological literature. Performance profiles compare solvers only after defining a common problem set and success measure (Dolan and Moré 2002); feature-balanced benchmark construction, as in MIPLIB 2017, is designed to avoid overrepresenting a narrow instance family (Gleixner et al. 2021). These principles expose limits of the present locked experiment rather than repairing it after the fact. Its methods have unequal success targets, some infeasible generator cells, and a structured block dominated by complete preprocessing. A future registered benchmark must therefore balance realized residual structure and separate feasibility, quality-qualified, and exact-completion targets before using comparative profiles.

Adjacent covering models optimize different objects. Budgeted maximum coverage and monotone submodular maximization seek the largest attainable value under a cardinality or resource constraint (Nemhauser et al. 1978). Submodular cover minimizes cost to reach the full value of a normalized, monotone, integer-valued submodular function (Wolsey 1982). Adaptive submodularity concerns sequential choices whose outcomes are revealed over time and requires adaptive monotonicity and diminishing returns (Golovin and Krause 2011). Exact innovation-safe compression instead minimizes positive active-strategy burden subject to complete coverage of fixed tagged requirements. The maximum-coverage, submodular-cover, and adaptive guarantees do not transfer unless their distinct assumptions are separately proved.

2.4 Resource-constrained and metareasoning decisions

Operations research has long treated information, search, computation, and research budgets as scarce resources. Weitzman (1979) optimizes costly sequential search among alternatives. Baker et al. (1976) allocate an organizational research-and-development budget across competing activities, and Russell and Wefald (1991) value computational actions by their effect on a subsequent decision. These literatures motivate the economic

importance of research resources without supplying a preservation condition for a maintained strategy library.

The distinction is between choosing what to investigate and choosing which representation to retain. Exact innovation-safe compression is a one-time outer decision that minimizes additive burden while holding the source frontier and source closure fixed. The paper’s capacity- and price-constrained variants relax that equality and ask which productive state is worth buying; they are closer to search, metareasoning, and research-budget allocation. Neither variant turns held-out financial performance into a retention score. This separation also explains why a locally safe deletion certificate answers a feasibility question, whereas global minimum-burden retention answers a resource-allocation question.

2.5 Maintained generative libraries

Library learning and program synthesis establish that reusable components can shape later construction. For example, [Bowers et al. \(2023\)](#) synthesize libraries of reusable abstractions rather than treating candidate programs as independent outputs. Its decision unit is internal program structure, and its preserved object is reusable abstraction within synthesis. The present problem instead selects a sublibrary of whole strategies that must jointly preserve a belief-indexed operating envelope and parent-conditioned generation. Innovation-safe compression is therefore not a substitute for procedural compression, provenance, repair, versioning, or rollback controls.

2.6 Automated financial strategy construction and governance

Automated strategy construction predates contemporary agent systems. [Allen and Karjalainen \(1999\)](#) use genetic search to construct technical trading rules, and [Moody and Saffell \(2001\)](#) learn trading policies by direct reinforcement. These studies optimize or evaluate generated strategies; they do not pose minimum-burden retention of a persistent parent library. Likewise, the model-risk literature studies the consequences of model uncertainty and robust risk measurement ([Cont 2006](#); [Glasserman and Xu 2014](#)). Supervisory guidance documents expectations for model inventories, validation, monitoring, and governance ([Board of Governors of the Federal Reserve System and Office of the Comptroller of the Currency 2011](#)). This supports the existence of organizational burdens, not the paper’s optimization model or empirical findings.

The finance literature also gives a direct warning about the interpretation of large rule catalogs. [Sullivan et al. \(1999\)](#) evaluate technical-rule performance while accounting for the universe from which the rules were selected, and [White \(2000\)](#) develops a general reality-check framework for data snooping. Multiple-testing concerns are especially acute in large return- predictor collections ([Harvey et al. 2016](#)), while post-publication evidence shows that reported anomaly returns can attenuate out of sample ([McLean and Pontiff 2016](#)). The article’s ex post maximum over enabled descendants is not a corrected performance test under any of these frameworks. It is used only to witness the mechanical implication of preserving or losing the declared catalog, and no significance or investment claim is attached to it.

An accepted recent financial comparator, ReCAP, maintains regime-adaptive portfolio policies (Pan et al. 2026). Its optimized object is regime-specific portfolio performance, not minimum-burden preservation of a source frontier and generative closure. The present paper begins after a finite generation–verification–admission law has been declared and asks which source entries can be removed without changing those two objects. Its financial audits assess retention burden and semantic preservation, not improved returns, forecasting ability, alpha, or deployable performance.

Claim boundary. The paper claims novelty neither for state, value, policy, action-set, control-library, or internal procedural compression nor for weighted cover, metareasoning, maintained skill libraries, or automated strategy generation. The narrower contribution is that, under the declared finite dynamic model, the operating frontier and generative closure jointly define the object that safe retention must preserve; under identity closure, those semantics induce an exact tagged weighted-cover subclass and source-relative certificates. None of the verified comparisons reviewed here establishes that joint formulation. This is a bounded audit conclusion, not a priority claim over all related library work.

3 Financial strategy-library model and sufficient state

This section states a conditional representation theorem for a declared information architecture. The controller observes a filtered belief, operates from a verified strategy library, and may initiate positive-duration research that generates and verifies a candidate before admission. The theorem does not derive library invariance from finance primitives; it identifies the conclusion that follows when the primitives pass the explicit factorization checks below. The retention decision is made once, before the controlled process is evaluated.

3.1 Hidden state, filtered belief, and operating profiles

Let X be a finite hidden market-state set with at least two elements and let B be a finite nonempty set of filtered beliefs. A belief $b \in B$ has an exact rational interpretation $\mu_b \in \Delta_{\mathbb{Q}}(X)$; the already-filtered belief evolves under an exact rational Markov kernel P_B . The controller observes b , not the hidden state $x \in X$.

The finite strategy catalog is S , with distinguished inactive strategy s_0 . Strategy s has immutable state-contingent operating payoff $u_s : X \rightarrow \mathbb{Q}$ and belief-indexed operating profile

$$j_s(b) := \sum_{x \in X} \mu_b(x) u_s(x), \quad j_{s_0}(b) = 0. \quad (3.1)$$

Thus $j_s(b)$ is the payoff from operating s now; research costs and continuation values are separate objects. The formal projection theorem uses the resulting exact profile table as primitive data, so it does not claim a kernel-verified derivation of that table from a hidden-state model.

Partial information matters to the controlled projection through the filtered belief and its transition law. It is not required by the later tagged-cover equivalence, which needs only a finite index set of operating-profile rows. The financial cases in Section 8

use observed SPY-signal bins as such rows; they do not estimate the hidden-state filter in this subsection.

3.2 Modules, closure, and the raw retained library

Let M be a finite module set. Each strategy carries an immutable set $\text{mods}(s) \subseteq M$, with $\text{mods}(s_0) = \emptyset$. A module is a declared reusable capability tag for later research, not a direct operating payoff or necessarily a separately owned software artifact. The declared closure operator

$$\text{cl} : 2^M \longrightarrow 2^M$$

is extensive, monotone, and idempotent. These properties allow capabilities to be generated from combinations of raw modules; identity closure is the special case $\text{cl}(A) = A$.

A raw retained library is a source subset $L \subseteq S$ containing s_0 . Its operating frontier, raw module inventory, and generative closure are

$$F_L(b) := \max_{s \in L} j_s(b), \quad U_L^{\text{raw}} := \bigcup_{s \in L} \text{mods}(s), \quad (3.2)$$

$$C_L := \text{cl}(U_L^{\text{raw}}). \quad (3.3)$$

Finiteness and mandatory inactivity make the maximum attained and imply $F_L(b) \geq 0$. The raw decision state is (b, L) : continuing earns the frontier payoff, advances the belief once, and leaves the library unchanged.

3.3 Generation, verification, admission, and timing

Let Q be a finite project set. At project q , belief b , and closure C , candidate generation follows an exact law

$$G(q, b, C) \in \Delta_{\mathbb{Q}}(\{\text{none}\} \dot{\cup} S).$$

A generated candidate s passes verification with exact probability $\nu(q, b, C, s)$. The admitted law is derived rather than separately chosen:

$$\Gamma(q, b, C)(s) = G(q, b, C)(s)\nu(q, b, C, s), \quad (3.4)$$

and all failed-generation and failed-verification mass is assigned to **none**. Successful admission adds the verified catalog row, whereas every other outcome leaves the source unchanged:

$$L \oplus \text{none} = L, \quad L \oplus s = L \cup \{s\}. \quad (3.5)$$

Each available project has an initiation cost, an operation/suspension flag, and an integer duration $d_q \geq 1$. Its exact joint completion law couples the intervening belief path with the admitted outcome; the two components need not be independent. The incumbent library remains in force until completion, then (3.5) is applied. Full probability-table definitions, null-outcome normalization, path and outcome marginals, and pushforward identities are assigned to Online Resource 1, Section S1.

3.4 The productive library state

Definition 3.1 (Frontier-closure library state). The productive state of a retained library is

$$K_L := (F_L, C_L). \quad (3.6)$$

It is the *library component* of the controlled state. The filtered belief remains part of the decision state, and an unfinished project also requires its identity and remaining duration.

For $K = (F, C)$, define the local compressed update

$$\text{addK}(K, \text{none}) = K, \quad \text{addK}(K, s) = (b \mapsto \max\{F(b), j_s(b)\}, \text{cl}(C \cup \text{mods}(s))).$$

Extensivity and idempotence of closure give

$$K_{L \oplus o} = \text{addK}(K_L, o). \quad (3.7)$$

Assumption 3.2 (Finite library-invariant research process). The sets B, S, M, Q are finite, B and every admissible library are nonempty, and all displayed profiles and probability tables are exact. The inactive strategy has zero profile and no modules; closure is extensive, monotone, and idempotent. Generation and primitive verification depend on the incumbent library only through (q, b, C_L) . Project availability, initiation cost, duration, operation flag, and the joint completion law depend on the library only through (b, K_L) . Each duration is positive, the joint law has the declared belief-path and admitted-outcome marginals, and discount satisfies $0 \leq \beta < 1$. Stationary conclusions additionally assume the recorded raw and compressed Bellman operators are contractions.

Operational factorization audit.

The assumption can be checked table by table. Project availability, cost, duration, operation while active, and the joint completion law must be constant whenever (b, F_L, C_L) is constant; candidate generation and verification must be constant whenever (q, b, C_L) is constant; and admission must update the compressed state through addK . A nontrivial finite class satisfying these conditions consists of module-gated projects whose availability is a predicate of (q, b, C_L) , whose candidate and verification tables are indexed by (q, b, C_L) , and whose duration, cost, and completion tables are indexed by (q, b, F_L, C_L) . Correlation between terminal belief and admission remains unrestricted within each indexed joint table.

The condition is substantive. Let two one-active-strategy libraries $L_a = \{s_0, a\}$ and $L_b = \{s_0, b\}$ have identical zero profiles and no modules. Then $K_{L_a} = K_{L_b}$. If a zero-cost, one-period project with deterministic positive completion value is available only when the raw identity a is retained, their feasible actions and values differ. The projection therefore fails. The same failure occurs if a primitive depends directly on raw cardinality, provenance, owner, or archival status. These variables must either be included in the compressed state or ruled out by the declared information architecture.

Theorem 3.3 (Raw-to-compressed controlled Markov projection). *Under Assumption 3.2, the update in (3.7) induces a normalized transition on realizable states K_L . If $K_L = K_{\bar{L}}$, the two raw libraries have identical project availability, cost, duration, incumbent operating reward blocks, and joint terminal-belief-next-compressed-state laws after projection. Hence (b, K_L) is a controlled semi-Markov projection at embedded decision epochs and, for every finite calendar horizon h ,*

$$V_h^{\text{raw}}(b, L) = \bar{V}_h(b, K_L).$$

Under the additional contraction clause, stationary fixed-point values agree under the same projection and a maximizing stationary compressed selector lifts to a Bellman-optimal raw selector.

Proof sketch. Equation (3.7) makes the next compressed state a deterministic pushforward of the admitted raw update. The library-invariance assumptions make this law, every feasible action, and its payoff block constant on each fiber of $L \mapsto K_L$. Strong induction on calendar horizon is well founded because every research duration is positive; corresponding raw and compressed action values, and then their finite maxima, agree. Bellman intertwining and contraction give the stationary clauses. The complete normalization and projection proof is in Online Resource 1, Sections S1–S2. The result permits belief-admission dependence and makes no generic quotient-minimality claim.

3.5 Retention burden and exact source-relative compression

Let $w_{s_0} = 0$ and give every active strategy an exact rational retention weight $w_s > 0$. The one-time burden is

$$W(L) := \sum_{s \in L} w_s. \tag{3.8}$$

It may index storage, validation, monitoring, active maintenance, retrieval, governance, and human attention. Additivity is a modeling assumption. A weight is not automatically an observed monetary or labor cost, and it is not operating payoff or project cost. Burden is not a component of K_L : equal productive states can have different burdens. In the financial cases, “retained” means retained in the active maintenance and validation set; mandatory archival records, code, data, ownership, and reactivation controls remain outside this binary decision.

For a fixed verified source L , exact source-relative compression chooses an inactive-containing sublibrary $L' \subseteq L$ with $K_{L'} = K_L$ and minimum $W(L')$. By Theorem 3.3, this outer retention decision changes only the raw representation, not the declared productive process. Section 4 defines its feasible family and optimization model. Secondary dynamic comparative statics are not needed for this reduction and are assigned to Online Resource 1, Section S3.

4 Exact compression, covering representation, and complexity

Fix a verified finite source library L . Safety is source-relative: compression may delete source entries but may neither import a replacement nor change the declared profiles, modules, or closure operator.

4.1 Safe feasibility and productive preservation

The safe-feasible family and its global minimum-weight correspondence are

$$\mathcal{P}_{\text{safe}}(L) := \{L' \subseteq L : s_0 \in L', F_{L'} = F_L, C_{L'} = C_L\}, \quad (4.1)$$

$$\mathcal{M}_{\text{safe}}(L) := \arg \min_{L' \in \mathcal{P}_{\text{safe}}(L)} W(L'). \quad (4.2)$$

Equality is exact at every declared belief and for the entire declared generative closure. In particular, a currently dominated strategy is not safe to remove merely because it contributes no current frontier value.

Theorem 4.1 (Minimum-resource innovation-safe representation). *Let L be a finite source library containing s_0 , let the catalog and closure satisfy Assumption 3.2, and let $w_{s_0} = 0$ with exact rational $w_s > 0$ for every active source strategy. Then $\mathcal{P}_{\text{safe}}(L)$ is finite and nonempty, $\mathcal{M}_{\text{safe}}(L) \neq \emptyset$, and every $L' \in \mathcal{P}_{\text{safe}}(L)$ has the same finite-horizon productive values as L . Under the contraction clause of Assumption 3.2, it also has the same stationary productive value and complete optimal stationary-action correspondence.*

Proof. The source is feasible relative to itself and has only finitely many sublibraries, so an exact burden minimum is attained. Feasibility is exactly $K_{L'} = K_L$; the value and action conclusions follow from Theorem 3.3. Neither optimizer uniqueness nor a particular deletion path follows.

For comparison with observable process behavior, write $L \sim_{\text{DI}} L'$ when the libraries have equal frontier and equal availability-tagged project cost, duration, incumbent-reward, and joint terminal observations at every belief. Raw closure is *detectable* when equal frontiers but unequal closures necessarily change at least one such observation after generation, verification, admission, and projection.

Theorem 4.2 (Frontier-closure characterization). *Under Assumption 3.2, $F_L = F_{L'}$ and $C_L = C_{L'}$ imply $L \sim_{\text{DI}} L'$. If raw closure is additionally detectable in the preceding sense, the implication is an equivalence. The forward direction, which supplies productive preservation for every safe-feasible library, does not require detectability.*

Proof sketch. Equal frontier-closure states make every library-invariant primitive and its projected transition equal. Conversely, detectability rules out an unequal closure once all availability-tagged observations agree. Complete quotient and assumption-boundary arguments are in Online Resource 1, Section S2.

4.2 Identity closure induces a tagged cover

The reduction in this subsection is the computational consequence of the preservation semantics. Once the tagged universe is constructed, the resulting optimization problem is ordinary weighted set cover: its generic binary model, NP-hardness antecedent, greedy approximation theorem, mask recurrence, and standard cover preprocessing are not claimed as new combinatorial-optimization theory. The application-specific content is the exact mapping from frontier and closure equality to tagged coverage, together with its identity-closure limit.

IDENTITY-CLOSURE ALGORITHMIC BOUNDARY. The tagged rows, binary covering model, preprocessing rules, MIP, and all algorithms in Section 5 require $\text{cl}(A) = A$. The general semantic problem remains $F_{L'} = F_L$ and $\text{cl}(\bigcup_{s \in L'} \text{mods}(s)) = C_L$; it does not become an independent module-row cover when closure contains complementarities.

Assume throughout this subsection that closure is identity. Define the disjoint tagged requirement universe

$$\mathcal{U}_L^{\text{tag}} := \{\text{frontier}(b) : b \in B\} \dot{\cup} \{\text{module}(m) : m \in C_L\}, \quad (4.3)$$

and, for each $s \in L$, its source-relative coverage

$$\mathcal{R}_s := \{\text{frontier}(b) : j_s(b) = F_L(b)\} \cup \{\text{module}(m) : m \in \text{mods}(s)\}. \quad (4.4)$$

Frontier and module tags remain distinct even when their underlying identifiers print alike. Exact equality in the first term means that every tied source attainer covers the belief row.

Theorem 4.3 (Tagged-cover equivalence under identity closure). *Let B, M, S be finite, let $B \neq \emptyset$, and let $s_0 \in L' \subseteq L \subseteq S$. Profiles are exact rational functions $j_s : B \rightarrow \mathbb{Q}$; $j_{s_0} = 0$, $\text{mods}(s_0) = \emptyset$, maxima are attained, and closure is identity on 2^M . Then*

$$F_{L'} = F_L \text{ and } C_{L'} = C_L \iff \mathcal{U}_L^{\text{tag}} \subseteq \bigcup_{s \in L'} \mathcal{R}_s. \quad (4.5)$$

No unique frontier attainer or unique module carrier is assumed. If $F_L(b) = 0$, mandatory s_0 itself covers $\text{frontier}(b)$.

Proof. If L' preserves the frontier, a finite maximizer in L' covers each frontier tag. Under identity closure, $C_{L'} = C_L$ is equality of raw module unions, so some retained carrier covers every module tag. Conversely, source inclusion gives $F_{L'} \leq F_L$ and $C_{L'} \subseteq C_L$. Coverage supplies a retained source attainer at every belief and a retained raw carrier for every source module, giving the reverse inclusions. Ties and multiple carriers are existential choices, while the zero-frontier row is witnessed by s_0 .

For exact rational weights, the equivalent binary model is

$$\min_x \sum_{s \in L} w_s x_s, \quad (4.6)$$

$$\text{s.t.} \quad \sum_{s \in L: j_s(b) = F_L(b)} x_s \geq 1, \quad b \in B, \quad (4.7)$$

$$\sum_{s \in L: m \in \text{mods}(s)} x_s \geq 1, \quad m \in C_L, \quad (4.8)$$

$$x_{s_0} = 1, \quad x_s \in \{0, 1\}, \quad s \in L. \quad (4.9)$$

The source restriction is built into the columns. All zero-frontier rows may be removed after fixing s_0 , but retaining them in $\mathcal{U}_L^{\text{tag}}$ keeps the semantic construction literal.

The boundary is substantive. If $\text{cl}(\{a, b\}) = \{a, b, c\}$ although no strategy carries c , an independent row for each member of C_L rejects even the source library. General closure therefore requires direct closure equality or a separately proved extended formulation. Frontier rows alone and module rows alone are likewise insufficient; exact counterexamples are in Online Resource 1, Section S4.

4.3 Baseline mixed-integer formulation

Equations (4.6)–(4.9) are the baseline MIP. Its input is a validated, canonically ordered identity-closure instance with exact weights and declared mandatory columns. Exact preprocessing may fix columns and remove satisfied rows before JuMP and HiGHS receive the residual binary cover. Exact weights are scaled to an integer objective only after the scaling identity and solver range are verified.

The output is one incumbent in canonical original-strategy order together with the solver version, controls, termination/primal/dual statuses, objective, bound, reported gap, time, nodes when available, preprocessing audit, and complete log. The method makes no all-ties claim: a tied incumbent is one declared representative, and stable input ordering does not turn the solver’s choice into a theorem about optimizer identity. A returned vector is accepted only when every component is exactly 0 or 1; no fractional value is rounded. Reconstruction is followed by independent exact checks of mandatory retention, tagged coverage, the original frontier, the original identity closure, and burden. A HiGHS status of **OPTIMAL** remains solver evidence, not exhaustive or formal proof. When preprocessing leaves no residual row or column, the solver is recorded as not called rather than given a fabricated status.

4.4 Computational complexity

IDENTITY-CLOSURE SAFE COMPRESSION (ISC-ID) has the following explicit decision encoding. The input lists a finite source $L = \{s_0, s_1, \dots, s_n\}$, a finite nonempty belief set, exact binary-rational profiles, a finite module universe and incidence rows, identity closure, positive binary-integer active weights, and a nonnegative binary-integer threshold D . Strategy s_0 is mandatory, has zero profile, no modules, and zero

weight. The question is whether some $s_0 \in L' \subseteq L$ preserves the exact source frontier and closure and has $W(L') \leq D$.

Theorem 4.4 (Identity-closure safe-compression complexity). *Under the explicit encoding in the preceding paragraph, ISC-ID is NP-complete. It remains NP-complete on the restricted subclass with one belief, zero profiles for every strategy, a zero-weight module-free inactive strategy, identity closure, a source whose active module rows have full union, and unit active weights. Consequently, minimum-burden identity-closure safe compression is NP-hard, and any broader model class containing this subclass is NP-hard. No NP-membership conclusion for arbitrary nonidentity closure follows unless its encoding admits a polynomial exact closure-equality verifier.*

Proof sketch. A selected-strategy bit vector is a polynomial certificate: exact rational maxima, row unions, binary-integer burden, source inclusion, and mandatory retention are polynomially checkable in the explicit input length. For hardness, normalize Karp’s SET COVER decision problem (Karp 1972) to a nonempty universe and a full-union indexed family, using fixed yes/no outputs for the empty- or uncovered-universe branches. Give each universe element a module, each indexed set an active strategy carrying its modules, every active strategy unit weight and zero profile, and add one zero-profile inactive strategy. Use one belief and copy the budget. The inactive strategy preserves the trivial frontier, while identity-closure equality is exactly set-cover equality and burden is exactly selected-set cardinality. Incidence size and integer bit lengths are copied polynomially. The complete membership, normalization, forward, reverse, size, and threshold audit is in Online Resource 1, Section S4. No strong-hardness, inapproximability, or APX-hardness claim is made.

4.5 Exact preprocessing of the tagged cover

Fix the tagged universe from the source before reducing the model. After declared mandatory strategies are selected, let Q be the unsatisfied rows, A the available nonfixed columns, and $\mathcal{R}_s^Q = \mathcal{R}_s \cap Q$. Mandatory weights are nonnegative, optional weights are strictly positive exact rationals, and there are no identity-sensitive side constraints, groups, exact-cardinality conditions, or mutual exclusions beyond the pure cover. All incidence and weight comparisons are exact.

Proposition 4.5 (Exact preprocessing suite). *Under the assumptions in the preceding paragraph, the following fixed-point rules preserve feasibility, the exact optimum value including forced-weight offsets, and at least one liftable optimum:*

1. fix the unique available carrier of any residual requirement;
2. delete nonmandatory s when $\mathcal{R}_s^Q \subset \mathcal{R}_t^Q$ and $w_t \leq w_s$;
3. for identical residual coverage, retain a stable minimum-weight representative and record every equal-weight substitution;
4. after a forced selection, remove satisfied rows and recompute all rules;
5. delete a nonmandatory column with empty residual coverage; and
6. merge rows having identical carrier sets.

Strictly cheaper dominance, strictly heavier duplicates, empty positive-weight columns, forced columns, and duplicate rows preserve all original optimizer identities. Equal-weight duplicate columns recover all identities through the recorded substitution map. Equal-weight strict dominance need not preserve all optimizer identities and makes no such claim. Restarting the stable scan after each transformation terminates after finitely many row or column removals and yields a fixed point. A zero-carrier residual row is an infeasibility certificate, not an invitation to repair the instance.

Proof sketch. A unique carrier is forced by its covering row. A no-heavier coverage superset can replace a dominated selected column without losing a witness. Identical columns permit exact substitution; identical rows impose the same inequality; and an empty positive-weight column occurs in no optimum. Forced columns contribute a selection-independent offset, and propagation composes these correspondences. Every successful step removes a finite row or column, which proves termination. The strict/equal weight cases give precisely the optimizer-identity qualifications in the statement. Full propositions, adversarial examples, reconstruction maps, and the deterministic rule order are in Online Resource 1, Section S5.

5 Algorithms and guarantees

All methods in this section consume the same validated identity-closure instance. Strategies and tagged requirements have canonical original order; weights are exact rationals; mandatory flags, source profiles, raw module memberships, the source frontier, and the source closure are stored independently of the incidence matrix. Preprocessing supplies an exact objective offset and a reconstruction map. The number of residual optional strategies is n , the number of residual tagged requirements is r , and strategy i has residual mask R_i and positive weight w_i .

APPLICABILITY BOUNDARY. Enumeration, requirement-mask dynamic programming, the tagged-cover MIP, weighted greedy, and the journal preprocessing pipeline solve only the identity-closure subclass established in Section 4.2. They are not algorithms for arbitrary closure. Rechecked deletion is semantically valid for a general declared closure only when every trial deletion is evaluated by a separate exact general-closure checker; the common implementation and benchmark use identity closure.

5.1 Common output and certificate

Every algorithm returns original strategy identifiers in canonical order, the exact burden, its declared completion status, its tie rule, an algorithm trace or search counters, and any required random seed. A result is reportable only after an independent certificate recomputes from the original catalog:

1. every mandatory strategy is retained;
2. every tagged requirement is covered;
3. the complete source operating frontier is preserved exactly;

4. the complete source identity closure is preserved exactly; and
5. the exact burden agrees with the returned burden.

Passing this certificate establishes feasibility and burden for the returned library. It does not establish heuristic optimality, convert a solver status into a proof, or show that two implementations are independent. Complete-tie output is permitted only when the preprocessing map declares every optimizer identity reconstructable; otherwise the result is one explicitly declared representative.

Table 1: Algorithm guarantees and claim boundaries. Guarantees originate in human proofs or transferred classical theorems, not benchmark timings.

algorithm	guarantee	evidence class	scope and limit
Enumeration	Exact optimum after complete finite search	Human proof + exact finite computation	Small finite identity-closure instances; exponential in optional strategies
Mask DP	Exact optimum; $O(n 2^r)$ recurrence operations	Human proof + exact finite computation	FPT in residual requirements r , not polynomial in full input size
HiGHS MIP	Exactly rechecked feasible incumbent; solver-optimal status reported separately	Mixed-integer solver evidence + exact finite post-check	OPTIMAL is neither formal proof nor exhaustive enumeration
Weighted greedy	$H(d) \leq H(r)$ weighted-cover approximation	Transferred primary-source theorem	Positive active weights and identity-closure tagged-cover equivalence only
Weighted greedy + reverse deletion	$H(d)$ guarantee inherited; deletion cannot increase burden	Transferred theorem + exact deletion check	Same identity-closure scope; reverse deletion adds no general-closure guarantee
Cardinality greedy	No burden approximation guarantee for heterogeneous weights	Algorithmic limitation	Unweighted score is reported as a comparator
Heaviest safe first	Exact feasibility and irreducibility only; registered family proves an unbounded burden ratio	Exact finite certification + human-proof boundary	Identity closure for tagged checks; every accepted deletion independently rechecked
Lightest safe first	Exact feasibility and endpoint irreducibility only; no global burden factor	Exact finite certification + human-proof boundary	Identity closure for tagged checks; every accepted deletion independently rechecked
Maximum immediate release	Exact feasibility and endpoint irreducibility only; no global burden factor	Exact finite certification + human-proof boundary	Identity closure for tagged checks; every accepted deletion independently rechecked
Minimum unique-carrier exposure	Exact feasibility and endpoint irreducibility only; no global burden factor	Exact finite certification + human-proof boundary	Identity closure for tagged checks; every accepted deletion independently rechecked
Declared source order	Exact feasibility and endpoint irreducibility only; no global burden factor	Exact finite certification + human-proof boundary	Identity closure for tagged checks; every accepted deletion independently rechecked

algorithm	guarantee	evidence class	scope and limit
Random-order deletion	Exact feasibility and endpoint irreducibility only; no global burden factor	Exact finite certification + human-proof boundary	Identity closure for tagged checks; every accepted deletion independently rechecked
32-start random deletion	Best of 32 certified irreducible endpoints; no approximation factor	Synthetic algorithmic evidence + exact finite certification	Registered seeds and fixed 32 starts; not adaptive

5.2 Independent exact methods

5.2.1 Complete enumeration

Pseudocode: complete enumeration.

1. **Input:** a validated residual cover, forced selections and exact offset W_0 , a tie mode, and a finite output cap.
2. Visit every residual binary selection in increasing canonical mask order; test full tagged coverage and sum its weight exactly.
3. Keep the lower burden. On equality, retain every selection in complete mode or the minimum selected-index list in representative mode.
4. Lift forced selections and reconstruction classes, run the independent certificate, and return the optimizer set or declared representative, exact burden, number of candidates, tie status, and certificates.

Proposition 5.1 (Correctness of complete enumeration). *Let the input be a valid finite identity-closure tagged instance with exact nonnegative mandatory weights, strictly positive optional weights, at least one carrier for every residual requirement, and a valid preprocessing reconstruction map. Complete enumeration returns a minimum-burden safe source sublibrary after checking all 2^n residual selections. If complete ties are requested, the map preserves all optimizer identities, and the output cap is not exceeded, it returns every original optimizer; otherwise it returns the declared stable representative.*

Proof. Every admissible library is a forced set plus one residual subset. Enumeration evaluates every such subset and takes the exact minimum; the fixed offset and valid lift preserve burden and feasibility. The final certificate maps tagged feasibility back to the original source semantics through Theorem 4.3. Complete reconstruction follows only under the stated map and output-cap conditions. Time is $2^n \text{poly}(|I|)$, so enumeration is exact but exponential in the number of optional strategies.

5.2.2 Requirement-mask dynamic programming

Encode a residual requirement subset by an r -bit mask. With $D_i(q)$ the least exact burden that attains mask q using the first i strategies, the recurrence is

$$D_i(q) = \min \left\{ D_{i-1}(q), \min_{p: \text{ORR}_i = q} (D_{i-1}(p) + w_i) \right\}. \quad (5.1)$$

Pseudocode: requirement-mask DP.

1. **Input:** the same validated residual cover, exact offset, tie mode, and output cap used by enumeration.
2. Initialize $D_0(0) = 0$ and every other mask as unreachable.
3. For strategies in canonical order and masks in increasing order, emit the skip transition before the take transition $q \mapsto q \text{ OR } R_i$. Keep the lower exact burden.
4. On an exact tie, store the first predecessor under this traversal or every predecessor in complete-tie mode.
5. Read the all-ones mask, reconstruct, add W_0 , lift to original identifiers, certify, and return burden, selections, visited states, transitions, tie status, and certificates.

Theorem 5.2 (Exact requirement-mask dynamic programming). *Let the input satisfy every assumption of Proposition 5.1. The recurrence in (5.1) reconstructs at least one minimum-burden safe source sublibrary. Representative mode uses $O(n2^r)$ recurrence operations, $O(2^r)$ exact cost states, and $O(n2^r)$ predecessor records when full reconstruction is retained. Value-only two-layer storage uses $O(2^r)$ memory. Materializing T tied optimizers adds $O(Tn)$ output bits and may be exponential in n . Its bit time is $2^r \text{ poly}(|I|)$: the method is fixed-parameter tractable in r , but it is not polynomial in unrestricted input size.*

Proof sketch. Induct on i . At $i = 0$, only the empty subset attains mask zero. Every subset of the first i strategies either skips i , or removes i to give a predecessor mask and then takes it. The two transitions are exhaustive and construct valid subsets, so the recurrence stores the least exact burden for each mask. The all-ones mask is precisely residual feasibility. Offset, lifting, and Theorem 4.3 yield a safe global optimum. The state, transition, memory, and output counts give the displayed bounds. The complete induction and reconstruction proof is in Online Resource 1, Section S6.

5.3 Weighted greedy construction

After selecting every mandatory or forced strategy, let U^0 be the remaining tagged requirements and let

$$d := \max_i |R_i|, \quad H(d) := \sum_{j=1}^d \frac{1}{j}.$$

Pseudocode: weighted greedy and reverse deletion.

1. **Input:** a valid residual identity-closure cover and a Boolean reverse-deletion flag.
2. While requirements remain, choose a strategy minimizing $w_i/|R_i \cap U|$, comparing exact rationals; break ties by the lowest canonical original strategy index. Add it and delete its newly covered rows. If no carrier exists, return an explicit infeasibility failure.
3. If requested, examine the greedily selected optional strategies in reverse selection order and delete one only when complete tagged coverage and the original semantic checks remain true.

4. **Output:** the selected original library, exact burden, step scores, uncovered counts, reverse-deletion trace, tie declaration, and final independent certificate.

Theorem 5.3 (Transferred weighted-greedy guarantee). *Let the source be a finite identity-closure instance satisfying Theorem 4.3; let every residual requirement have a carrier; let mandatory weights be nonnegative and every optional weight be positive; and apply only mandatory/forced preprocessing with a valid exact offset and lift. If $U^0 \neq \emptyset$, weighted greedy returns a safe library of burden W_G satisfying*

$$W_G \leq H(d) \text{OPT} \leq H(r) \text{OPT}.$$

Reverse deletion returns a safe, inclusion-wise irreducible library with $W_{GR} \leq W_G$, so it satisfies the same bound. If $U^0 = \emptyset$, the forced library is optimal and the factor is one. The guarantee does not extend to arbitrary nonidentity closure.

Proof sketch. On the residual tagged universe, the rule is precisely the positive-cost weighted set-cover greedy algorithm of Chvátal (1979), which gives $G \leq H(d) \text{OPT}_R$. Adding a nonnegative fixed burden W_0 preserves the displayed factor. Reverse deletion accepts only feasibility-preserving removals of positive-weight strategies and therefore cannot increase burden. Later removals cannot restore coverage lost by an earlier rejected removal, so one reverse pass is irreducible. The theorem transfers the classical guarantee through Theorem 4.3; it does not reprove the classical set-cover theorem. The cardinality variant replaces w_i by one in the score and has no retention-burden approximation guarantee when weights differ.

With straightforward incidence scans, greedy and reverse deletion each use a polynomial number of exact operations (at most n selection/deletion rounds and $O(nr)$ incidence work per round). This implementation bound does not alter the approximation factor.

5.4 Certified rechecked deletion

Rechecked deletion starts from the full source and never infers safety from a stale certificate. At every scan, each nonmandatory trial deletion is checked against the original source frontier and closure; after an accepted deletion, all candidate certificates are discarded and the scan restarts.

Pseudocode: rechecked deletion.

1. **Input:** a validated source library, one declared deterministic score or a seeded stable random order, and an exact feasibility checker.
2. Recompute the safety of every currently retained nonmandatory one-strategy deletion. If none is safe, stop.
3. Choose among safe candidates by the declared score and tie rule, append the exact check to the trace, accept the deletion, and restart.
4. Perform a complete final scan. Return the endpoint, exact burden, trace, frontier/closure check counts, complete-scan count, seed and order when stochastic, feasibility certificate, and irreducibility certificate.

The deterministic scores are:

- heaviest-safe-first, with exact weight ties broken by canonical index;
- lightest-safe-first, with the same tie rule;
- maximum immediate burden release, which equals heaviest-safe-first under additive burden;
- minimum remaining unique-carrier exposure, defined for safe deletion s from retained set A by

$$E(s \mid A) := \left| \left\{ u : \sum_{t \in A \setminus \{s\}} \mathbf{1}\{u \in \mathcal{R}_t\} = 1 \right\} \right|,$$

with ties resolved by greater exact burden release and then canonical index; and

- declared source order, using a complete caller-supplied permutation or canonical order by default.

The exposure score uses only source incidence and no held-out financial quantity. Random-order deletion uses the first safe strategy in a recorded stable seeded permutation.

Maximum immediate release is retained as an API alias and as a historical label in the locked benchmark because those raw records are immutable. It is not an independent scientific baseline, and coincident results under the two names are not corroborating replications.

Proposition 5.4 (Certified deletion endpoint). *Let the source be finite and safe relative to itself, let all mandatory strategies be protected, and suppose every accepted trial deletion is recomputed exactly and shown to preserve the original frontier, declared closure, and burden identity. Then every deterministic or seeded-random rechecked-deletion variant above terminates at a safe source sublibrary. A complete exact final scan certifies that no retained nonmandatory strategy can be safely deleted. This is a one-deletion irreducibility certificate, not a global-optimality or approximation certificate.*

Proof. The source is safe. Exact certification of each accepted trial preserves safety inductively, and each step removes one strategy, so the trace is finite. The final scan checks every remaining eligible deletion. The score and random order choose only among already certified candidates and therefore do not change the argument. If closure is nonidentity, this proposition requires the separate exact closure checker stipulated in its assumptions.

5.4.1 Multi-start random deletion

Pseudocode: multi-start deletion. Run rechecked random-order deletion for a declared number p of independently derived, recorded seeds. Retain the certified endpoint of lowest exact burden; break endpoint ties by the lexicographically smallest selected-index tuple and then the earliest start. Return every seed and endpoint summary, the winning trace, aggregate check counts, and the winner’s feasibility and irreducibility certificates. The method costs p certified deletion runs and provides no

approximation factor; additional starts cannot worsen the best observed burden, but they do not prove optimality.

5.5 Worst-case local–global separation

Theorem 5.5 (Unbounded gap for heaviest-safe-first deletion). *Let $k \geq 2$ be an integer and let $\epsilon \in \mathbb{Q}$ satisfy $0 < \epsilon < k - 1$. Use one belief, identity closure, modules $m_1, \dots, m_k$, and a mandatory inactive strategy with zero profile, zero burden, and no modules. For each i , let singleton s_i have zero profile, weight one, and module set $\{m_i\}$. Let bundle s_B have zero profile, weight $1 + \epsilon$, and all k modules. Rechecked heaviest-safe-first deletes the uniquely heaviest safe bundle, terminates at the k -singleton library of burden k , and returns an inclusion-wise irreducible endpoint. The unique global optimum retains only the bundle and has burden $1 + \epsilon$. The exact ratio is*

$$\frac{k}{1 + \epsilon},$$

so heaviest-safe-first and one-deletion irreducibility alone have no constant burden guarantee over this family. The conclusion does not apply to every deletion order or every certified deletion heuristic.

Proof. All profiles are zero, so safety is exactly coverage of all modules. At the source, the bundle makes every singleton deletable and all singletons together make the bundle deletable. Since $\epsilon > 0$, the bundle is uniquely heaviest and is deleted first; afterward each singleton uniquely carries its module. Any safe library without the bundle must therefore retain all singletons and costs k , whereas the bundle alone is safe and costs $1 + \epsilon < k$; positive weights make it the unique optimum. Fixing, for example, $\epsilon = 1/2$ makes the ratio unbounded as k grows. A singleton-first order instead reaches the bundle optimum, proving the stated order boundary. Full finite fixtures and the reduction to the earlier small example are in Online Resource 1, Section S7.

5.6 Guarantee and evidence boundary

Enumeration and mask DP are exact finite methods under their stated identity-closure assumptions; the former is exponential in n , and the latter is exponential in r . Weighted greedy is approximate but has the transferred $H(d)$ bound; reverse deletion preserves that bound. Rechecked deletion and multi-start return exactly certified feasible, irreducible endpoints but remain heuristics. The MIP in Section 4.3 supplies solver evidence plus an exact returned-library postcheck. Complete proofs and machine-readable certificate schemas are in Online Resource 1; benchmark runtimes and method comparisons are experimental evidence reported separately in Section 7.

The suite does not include a certified multi-exchange neighborhood or an LP relaxation baseline. Accordingly, the local–global theorem isolates the named heaviest-safe-first rule and one-deletion irreducibility; it says nothing about stronger local search. Adding those baselines after seeing the locked benchmark would be post hoc and is reserved for a separately registered study.

6 Economic consequences of unsafe and capacity-constrained retention

The preservation constraint has an economic purpose beyond reproducing a static payoff menu. Frontier equality protects current operating opportunities; closure equality protects the inputs from which future opportunities can be generated. This section first separates those two value channels and then studies two outer resource decisions: retention under a hard capacity and retention under a unit burden price. Neither outer problem changes the generation, verification, admission, or positive-duration timing inside the productive process of Section 3.

6.1 Operational and generative value

For a finite horizon, let $P_n(b, L)$ be the value when the library is frozen and research is unavailable, and let $U_n(b, L)$ be the corresponding full value when every feasible positive-duration project is available. Thus

$$\begin{aligned} P_0(b, L) &= 0, \\ P_{n+1}(b, L) &= F_L(b) + \beta \sum_{b'} P_B(b, b') P_n(b', L), \end{aligned} \quad (6.1)$$

where P_B is the filtered-belief transition under continued operation. Write $\Omega_n = U_n - P_n$ for the research-option premium. For inserting a candidate s , define total, operational, and generative increments by

$$\begin{aligned} \mathcal{I}_n(s \mid b, L) &:= U_n(b, L \cup \{s\}) - U_n(b, L), \\ \Delta_n^{\text{op}}(s \mid b, L) &:= P_n(b, L \cup \{s\}) - P_n(b, L), \\ \Delta_n^{\text{gen}}(s \mid b, L) &:= \Omega_n(b, L \cup \{s\}) - \Omega_n(b, L). \end{aligned} \quad (6.2)$$

Proposition 6.1 (Operational–generative decomposition). *For every finite horizon n , filtered belief b , finite raw library L , and admissible candidate s ,*

$$\mathcal{I}_n(s \mid b, L) = \Delta_n^{\text{op}}(s \mid b, L) + \Delta_n^{\text{gen}}(s \mid b, L). \quad (6.3)$$

If $F_{L \cup \{s\}} = F_L$, then the operational increment is zero. If in addition $C_{L \cup \{s\}} = C_L$, then the total increment is zero under the projection assumptions of Theorem 3.3.

The equality is an accounting identity, not a sign theorem. A strategy may be silent in the passive operating problem and valuable because it carries a module that enables research; conversely, the generative residual need not be positive if insertion changes costs or the relative appeal of projects. The frontier–closure projection makes the two channels operationally useful: frontier silence identifies zero current operating value, whereas joint frontier and closure silence identifies zero productive value within the declared process.

6.2 Sharp loss from frontier-only retention

Consider a dominated bridge strategy that uniquely carries the capability needed for one descendant project. Its deletion leaves the source frontier unchanged but makes the distinguished descendant unreachable. The following statement fixes the scope of the loss calculation: the incumbent operating block is common, there are no other differential projects or continuation gains, and the terminal continuation gain is bounded by C .

Theorem 6.2 (Sharp normalized frontier-only bridge loss). *Fix exact rationals $0 \leq \beta < 1$, $0 \leq \rho, \pi \leq 1$, a reward cap $C \geq 0$, a project cost $\kappa \geq 0$, and an integer delay $d \geq 1$. Suppose the bridge uniquely enables the descendant, retention gives survival mass ρ^d and conditional admission probability π , deletion makes that descendant mass zero, and $\kappa \leq \beta^d \rho^d \pi C$. Then the retained-minus-deleted value loss is*

$$\text{Loss}_d = \beta^d \rho^d \pi C - \kappa. \quad (6.4)$$

For fixed $(\beta, \rho, \pi, d, \kappa)$, this is the sharp upper bound over descendant rewards in $[0, C]$, attained at C . Whenever the denominator is positive, the loss divided by the attainable net bridge value equals one. If $C \leq 1$, the additive loss is at most one; arbitrarily large additive loss requires reward scaling.

The proof expands the one available project branch. Retention contributes the discounted joint survival-and-admission mass times the capped continuation gain, less cost; deletion contributes zero. Maximization over the reward interval is attained at its upper endpoint. The factor $\rho^d \pi$ comes from sequential generation and conditional admission, not from an independence assumption. Common operating value cancels under frontier equality; if operation differs while research is in progress, its exact discounted difference must be added. Online Resource 1 gives the full fixture, proof, and delay qualifications.

The result explains why operational redundancy is not generative redundancy. A frontier-only rule can be exact for today's operating menu and still remove the entire positive net value of a future bridge. It does not assert an unbounded normalized loss or a general value formula for arbitrary deleted libraries.

6.3 Calibrated closure-option mechanism experiment

The theorem is universal under its assumptions, but it does not say whether a finite-data policy can recognize a valuable bridge. We therefore isolate that question in a design-locked synthetic experiment. The innovation-safe and frontier-only libraries have identical current frontiers and identical retention burdens. Only the safe library carries the capability needed to run one declared research project. For delay (d), survival probability ρ , admission probability π , descendant gain G , and research cost κ , the registered treatment margin is

$$M = \beta^d \rho^d \pi G - \kappa.$$

Regime	True margin	Oracle mean [95% CI]	Learned mean [95% CI]	Adoption
Adverse margin	-0.0270	0.0000 [0.0000, 0.0000]	-0.0000 [-0.0000, 0.0000]	0.05%
Null margin	0.0000	0.0000 [0.0000, 0.0000]	-0.0002 [-0.0003, 0.0000]	5.76%
Low positive dose	0.0025	0.0026 [0.0019, 0.0032]	0.0001 [-0.0001, 0.0002]	7.35%
Powered positive dose	0.0540	0.0541 [0.0529, 0.0554]	0.0512 [0.0500, 0.0525]	94.17%

Table 2 Calibrated closure-option mechanism experiment, with 4096 synthetic worlds per prespecified regime. Current frontier and burden are equal by construction; only the innovation-safe library can admit the designated bridge descendant. Intervals are registered finite-simulation uncertainty summaries. This is synthetic mechanism evidence, not a market-return, alpha, forecasting, or causal estimate.

The safe oracle researches exactly when $M > 0$; the frontier-only project is infeasible. A learner observes an independent noisy proposal signal and adopts only when it exceeds a threshold calibrated under the null.

The noise law uses only aggregate nuisance estimates from 2479 complete proposal paths and 623647 within-cell residual observations in the point-in-time financial study. Cross-strategy date means are removed before estimating volatility, lag-one dependence, and tail thickness. These quantities calibrate noise and dependence, not the sign of any regime. The test-world seeds are disjoint from calibration, and the powered margin is fixed at twice the null threshold before test simulation. Thus its size is a power-design dose, not an estimated return premium.

Table 2 reports all four prespecified regimes. At the low positive dose, the oracle value is positive but the learned-value interval crosses zero and adoption is only 7.35%. At the powered dose $M = 0.0540$, the learned mean is 0.0512 and adoption is 94.17%. The null point adoption rate, 5.76%, satisfies the registered 6.5% finite-test gate; its Wilson upper endpoint is 6.52%, so the interval is not wholly below that threshold. Every world passes the equal-frontier, equal-burden, and bridge-availability structural checks.

This evidence makes the option/exercise distinction concrete. Exact closure preservation can create access to a genuinely positive project while a conservative learner rationally fails to exercise a small option at the available signal-to-noise ratio. Conversely, the powered row shows recovery only for a deliberately larger treatment. The experiment validates the declared bridge mechanism under calibrated synthetic noise. It is not a historical market test, a causal return estimate, an alpha estimate, or a claim that the powered margin is empirically prevalent.

6.4 Capacity and price extensions

Exact source-relative compression holds productive state fixed. A different managerial problem allows productive value to vary under a hard maintenance capacity or a unit burden price. For a finite eligible family, the hard-capacity value is a nondecreasing step function of capacity; complementarity can make it nonconcave and can leave a capacity-optimal library unsupported by any scalar price. The price-penalized value is the upper envelope of finitely many affine branches, hence is continuous, nonincreasing, convex, and piecewise affine, with optimizer burden weakly decreasing in price.

These facts are standard finite resource-allocation geometry rather than a second principal contribution. Complete assumptions, theorem statements, proofs, and counterexamples are moved to Online Resource 1, Section S9. The main article uses only their implication: local safety, global minimum burden, and value under an insufficient capacity are three different questions.

7 Registered computational study

Registered Algorithmic Compression Benchmark v2 is a finite synthetic study of tagged identity-closure instances. It can test exact-method agreement, certified burden quality, preprocessing prevalence, and the declared adversarial mechanisms. It cannot establish average-case complexity, steady-state runtime rankings, or a population law for financial libraries.

7.1 Registered design, execution, and audit

The final registry contains 173 analysis instances: 24 small exact, 128 structured, and 21 adversarial. Family B requested variation in strategy and tagged-row counts, coverage density, module overlap, bundle prevalence, unique-carrier frequency, weight dispersion, and frontier-module association. Family C contains the predeclared gap, tie, duplicate, dominance, rare-carrier, symmetry, and bundle mechanisms. Failed generation remains a terminal result; no failed row is silently repaired or dropped.

The algorithm grid contains enumeration, requirement-mask DP, two registered HiGHS formulations, weighted and cardinality greedy, greedy plus reverse deletion, rechecked deletion orders, seeded random deletion, and 32-start deletion. Maximum immediate additive burden release is exactly the heaviest-safe-first ordering. Both labels remain in the immutable registered records, but they are one method rather than independent scientific baselines.

The audit passed all 3907 registered run units and accepted 3526 returned libraries after original-coordinate mandatory, coverage, frontier, identity-closure, and exact-burden checks. It retained 381 unsuccessful records in their terminal categories. HiGHS OPTIMAL remains solver evidence plus an exact incumbent check, not formal or exhaustive proof.

7.2 Exact agreement and applicability

Among 38 registry instances, all required exact-method burdens were available and equal on 31; 7 did not have a complete comparison. 6 adversarial enumeration jobs are recorded as IMPLEMENTATION_ERROR, but their substantive cause is known: the residual optional count exceeded the declared enumeration cap. Their historical statuses are not rewritten. They are enumeration-inapplicable under that cap, and exact agreement is unavailable rather than inferred. Optimizer identities differed at equal burden on 9 complete comparisons, which records valid tied optima rather than disagreement.

These are exact finite facts about specified instances. The medium structured references are HiGHS-optimal incumbents that pass exact feasibility and burden rechecks; no second solver was pinned.

Algorithm	Attained/reference	Mean gap	Median	q90	Worst
<i>Family A: exact finite reference</i>					
Weighted greedy	16/23	0.08684	0	0.25	0.5
Cardinality greedy	13/23	0.6375	0	1.4	7
Weighted greedy + reverse deletion	20/23	0.02923	0	0.16	0.25
Heaviest safe first	22/23	0.008696	0	0	0.2
Lightest safe first	5/23	1.586	1	3.32	7
Maximum immediate release	22/23	0.008696	0	0	0.2
Minimum unique-carrier exposure	17/23	0.2303	0	0.2686	3
Declared source order	11/23	0.6965	0.2857	2	3
Random-order deletion	8/23	1.384	0.6667	3.3	7
32-start random deletion	23/23	0	0	0	0
<i>Family B: HiGHS-OPTIMAL, exactly rechecked reference</i>					
Weighted greedy	82/111	0.02982	0	0.06452	0.5
Cardinality greedy	24/111	0.8904	0.25	2	7.5
Weighted greedy + reverse deletion	103/111	0.01038	0	0	0.5
Heaviest safe first	94/111	0.09983	0	0.2	3
Lightest safe first	11/111	4.128	0.7143	15	63
Maximum immediate release	94/111	0.09983	0	0.2	3
Minimum unique-carrier exposure	98/111	0.04435	0	0.05556	1
Declared source order	36/111	1.533	0.1818	5	20
Random-order deletion	28/111	1.753	0.2222	6.8	17
32-start random deletion	76/111	0.1324	0	0.5	2.667

Table 3 Registered heuristic burden quality. Relative gap is $(W - W^*)/W^*$. Attainment denominators include every registered row with an available reference; gap moments use exactly feasible returned libraries. Family A and Family B have different reference evidence and are deliberately not pooled.

7.3 Certified burden quality

Table 3 separates Family A exact finite references from Family B exactly rechecked HiGHS-optimal references. Within those distinct evidence classes, weighted greedy plus reverse deletion and heaviest-safe-first often attain the available reference, while several other deletion orders have larger observed tails. Multi-start attains every available Family A reference but not every Family B reference. These are descriptive outcomes of the registered instances, not approximation guarantees or an algorithm ranking beyond the design.

The registered heaviest-safe-first gap family gives a different kind of evidence. Its exactly rechecked finite ratios match Theorem 5.5 at every registered scale. The theorem establishes the universal family statement; the experiment checks the implementation on the named fixtures.

7.4 Preprocessing and realized design limits

Full fixed-point preprocessing has a strong realized effect. In every structured size cell, median variable and requirement reduction is complete among available records. This is evidence that forced-carrier propagation is important for these generators. It is simultaneously a limitation: most structured instances do not exercise a difficult residual MIP or DP search, so the study cannot identify general residual-search scaling.

Scale	Parameter	Reference evidence	Observed ratio	Theoretical ratio
small	$k = 4, \epsilon = \frac{1}{2}$	exact finite	$\frac{8}{3}$	$\frac{8}{3}$
medium	$k = 16, \epsilon = \frac{1}{2}$	HiGHS-OPTIMAL, rechecked	$\frac{32}{3}$	$\frac{32}{3}$
large	$k = 64, \epsilon = \frac{1}{2}$	HiGHS-OPTIMAL, rechecked	$\frac{128}{3}$	$\frac{128}{3}$

Table 4 Registered adversarial instances for heaviest-safe-first. Equality between observed and theoretical ratios is a synthetic fixture check of the human-proved family, not an empirical approximation guarantee.

Size	Preprocess n	Median var. red.	Median row red.	Median log PAR-2 ratio	32-start complete	Improved
N64-R32	28/32	1	1	1.187	28/32	20/28
N64-R64	28/32	1	1	1.29	28/32	19/28
N192-R32	28/32	1	1	2.031	15/32	12/15
N192-R64	28/32	1	1	2.243	27/32	21/27

Table 5 Registered preprocessing and multi-start evidence on structured instances. Reductions are shares removed. The log ratio is $\log(\text{PAR2}_{\text{full}}/\text{PAR2}_{\text{mandatory}})$, so a positive value favors the mandatory-only pipeline in elapsed penalized runtime despite the model-size reduction. Unavailable multi-start endpoints are not coded as zero improvement.

The registered generator also contains infeasible requested cells. All 352 run units arising from 16 structured source-instance cells are retained, and the artifact audit shows that the failures remove complete combinations of low density and bundle/unique-carrier settings rather than occurring at random. Effects and interactions requiring those absent cells are not estimable from this design.

Multi-start deletion completes all registered starts on 143 instances and improves on the first start on 96. Memory-limited primary runs remain failures; successful sensitivity records are not substituted into their denominators.

7.5 Why runtime and registered regressions are not inferential evidence

The v2 worker timed first-call Julia execution without an algorithm-specific warm-up. Compilation can constitute nearly all recorded algorithm time in representative raw records. Execution also used 8 concurrent one-thread subprocesses on a dirty worktree, so cache, memory-bandwidth, launch wave, and compilation effects are part of each observation. The saved wall times remain auditable cold-start concurrent records, but the article does not use them to rank algorithms or infer steady-state scaling.

The registered performance profile also combines unlike success events: heuristics need an exactly feasible candidate, whereas exact methods must complete search. It is therefore excluded from the submission’s comparative evidence. A valid future study must separate feasibility construction, quality-qualified attainment, and exact completion.

The prespecified regressions used HC3 uncertainty on repeated algorithm and preprocessing rows from the same instance. HC3 addresses heteroskedasticity, not within-instance dependence. Complete generator cells are also absent. Accordingly, their

coefficients, intervals, and adjusted probabilities are preserved in Online Resource 1 as an executed-analysis record but are not used for inferential or structural-driver claims. No unregistered clustered model is substituted after seeing the results.

7.6 Failure handling and next experimental boundary

No final run ended at `TIME_LIMIT`, so v2 supplies no empirical distribution of timeout incumbent gaps. Generation failures, memory limits, interruptions, missing primals, and nonbinary-candidate rejections remain in the machine-readable census and all applicable denominators.

A stronger performance study requires a new registration rather than an amendment to v2: warm-up outside timing, repeated blocked measurements on a clean tree, prospective generator-feasibility checks, nonempty residual cover families, common success targets, instance-level paired or clustered analysis, prevalidated method applicability, and bounded-exchange and LP-bound baselines. No outcome from such a study is claimed here.

8 Financial evidence: point-in-time portfolio-library retention

The financial study asks whether innovation-safe compression can reduce the active validation burden of portfolio-strategy libraries while preserving a larger menu of admissible future research options than a budget-matched frontier-only library. It is a retrospective development-facing panel, not an independent confirmatory market test. Its contribution is the combination of point-in-time source construction, an explicit portfolio grammar, protected proposal decisions, exact structural rechecks, and a transparent held-out null result.

8.1 Sequential information boundary

At each annual origin y , early historical blocks construct the source portfolio-strategy library and its operating frontier; the subsequent predecision block estimates compression profiles and solves the safe and frontier-only arms; year $y + 1$ enumerates closure-enabled proposal options and freezes both policy choices; and year $y + 2$ supplies the held-out portfolio return paths. Proposal outcomes do not enter compression. Evaluation outcomes do not enter compression, option generation, uncertainty estimation, adoption, or any policy choice. The full candidate ledger retains unavailable and rejected candidates rather than conditioning the analysis on successful selection.

The primary universe is formed from the most liquid eligible US common equities at each compression origin. A separate plain-ETF universe is the registered replication when its point-in-time eligibility gate passes. Eligibility never requires survival into the proposal or evaluation window; observed delisting returns are retained and a terminated position subsequently moves to cash. The single failed ETF universe gate remains in the registered denominator. Raw security identifiers, prices, and returns remain inside the licensed boundary.

Universe	Origins	Mean source burden	Mean safe burden	Mean reduction	Added options	Held-out CE difference
Common equity	19	8169.2	2183.8	73.1%	65.95	0.0000
Plain ETF	18	8909.2	2231.7	74.7%	66.06	-0.0022

Table 6 Point-in-time portfolio-library retention results. Burden is the registered validation-computation index, not currency. Added options are safe-library proposal actions absent from the budget-matched frontier-only library. The held-out column is innovation-safe minus frontier-only certainty-equivalent difference: all 19 common-equity choices coincide, while one of 18 complete ETF choices differs. Safe solutions pass exact original-instance frontier, closure, inactive-retention, and burden rechecks; solver-reported optimality remains solver evidence.

Each source library applies a fixed factorial grammar of portfolio-level signals, filters, holding horizons, allocation rules, gross-exposure targets, and exits to the same contemporaneous universe. Closure consists of 31 declared rule, interface, data, corporate-action, portfolio-aggregation, and cost/capacity capabilities. These declarations are more operationally concrete than free-form module tags, but they remain a modeled grammar: they do not prove that tacit knowledge or every institutional artifact is frictionlessly recoverable.

8.2 Compression and protected proposal policy

The innovation-safe arm minimizes the registered validation-computation burden subject to exact source-frontier and full declared-closure equality. Its frontier-only comparator is augmented under the safe arm’s realized burden cap using a frozen preproposal order. The comparator action set is a subset of the safe action set in every gate-passed cell. Cash is always feasible, and the safe policy begins with the exact comparator choice. It can replace that choice only when a simultaneous moving-block lower confidence bound for incremental proposal value clears the registered economic hurdle. The policy therefore treats closure as an option, not a forced trade.

Across the gate-passed origins, 3 safe solutions are resolved by exact preprocessing and 34 are returned with HiGHS OPTIMAL status. Every result is independently reconstructed and rechecked for mandatory inactive retention, exact table frontier equality, exact identity-closure equality, and exact integer burden. Those rechecks certify feasibility and burden; for the solver-resolved cells, global optimality remains mixed-integer solver evidence rather than a formal or exhaustive proof.

Table 6 gives the aggregate structural and held-out results. Innovation-safe retention reduces the mean source burden index by 73.1% in common equities and 74.7% in ETFs while preserving all registered capabilities. Relative to the budget-matched frontier-only library, it exposes on average 65.95 additional common-equity proposal actions and 66.06 additional ETF actions.

8.3 Held-out result: options are not exercises

The protected policy makes 0 common-equity replacements. Consequently, every common-equity held-out innovation-safe-minus-frontier-only contrast is exactly zero because the two policies choose the same action. This is not evidence that the two

libraries are economically equivalent; it is evidence that none of the additional options clears the proposal-time uncertainty rule.

The ETF replication makes 1 replacement. That replacement realizes a held-out certainty-equivalent difference of -0.0390, producing a mean of -0.0022 over the complete ETF origins. All other complete ETF pairs choose the same action. The negative observation is retained and is not offset by reporting a favorable post hoc selector. Full origin-level results, the prespecified robustness grid, negative-control policies, search-bias diagnostics, and capacity records appear in Online Resource 1.

The contrast is a postdecision portfolio certainty-equivalent diagnostic under the registered risk and cost rule. It is not an alpha estimate, a forecast, or a deployable-performance claim. In particular, the many structural options do not count as independent financial observations, and the safe policy’s abstention cannot be reinterpreted as positive performance. The calibrated mechanism experiment in Section 6.3 supplies the separate finite-simulation result: a small positive bridge can be structurally real yet statistically unrecoverable, while a prospectively powered bridge is recovered.

8.4 Robustness, licensed-data boundary, and managerial implication

The proposal robustness analysis retains the full prespecified grid, including unavailable cells. Common-equity safe and comparator choices remain identical throughout that grid. ETF differences remain confined to the same origin and disappear at higher transaction-cost settings. A closure-sham analysis that was not frozen before proposal outcomes became known is explicitly unavailable, not reconstructed post hoc. Capacity analysis scores frozen choices without reselection and retains missing-source, incomplete-path, and hard participation- cap failures in its denominators.

Licensed daily panels were used locally to construct the point-in-time universes and portfolio paths. The public artifact redistributes only aggregate structural manifests, cryptographic seals, permitted summaries, and synthetic records. It contains no raw CRSP/WRDS rows, security identifiers, or return paths. Independent replication of the licensed stages therefore requires separate data access, whereas all published table values can be regenerated from the committed public-safe manifests.

For research governance, the result is useful precisely because it is not a success story about returns. A manager can remove substantial declared validation burden while exactly preserving current opportunities and a much larger research-option menu. Whether any added option should be exercised is a separate statistical decision, and a protected selector can rationally abstain. Removal from the active maintenance set also does not authorize deletion from legal, compliance, or reproducibility archives. A stronger empirical claim would require a newly locked market vintage or independent panel, documented real capability inventories, and monetary maintenance costs calibrated without using postdecision outcomes.

9 Conclusion

Financial strategy libraries can be treated as productive inventories rather than merely lists of currently attractive rules. Under the article’s explicit factorization assumption

for a declared finite partial-information process, the operating frontier and generative closure are a sufficient library coordinate. Their exact equality then provides a precise source-relative preservation contract. The result is conditional: if later transitions, admission, duration, or rewards depend on raw provenance or identities beyond those objects, the projection does not apply.

That semantic reduction leads to a distinct operations-research problem. Under identity closure, safe compression is a tagged weighted cover; even a restricted decision subclass is NP-complete. This complexity result does not make set cover novel, nor does it extend the tagged formulation to arbitrary closure. It explains why globally minimum retention burden cannot generally be inferred from local deletion safety. The parameterized gap family sharpens that distinction for a named heaviest-safe-first rule without claiming that every deletion order fails.

The algorithmic response is deliberately plural. Enumeration and requirement-mask dynamic programming provide independent exact finite methods on small instances; exact preprocessing can expose forced structure; the MIP workflow supplies auditable solver evidence; weighted greedy transfers a classical covering guarantee within the identity-closure boundary; and certified deletion methods provide feasible, irreducible endpoints without an exactness claim. Every route ends with an original-instance certificate, so feasibility does not depend on trusting a heuristic or solver status.

The registered synthetic benchmark supplies exact-method agreement, certificate, heuristic-burden, failure, and preprocessing evidence. Its cold Julia startup timings, unequal success targets, sparse structured grid, and within-instance dependence do not support algorithm-speed rankings or general scaling inference. These are locked finite-design findings under one concurrent, dirty-worktree execution environment, not average-case complexity or population evidence about financial libraries.

The retrospective point-in-time financial study illustrates the burden and option implications for an explicit portfolio grammar. Innovation-safe retention preserves every declared capability while materially reducing the validation-computation index and exposing many proposal actions unavailable to a budget-matched frontier-only library. Yet the protected policy never replaces the common-equity comparator, and its one ETF replacement loses held-out value. The calibrated synthetic experiment resolves the apparent tension: a small positive bridge may be structurally present but statistically unrecoverable, whereas a prospectively powered bridge is recovered under the declared noise law. The first result is retrospective financial evidence; the second is synthetic mechanism evidence. Neither supports a causal, forecasting, alpha, or deployable-performance claim, and licensed source rows remain outside the public artifact.

Two extensions are now most important. Algorithmically, nonidentity closure requires a representation that preserves derived capability interactions without replacing them by independent raw rows. Empirically, the next study should use a newly locked market vintage or an independent panel, richer documented institutional capability artifacts, and monetary maintenance costs calibrated independently of postdecision outcomes. Neither extension can be obtained by relabeling the present locked designs. Both should retain the separation among human proof, formal verification, exact computation, solver evidence, synthetic experiments, and retrospective financial evidence used here.

Supplementary information

Online Resource 1, entitled *Proofs, Extended Results, Computational Records, and Reproducibility Materials*, contains the complete supporting proofs, extended theory and counterexamples, canonical numerical records, the historical frozen $N = 1024$ study, the distinct registered algorithmic benchmark, point-in-time financial and calibrated mechanism records, solver and exactness records, theorem–evidence correspondence, and reproducibility commands. It compiles independently from `journal/aor/online_resource/main.tex`; raw licensed data are excluded. The complete SSRN upload PDF appends this resource after the article’s references; its pages and numbered results use an S prefix. Separate article and supplement PDFs are also provided in the repository’s `release/v0.2.0-ssrn/` directory.

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Data availability. Exact fixtures, synthetic-study inputs and public-safe aggregate outputs, generated tables, and generated figures supporting the journal version are provided in the project repository. The retrospective financial audits use licensed CRSP/WRDS source data. Raw and row-level licensed observations cannot be redistributed. Researchers with independent licensed access can reproduce the licensed-data stages using the documented input contracts. Public materials and data-access instructions are available at <https://github.com/dramirez37/financial-strategy-library-compression>. Original manuscript and documentation materials are licensed under CC BY 4.0, subject to the licensed-data and third-party exclusions in the repository’s LICENSE file.

Code availability. The project repository contains Julia 1.12.6 and Lean 4.32.0 source, pinned environments, exact fixtures, registered configurations, and reproducibility commands. The editable article and Online Resource source trees include every public-safe table, figure, and canonical-data input they consume; public-safe Julia and Lean source remains in the project repository linked above. Original software is licensed under the MIT License. The `release/v0.2.0-ssrn/` directory records PDF checksums, source provenance, and submission metadata; no archival DOI is claimed.

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References

- Allen, F., Karjalainen, R.: Using genetic algorithms to find technical trading rules. *Journal of Financial Economics* **51**(2), 245–271 (1999) [https://doi.org/10.1016/S0304-405X\(98\)00052-X](https://doi.org/10.1016/S0304-405X(98)00052-X)
- Board of Governors of the Federal Reserve System, Office of the Comptroller of the Currency: Supervisory Guidance on Model Risk Management. Supervision and Regulation Letter SR 11-7, attachment. Official supervisory guidance (2011). <https://www.federalreserve.gov/boarddocs/srletters/2011/sr1107a1.pdf>
- Bowers, M., Olausson, T.X., Wong, L., Grand, G., Tenenbaum, J.B., Ellis, K., Solar-Lezama, A.: Top-down synthesis for library learning. *Proceedings of the ACM on Programming Languages* **7**(POPL), 1–32 (2023) <https://doi.org/10.1145/3571234>
- Baker, N.R., Souder, W.E., Shumway, C.R., Maher, P.M., Rubenstein, A.H.: A budget allocation model for large hierarchical R&D organizations. *Management Science* **23**(1), 59–70 (1976) <https://doi.org/10.1287/mnsc.23.1.59>
- Chvátal, V.: A greedy heuristic for the set-covering problem. *Mathematics of Operations Research* **4**(3), 233–235 (1979) <https://doi.org/10.1287/moor.4.3.233>
- Cassandra, A.R., Littman, M.L., Zhang, N.L.: Incremental pruning: A simple, fast, exact method for partially observable markov decision processes. In: *Proceedings of the Thirteenth Conference on Uncertainty in Artificial Intelligence*, pp. 54–61. Morgan Kaufmann Publishers Inc., San Francisco, CA, USA (1997). <https://arxiv.org/abs/1302.1525>
- Cont, R.: Model uncertainty and its impact on the pricing of derivative instruments. *Mathematical Finance* **16**(3), 519–547 (2006) <https://doi.org/10.1111/j.1467-9965.2006.00281.x>
- Dey, D., Liu, T.Y., Sofman, B., Bagnell, J.A.: Efficient optimization of control libraries.

- Proceedings of the AAAI Conference on Artificial Intelligence **26**(1), 1983–1989 (2012) <https://doi.org/10.1609/aaai.v26i1.8383>
- Dolan, E.D., Moré, J.J.: Benchmarking optimization software with performance profiles. *Mathematical Programming* **91**(2), 201–213 (2002) <https://doi.org/10.1007/s101070100263>
- Dai, M., Zhang, Q., Zhu, Q.J.: Trend following trading under a regime switching model. *SIAM Journal on Financial Mathematics* **1**(1), 780–810 (2010) <https://doi.org/10.1137/090770552>
- Givan, R., Dean, T., Greig, M.: Equivalence notions and model minimization in Markov decision processes. *Artificial Intelligence* **147**(1–2), 163–223 (2003) [https://doi.org/10.1016/S0004-3702\(02\)00376-4](https://doi.org/10.1016/S0004-3702(02)00376-4)
- Gleixner, A., Hendel, G., Gamrath, G., Achterberg, T., Bastubbe, M., Berthold, T., Christophel, P.M., Jarck, K., Koch, T., Linderoth, J., Lübbecke, M., Mittelman, H.D., Ozyurt, D., Ralphs, T.K., Salvagnin, D., Shinano, Y.: MIPLIB 2017: Data-driven compilation of the 6th mixed-integer programming library. *Mathematical Programming Computation* (2021) <https://doi.org/10.1007/s12532-020-00194-3>
- Golovin, D., Krause, A.: Adaptive submodularity: Theory and applications in active learning and stochastic optimization. *Journal of Artificial Intelligence Research* **42**, 427–486 (2011) <https://doi.org/10.1613/jair.3278>
- Glasserman, P., Xu, X.: Robust risk measurement and model risk. *Quantitative Finance* **14**(1), 29–58 (2014) <https://doi.org/10.1080/14697688.2013.822989>
- Harvey, C.R., Liu, Y., Zhu, H.: ... and the cross-section of expected returns. *The Review of Financial Studies* **29**(1), 5–68 (2016) <https://doi.org/10.1093/rfs/hhv059>
- Karp, R.M.: Reducibility among combinatorial problems. In: Miller, R.E., Thatcher, J.W., Bohlinger, J.D. (eds.) *Complexity of Computer Computations*, pp. 85–103. Springer, Boston, MA (1972). https://doi.org/10.1007/978-1-4684-2001-2_9. https://link.springer.com/chapter/10.1007/978-1-4684-2001-2_9
- Lakner, P.: Optimal trading strategy for an investor: The case of partial information. *Stochastic Processes and their Applications* **76**(1), 77–97 (1998) [https://doi.org/10.1016/S0304-4149\(98\)00032-5](https://doi.org/10.1016/S0304-4149(98)00032-5)
- Li, K., Nyström, K., Olofsson, M.: Optimal switching problems under partial information. *Monte Carlo Methods and Applications* **21**(2), 91–120 (2015) <https://doi.org/10.1515/mcma-2014-0013>
- Mutti, M., Del Col, S., Restelli, M.: Reward-free policy space compression

- for reinforcement learning. In: Proceedings of the 25th International Conference on Artificial Intelligence and Statistics. Proceedings of Machine Learning Research, vol. 151, pp. 3187–3203. PMLR, A Virtual Conference (2022). <https://proceedings.mlr.press/v151/mutti22a.html>
- McLean, R.D., Pontiff, J.: Does academic research destroy stock return predictability? *The Journal of Finance* **71**(1), 5–32 (2016) <https://doi.org/10.1111/jofi.12365>
- Moody, J.E., Saffell, M.: Learning to trade via direct reinforcement. *IEEE Transactions on Neural Networks* **12**(4), 875–889 (2001) <https://doi.org/10.1109/72.935097>
- Nemhauser, G.L., Wolsey, L.A., Fisher, M.L.: An analysis of approximations for maximizing submodular set functions—i. *Mathematical Programming* **14**, 265–294 (1978) <https://doi.org/10.1007/BF01588971>
- Pan, C., Ren, L., Xiong, L., Li, Y., Wei, W., Yang, X.: Regime-adaptive continual learning for portfolio management. In: Proceedings of the ACM SIGKDD Conference on Knowledge Discovery and Data Mining (2026). Accepted for publication; arXiv:2606.00143. <https://arxiv.org/abs/2606.00143>
- Russell, S.J., Wefald, E.: Principles of metareasoning. *Artificial Intelligence* **49**(1–3), 361–395 (1991) [https://doi.org/10.1016/0004-3702\(91\)90015-C](https://doi.org/10.1016/0004-3702(91)90015-C)
- Sondik, E.J.: The optimal control of partially observable markov processes over the infinite horizon: Discounted costs. *Operations Research* **26**(2), 282–304 (1978) <https://doi.org/10.1287/opre.26.2.282>
- Smallwood, R.D., Sondik, E.J.: The optimal control of partially observable markov processes over a finite horizon. *Operations Research* **21**(5), 1071–1088 (1973) <https://doi.org/10.1287/opre.21.5.1071>
- Sullivan, R., Timmermann, A., White, H.: Data-snooping, technical trading rule performance, and the bootstrap. *The Journal of Finance* **54**(5), 1647–1691 (1999) <https://doi.org/10.1111/0022-1082.00163>
- Weitzman, M.L.: Optimal search for the best alternative. *Econometrica* **47**(3), 641–654 (1979) <https://doi.org/10.2307/1910412>
- White, C.C.: Procedures for the solution of a finite-horizon, partially observed, semi-markov optimization problem. *Operations Research* **24**(2), 348–358 (1976) <https://doi.org/10.1287/opre.24.2.348>
- White, H.: A reality check for data snooping. *Econometrica* **68**(5), 1097–1126 (2000) <https://doi.org/10.1111/1468-0262.00152>
- Wolsey, L.A.: An analysis of the greedy algorithm for the submodular set covering problem. *Combinatorica* **2**, 385–393 (1982) <https://doi.org/10.1007/BF02579435>

Online Resource 1: Proofs, Extended Results, Computational Records, and Reproducibility Materials

for “Innovation-Safe Compression of Financial
Strategy Libraries: Semantics, Complexity, and
Algorithms”

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Abstract

Online Resource 1 for the accompanying research article. It contains complete supporting proofs, extended theoretical results and counterexamples, canonical numerical records, the frozen preprint-era randomized-library study, the distinct registered algorithmic benchmark, point-in-time financial and calibrated mechanism records, solver and exactness records, theorem–evidence correspondence, and reproducibility commands. Raw licensed CRSP/WRDS observations are excluded.

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Evidence boundary: Human proof, Lean kernel verification, exact finite computation, floating-point numerical evidence, mixed-integer solver evidence, synthetic experimental evidence, and retrospective financial evidence are labeled separately throughout.

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S1 Exact probability normalization and projection identities

For a finite carrier A , every probability law in the model is an exact rational table

$$\Delta_{\mathbb{Q}}(A) := \left\{ p : A \rightarrow \mathbb{Q} : p(a) \geq 0 \text{ for every } a, \sum_{a \in A} p(a) = 1 \right\}.$$

Beliefs, belief-kernel rows, and candidate-generation rows use this convention. At fixed project q , belief b , and closure C , verification assigns

$$\Gamma(q, b, C)(s) = G(q, b, C)(s) \nu(q, b, C, s), \quad s \in S, \quad (\text{S1.1})$$

$$\Gamma(q, b, C)(\text{none}) = G(q, b, C)(\text{none}) + \sum_{s \in S} G(q, b, C)(s) (1 - \nu(q, b, C, s)). \quad (\text{S1.2})$$

Every term is nonnegative. Summing (S1.1)–(S1.2) cancels the verification split and recovers the unit mass of G , so Γ is normalized.

Conditional on initial belief b , productive state $K = (F, C)$, and project q of duration d_q , let $\Lambda_q(\mathbf{b}, o \mid b, K)$ be the exact joint law of the belief path $\mathbf{b} = (b_0, \dots, b_{d_q})$ and admitted outcome o . Its specified marginals are

$$\sum_o \Lambda_q(\mathbf{b}, o \mid b, K) = \mathbf{1}\{b_0 = b\} \prod_{t=0}^{d_q-1} P_B(b_t, b_{t+1}), \quad (\text{S1.3})$$

$$\sum_{\mathbf{b}} \Lambda_q(\mathbf{b}, o \mid b, K) = \Gamma(q, b, C)(o). \quad (\text{S1.4})$$

These equations do not impose independence between the belief path and the admitted outcome.

For a raw source library L , deterministic update gives

$$\mathcal{Q}_q^{\text{raw}}(b', L' \mid b, L) := \sum_{\substack{\mathbf{b}, o: b_{d_q} = b' \\ L \oplus o = L'}} \Lambda_q(\mathbf{b}, o \mid b, K_L).$$

For realizable $K = (F, C)$, the two compressed pushforwards are

$$\overline{T}_q(K' \mid b, K) := \sum_{o: \text{addK}(K, o) = K'} \Gamma(q, b, C)(o), \quad (\text{S1.5})$$

$$\overline{\mathcal{Q}}_q(b', K' \mid b, K) := \sum_{\substack{\mathbf{b}, o: b_{d_q} = b' \\ \text{addK}(K, o) = K'}} \Lambda_q(\mathbf{b}, o \mid b, K). \quad (\text{S1.6})$$

Finite deterministic pushforward preserves nonnegativity and total mass. The local update identity $K_{L \oplus o} = \text{addK}(K_L, o)$ also makes every compressed successor realizable. These identities supply the normalization clause of the raw-to-compressed projection theorem; they do not supply the factorization, positive-duration induction, or contraction arguments by themselves.

S1.1 Complete projection proof

The journal article assumes a finite hidden state, finite filtered belief carrier, finite strategy catalog, finite project catalog, positive integer project durations, exact rational transition and completion laws, and a mandatory inactive strategy. Every operating reward, project availability indicator, project cost, duration, generation law, verification law, and joint terminal belief-admission law factors through the current belief and the productive library state $K_L = (F_L, C_L)$. Raw insertion and compressed insertion obey

$$K_{L \oplus o} = \text{addK}(K_L, o). \quad (\text{S1.7})$$

Stationary claims additionally assume a common discount factor $0 < \beta < 1$; finite-horizon claims do not.

Theorem S1.1 (Raw-to-compressed controlled semi-Markov projection). *Under the preceding assumptions, $K_L = (F_L, C_L)$, augmented during an unfinished project by project identity and remaining duration, is a controlled semi-Markov state for the raw library process. Raw and compressed finite-horizon values agree at every raw state, and the complete sets of maximizing actions agree under the natural action identification. Under the stationary discount clause, the raw and compressed Bellman operators have unique fixed points with the same values and maximizing stationary-action correspondence.*

Proof Fix a raw library L , belief b , and feasible action. For Continue, the operating reward and belief transition depend on L only through F_L , hence through K_L . For a project q , the factorization assumption makes feasibility, cost, duration, the incumbent operating-reward block, and the joint completion law functions of (b, K_L, q) . Equations (S1.1)–(S1.4) show that this law is normalized without imposing independence between the belief path and admission. Pushing it through the deterministic raw update and using (S1.7) gives exactly the compressed transition law (S1.6). Its support consists of realizable compressed states. Thus two raw libraries in the same K -fiber have the same feasible actions, current reward blocks, durations, and conditional laws of the next compressed decision state.

For a finite calendar horizon h , prove equality of every corresponding action value by strong induction on h . The claim is immediate at horizon zero. Continue invokes the induction hypothesis at $h - 1$; a feasible project invokes it after its positive duration d_q , at $h - d_q < h$. Current rewards and transition laws already agree by the preceding paragraph. Each corresponding action value therefore agrees, as do the finite maxima and the complete maximizing-action sets. Finiteness makes every maximum attained.

For the stationary clause, cast the exact rational tables into the reals and use the sup norm on the finite state carrier. Conditional expectation is nonexpansive. Continue discounts continuation by β , and a project discounts it by $\beta^{d_q} \leq \beta$. Maximization over the common

finite action set preserves this modulus, so both Bellman operators are β -contractions. Bellman intertwining, established by the action-value identity above, maps the compressed fixed point to a raw fixed point. Banach uniqueness gives value equality; the same pointwise action-value identity gives the complete maximizing stationary-action correspondence. No closure detectability assumption enters this forward projection. $\square$

S2 Productive Preservation and the Frontier–Closure Quotient

For a finite library A containing the inactive strategy, write

$$F_A(b) = \max_{s \in A} j_s(b), \quad C_A = \text{cl} \left(\bigcup_{s \in A} \text{mods}(s) \right), \quad K_A = (F_A, C_A).$$

The forward preservation result uses only the process factorization and local update assumptions stated in Section S1. A converse needs a distinct detectability assumption: whenever two libraries have equal frontiers but unequal closures, at least one availability-tagged project cost, duration, incumbent-reward block, or joint terminal observation differs after the declared generation, verification, admission, and projection steps.

Theorem S2.1 (Frontier–closure characterization). *If $F_A = F_L$ and $C_A = C_L$, then A and L induce identical productive observations and identical finite-horizon values and maximizing actions. The stationary conclusions of Section S1 also agree under its discount clause. If closure is detectable in the preceding sense, equality of all productive observations together with frontier equality implies $C_A = C_L$.*

Proof Frontier–closure equality is $K_A = K_L$. The factorization assumption then identifies project availability, cost, duration, incumbent operating rewards, generation and verification laws, and the conditional joint completion law. The normalized admission map and deterministic pushforward from Section S1 therefore identify the complete productive transition observations. The finite-horizon and stationary value and action claims follow from Theorem S1.1. Conversely, suppose the frontiers and all named productive observations agree. If the closures differed, detectability would produce a named observation that differs, a contradiction. Hence the closures agree. Detectability is used only for this reverse implication. $\square$

Corollary S2.2 (Source-relative safe compression). *For $s_0 \in A \subseteq L$, exact equality $F_A = F_L$ and $C_A = C_L$ is sufficient for productive preservation. With exact positive active weights and $w_{s_0} = 0$, the finite safe-feasible family contains L and therefore has at least one minimum-burden member.*

Proof The productive claim is the forward direction of the theorem. The source is safe relative to itself. A finite source has finitely many sublibraries, so the nonempty feasible set has an attained minimum under exact weights. $\square$

The converse characterization is deliberately restricted. It is not a claim that K is a generic coarsest bisimulation or a full-abstraction theorem. It states that the declared finite process factors through K , and that the additional detectability clause is sufficient to recover closure equality from the named productive observations.

S2.1 Operational factorization audit

The assumption can be checked against a proposed application by asking whether, conditional on belief and active-project state, every operating reward, generation law, verification outcome, admission probability, duration law, completion observation, and next-library update depends on the raw library only through (F_A, C_A) . Provenance, cardinality, identity-specific switching cost, or a requirement for a named parent violates the assumption unless it is added to the compressed state. A finite module-gated construction satisfies the assumption when all projects are indexed only by requirements contained in C_A , incumbent rewards use only F_A , and all subsequent kernels use only belief, active-project variables, and those two objects.

The following exact countermodel shows why the checklist is substantive. Let s_0 be inactive and let active strategies a and b have identical zero profiles and the same single module. Then libraries $\{s_0, a\}$ and $\{s_0, b\}$ have identical frontier and closure. If a project is available only when the raw identity a is present, their project menus differ. The model therefore does not factor through (F, C) , and the projection theorem makes no claim about it. The countermodel changes no theorem: it falsifies an omitted assumption, not the stated conditional result.

S3 Belief Occupation and Coverage

The main paper’s operational–generative value decomposition separates the passive operating channel from the research premium. This section develops the passive channel through discounted belief occupation and then gives the associated finite coverage, threshold, and belief-dynamics results. These are supporting results: a one-shot cost-covering region is not an optimized Bellman research region.

Throughout, expectations and occupation rows are conditional on the displayed initial belief. Every horizon and state carrier is finite. Finiteness justifies the sum rearrangements and attained extrema below; no limiting argument, closure-detectability condition, stationary contraction, or path–outcome independence is used.

S3.1 Discounted gap sums and operational insertion

Definition S3.1 (Positive gap and discounted gap sum). For a fixed candidate s , define

$$\begin{aligned}\Delta_{s,L}(b) &:= \max\{j_s(b) - F_L(b), 0\}, \\ G_0(b; L, s) &= 0, \\ G_{n+1}(b; L, s) &= \Delta_{s,L}(b) + \beta \sum_{b'} P_B(b, b') G_n(b'; L, s).\end{aligned}\tag{S3.1}$$

Let

$$w_n(b, b') := \sum_{t=0}^{n-1} \beta^t \Pr(B_t = b' \mid B_0 = b)$$

be discounted expected occupation, not a normalized probability.

For the supporting frozen-library process, set $\Delta_n^{\text{op}, \text{aux}}(s \mid b, L) := P_n(b, L \cup \{s\}) - P_n(b, L)$, where P_n is its passive value.

Proposition S3.2 (Passive gap-sum identity). *For the supporting frozen-library process,*

$$\Delta_n^{\text{op}, \text{aux}}(s \mid b, L) = G_n(b; L, s) = \sum_{b'} w_n(b, b') \Delta_{s, L}(b'). \quad (\text{S3.2})$$

When the unified process has this frozen-library no-research counterfactual, combining the occupancy identity with the operational–generative value decomposition in the main paper gives

$$\mathcal{I}_n = \sum_{b'} w_n(b, b') \Delta_{s, L}(b') + \Delta_n^{\text{gen}}. \quad (\text{S3.3})$$

The occupancy formula shows why a zero current gap does not make a candidate worthless if the belief process later visits states with positive gaps.

Corollary S3.3 (Zero current gap, positive future operating value). *With a deterministic move to a gap-two belief, $n = 2$, and $\beta = 1/2$, operational insertion value is one.*

Proposition S3.4 (Diminishing marginal operational innovation). *If $L \subseteq L'$, then every fixed candidate satisfies*

$$\Delta_n^{\text{op}}(s \mid b, L') \leq \Delta_n^{\text{op}}(s \mid b, L).$$

For Proposition S3.2, insertion replaces the frontier pointwise by $\max\{F_L, j_s\}$. Subtracting F_L gives $\Delta_{s, L}$. Conditional on the displayed initial belief and frozen library, subtracting the two passive recursions and inducting on the horizon gives (S3.1). Unrolling that finite recursion gives the discounted-occupancy representation; only finite sums are rearranged. The delayed-coverage specialization in Corollary S3.3 follows by placing unit discounted mass on the future belief with gap two. For Proposition S3.4, library inclusion raises the frontier and lowers the pointwise positive gap, and nonnegative discounted expectation preserves the order.

Persistence itself is ambiguous because it may keep the process in, or away from, the candidate’s advantage region. The valid positive comparison is occupation alignment on positive-gap beliefs, developed below, not persistence in isolation.

S3.2 Coverage potential and occupation bounds

Coverage links a candidate’s operational advantage to the beliefs at which that advantage will matter.

Definition S3.5 (Certified gap and coverage potential). For a fixed project q , library K , candidate profile j_q , and operational frontier F_K , define the certified gap

$$\Delta(q, K, b) := \max\{j_q(b) - F_K(b), 0\}.$$

Let $\omega_t(b, b') \geq 0$ be the date- t occupation weight of future belief b' from initial belief b . For horizon H , discount β , and candidate-survival factor ρ_q , set

$$W_H^{\beta, \rho_q}(b, b') := \sum_{t < H} \beta^t \rho_q^t \omega_t(b, b'), \quad (\text{S3.4})$$

$$\Psi_H(q, K, b) := \sum_{b' \in B} W_H^{\beta, \rho_q}(b, b') \Delta(q, K, b'). \quad (\text{S3.5})$$

The potential Ψ_H is discounted exposure to beliefs at which the candidate improves the current frontier. The weight W_H^{β, ρ_q} need not itself be a probability: it combines belief occupation, discounting, and survival of the candidate's advantage.

Theorem S3.6 (Exact finite coverage representation). *For every finite horizon, occupation table, gap table, project, library, and initial belief,*

$$\Psi_H(q, K, b) = \sum_{t < H} \sum_{b' \in B} \beta^t \rho_q^t \omega_t(b, b') \Delta(q, K, b'). \quad (\text{S3.6})$$

The right-hand side is the date-by-date one-shot gross operational value of the supplied candidate.

With nonnegative inputs, the same finite order argument makes Ψ_H increasing in the gap, discount, survival, and occupation weights. A pointwise higher frontier lowers the positive-part gap of a fixed candidate and therefore weakly lowers its coverage potential.

Corollary S3.7 (Finite occupation bounds). *For any nonempty $A \subseteq B$, its attained minimum gap times its discounted occupation is a lower bound for Ψ_H . The attained global maximum gap times total discounted occupation is an upper bound. If the gap vanishes outside A , the upper bound may use occupation restricted to A .*

Example S3.8 (Delayed coverage). On a two-belief grid, let date zero occupy the current belief and date one the other belief. With $H = 2$, $\beta = 1/2$, $\rho_q = 1$, current gap zero, and future gap two,

$$\Delta(q, K, b_{\text{current}}) = 0, \quad \Psi_2(q, K, b_{\text{current}}) = 1 > 0.$$

Screening only at the current belief would discard this future advantage.

Fix (q, K, b, H) . Substitute (S3.4) into (S3.5). Since both index sets are finite, distributivity and finite-sum commutation give

$$\begin{aligned}\Psi_H(q, K, b) &= \sum_{b' \in B} \left[\sum_{t < H} \beta^t \rho_q^t \omega_t(b, b') \right] \Delta(q, K, b') \\ &= \sum_{b' \in B} \sum_{t < H} \beta^t \rho_q^t \omega_t(b, b') \Delta(q, K, b') \\ &= \sum_{t < H} \sum_{b' \in B} \beta^t \rho_q^t \omega_t(b, b') \Delta(q, K, b'),\end{aligned}$$

which is (S3.6). This proof uses no limiting argument.

For a nonempty region $A \subseteq B$, write

$$\underline{\Delta}_A := \min_{b' \in A} \Delta(q, K, b'), \quad \overline{\Delta} := \max_{b' \in B} \Delta(q, K, b').$$

All weights and gaps are nonnegative. Hence

$$\underline{\Delta}_A \sum_{b' \in A} W_H^{\beta, \rho_q}(b, b') \leq \Psi_H(q, K, b) \leq \overline{\Delta} \sum_{b' \in B} W_H^{\beta, \rho_q}(b, b').$$

If the gap vanishes outside A , the last sum can be restricted to A . The four supporting monotonicities follow term by term: raising a nonnegative gap, discount, survival factor, or occupation weight raises every affected summand. If $F_0 \leq F_1$, then $\max\{j_q - F_1, 0\} \leq \max\{j_q - F_0, 0\}$, proving frontier saturation for a fixed candidate.

S3.3 Monotone one-shot coverage

For a one-step project, let $p(b) \geq 0$ scale survival or admission and let $\kappa(b)$ be initiation cost.

Definition S3.9 (One-shot cost-covering region). On a finite linearly ordered belief grid with exact transition kernel P , define

$$G(b) := \beta p(b) \sum_{b' \in B} P(b, b') \Delta(b'), \quad \mathcal{C} := \{b \in B : \kappa(b) \leq G(b)\}.$$

The set $\mathcal{C}$ is the *one-shot cost-covering region*.

Theorem S3.10 (Monotone-gap upper-threshold theorem). *Assume that B is a nonempty finite linearly ordered grid; P is an exact nonnegative row-stochastic kernel that is first-order stochastically monotone; Δ is nonnegative and increasing; p is nonnegative and increasing; $\beta \geq 0$; and κ is antitone. Then G is increasing and $\mathcal{C}$ is an upper set. Consequently, either $\mathcal{C} = \emptyset$, or there is $b^* \in B$ such that*

$$\mathcal{C} = \{b \in B : b^* \leq b\}. \tag{S3.7}$$

In the latter case the region is order-connected on the finite grid.

The theorem is about the displayed one-shot cost-covering region. We reserve *optimal research region* for the action selected by the Bellman policy. Continuation-value differences need not inherit the order properties used above, so the theorem does not characterize that Bellman region.

For each current belief b , let

$$E_\Delta(b) := \sum_{b' \in B} P(b, b') \Delta(b').$$

Here $E_\Delta(b)$ is the conditional expected gap after one transition. First-order stochastic monotonicity of P and monotonicity of Δ make it increasing. Exact row stochasticity and $\Delta \geq 0$ make it nonnegative. If $p \geq 0$ is increasing, then for $b \leq \tilde{b}$,

$$p(b)E_\Delta(b) \leq p(\tilde{b})E_\Delta(b) \leq p(\tilde{b})E_\Delta(\tilde{b}).$$

Multiplication by $\beta \geq 0$ therefore makes G increasing.

Suppose $b \in \mathcal{C}$ and $b \leq \tilde{b}$. Antitonicity of cost and monotonicity of gross value yield

$$\kappa(\tilde{b}) \leq \kappa(b) \leq G(b) \leq G(\tilde{b}),$$

so $\tilde{b} \in \mathcal{C}$. Thus $\mathcal{C}$ is an upper set. A nonempty upper set in a finite linear order has a least element b^* and equals $\{b : b^* \leq b\}$; the empty case is separate.

The cutoff directions and the exact failures of the threshold assumptions are collected in Section [S9.1](#).

S3.4 Patience, survival, and belief dynamics

Theorem S3.11 (Finite patience–survival complementarity). *Let P be an exact finite row-stochastic matrix, $g \geq 0$, and*

$$\Psi_H(\beta, \rho) := \sum_{t=0}^{H-1} (\beta \rho)^t P^t g.$$

The potential is pointwise increasing in either nonnegative β or ρ . If $0 \leq \beta_0 \leq \beta_1$ and $0 \leq \rho_0 \leq \rho_1$, then, componentwise,

$$\Psi_H(\beta_1, \rho_1) + \Psi_H(\beta_0, \rho_0) \geq \Psi_H(\beta_1, \rho_0) + \Psi_H(\beta_0, \rho_1). \quad (\text{S3.8})$$

Conditional on an initial belief, row stochasticity implies that P^t is nonnegative, so $P^t g \geq 0$. Monotonicity in either nonnegative scalar follows term by term. For the four displayed corners,

$$\Psi_H(\beta_1, \rho_1) + \Psi_H(\beta_0, \rho_0) - \Psi_H(\beta_1, \rho_0) - \Psi_H(\beta_0, \rho_1)$$

$$= \sum_{t < H} (\beta_1^t - \beta_0^t)(\rho_1^t - \rho_0^t)P^t g \geq 0. \quad (\text{S3.9})$$

Indeed, the date- t coefficient factorizes exactly as

$$(\beta_1 \rho_1)^t + (\beta_0 \rho_0)^t - (\beta_1 \rho_0)^t - (\beta_0 \rho_1)^t = (\beta_1^t - \beta_0^t)(\rho_1^t - \rho_0^t).$$

Every factor is nonnegative. Finiteness of H justifies the displayed termwise expansion. No independence, inverse, infinite series, real derivative, detectability, or contraction enters the result.

A scalar persistence parameter does not order coverage. Consider

$$P(\theta) = \begin{pmatrix} \theta & 1 - \theta \\ 1 - \theta & \theta \end{pmatrix}, \quad 0 \leq \theta \leq 1.$$

Proposition S3.12 (Opposite persistence effects). *Let $H = 2$, effective discount $\alpha = 1/2$, initial belief 0, and compare $\theta_0 = 1/4$ with $\theta_1 = 3/4$. Exact evaluation gives*

g	$\Psi_2(\theta_0)$	$\Psi_2(\theta_1)$
(1, 0)	9/8	11/8
(0, 1)	3/8	1/8
(1, 1)	3/2	3/2.

Thus greater persistence can raise, lower, or leave coverage unchanged.

Persistence raises coverage when it retains occupation at an uncovered current belief, but lowers it when it traps the process away from uncovered beliefs elsewhere. The positive order must therefore be stated in terms of where occupation moves. Define

$$W_{H,\alpha}^P(b, x) := \sum_{t=0}^{H-1} \alpha^t (P^t)(b, x).$$

Theorem S3.13 (Advantage-region occupation alignment). *Let P_0, P_1 be finite row-stochastic kernels and $g \geq 0$. If, for every initial belief b and every x with $g(x) > 0$,*

$$W_{H,\alpha}^{P_1}(b, x) \geq W_{H,\alpha}^{P_0}(b, x),$$

then

$$\sum_{t=0}^{H-1} \alpha^t P_1^t g \geq \sum_{t=0}^{H-1} \alpha^t P_0^t g$$

componentwise.

Fix the initial belief b . Finite sum exchange yields

$$\begin{aligned}\Psi_{H,\alpha}^P(b) &= \sum_{t < H} \alpha^t \sum_x (P^t)(b, x) g(x) \\ &= \sum_x \left[\sum_{t < H} \alpha^t (P^t)(b, x) \right] g(x) \\ &= \sum_x W_{H,\alpha}^P(b, x) g(x).\end{aligned}$$

Outside the advantage region $g(x) = 0$. Inside it, the assumed occupation difference and the gap are nonnegative. Termwise comparison proves Theorem S3.13.

For the exact persistence boundary, $H = 2$ gives $\Psi_2 = g + \alpha P(\theta)g$. At initial belief zero and $\alpha = 1/2$,

$$\begin{aligned}g = (1, 0) &: \Psi_2 = 1 + \theta/2, \\ g = (0, 1) &: \Psi_2 = (1 - \theta)/2, \\ g = (1, 1) &: \Psi_2 = 3/2.\end{aligned}$$

Substitution of $\theta = 1/4$ and $3/4$ gives respectively $9/8 \rightarrow 11/8$, $3/8 \rightarrow 1/8$, and $3/2 \rightarrow 3/2$. Both matrices are exactly row stochastic. The sign changes because persistence reallocates occupation; it does not order occupation on the positive-gap region without an alignment condition.

S4 Tagged Cover, Complexity, and Boundary Counterexamples

This section gives the complete proofs supporting the journal article's identity-closure covering representation and complexity result. The semantic safe-feasibility condition is defined for arbitrary declared closure; only the tagged row reduction and the algorithms built on it assume identity closure.

S4.1 An exact tagged covering representation under identity closure

Fix a finite belief set B , a finite module carrier M , a finite strategy catalog S , and a source library $L \subseteq S$. Every admissible library contains the distinguished inactive strategy s_0 . Profiles are exact rational functions $j_s : B \rightarrow \mathbb{Q}$, $j_{s_0}(b) = 0$ for every b , and $\text{mods}(s_0) = \emptyset$. Each $\text{mods}(s)$ is a subset of M . For an admissible library A , let

$$F_A(b) := \max_{s \in A} j_s(b), \quad U_A^{\text{raw}} := \bigcup_{s \in A} \text{mods}(s), \quad C_A := \text{cl}(U_A^{\text{raw}}).$$

The maximum is attained because A is finite and nonempty. The inactive strategy also implies $F_A(b) \geq 0$.

In this subsection only, closure is the identity on 2^M . Thus $C_A = U_A^{\text{raw}}$ for every A . Define the source-relative tagged requirement universe

$$\mathcal{U}_L^{\text{tag}} := \{\text{frontier}(b) : b \in B\} \cup \{\text{module}(m) : m \in C_L\}. \quad (\text{S4.1})$$

The constructors are disjoint: a frontier tag and a module tag are different even if their underlying identifiers have the same printed label. For each source strategy $s \in L$, define

$$\mathcal{R}_s := \{\text{frontier}(b) : j_s(b) = F_L(b)\} \cup \{\text{module}(m) : m \in \text{mods}(s)\}. \quad (\text{S4.2})$$

Equality in the first set is exact. Consequently every tied source maximizer covers the frontier row; no tie-breaking rule or distinguished maximizer enters the construction.

Theorem S4.1 (Tagged-cover equivalence under identity closure). *Under the assumptions above, let L' be an admissible source sublibrary: $s_0 \in L' \subseteq L$. Then*

$$F_{L'} = F_L \text{ and } C_{L'} = C_L \iff \mathcal{U}_L^{\text{tag}} \subseteq \bigcup_{s \in L'} \mathcal{R}_s. \quad (\text{S4.3})$$

All belief rows are retained in (S4.3). If $F_L(b) = 0$, the mandatory inactive strategy attains that frontier and automatically covers $\text{frontier}(b)$.

Proof Suppose first that $F_{L'} = F_L$ and $C_{L'} = C_L$. Fix $b \in B$. The finite maximum defining $F_{L'}(b)$ has an attainer $s \in L'$. Frontier equality gives $j_s(b) = F_{L'}(b) = F_L(b)$, so $\text{frontier}(b) \in \mathcal{R}_s$. This argument is existential and therefore permits any number of tied attainers. Next fix $m \in C_L$. Identity closure and closure equality give

$$m \in C_{L'} = U_{L'}^{\text{raw}} = \bigcup_{s \in L'} \text{mods}(s).$$

Hence some $s \in L'$ carries m , so $\text{module}(m) \in \mathcal{R}_s$. Every tagged requirement is covered.

Conversely, suppose every tagged requirement is covered. Because $L' \subseteq L$, maximization over the smaller set gives $F_{L'}(b) \leq F_L(b)$ for every b . Coverage of $\text{frontier}(b)$ supplies some $s \in L'$ with $j_s(b) = F_L(b)$. Therefore $F_L(b) = j_s(b) \leq F_{L'}(b)$, proving frontier equality pointwise. This also covers the zero-frontier case: $s_0 \in L'$ is itself a valid attainer.

For modules, source inclusion implies $U_{L'}^{\text{raw}} \subseteq U_L^{\text{raw}}$. If $m \in U_L^{\text{raw}} = C_L$, coverage of $\text{module}(m)$ supplies a retained carrier, so $m \in U_{L'}^{\text{raw}}$. Thus the two raw unions are equal. Identity closure converts this equality into $C_{L'} = C_L$. A module may have one or many source carriers; the row constraint requires at least one and does not select a canonical carrier. $\square$

Weighted binary formulation.

Let the exact rational retention weights satisfy $w_{s_0} = 0$ and $w_s > 0$ for $s \neq s_0$. Introduce one variable $x_s \in \{0, 1\}$ for each $s \in L$, where $x_s = 1$ means that s is retained. The theorem gives the exact formulation

$$\min_x \sum_{s \in L} w_s x_s, \quad (\text{S4.4})$$

$$\text{s.t.} \quad \sum_{s \in L: j_s(b)=F_L(b)} x_s \geq 1, \quad b \in B, \quad (\text{S4.5})$$

$$\sum_{s \in L: m \in \text{mods}(s)} x_s \geq 1, \quad m \in C_L, \quad (\text{S4.6})$$

$$x_{s_0} = 1, \quad x_s \in \{0, 1\}, \quad s \in L. \quad (\text{S4.7})$$

The source-sublibrary restriction is built into the column set L . The selected library is $L_x = \{s \in L : x_s = 1\}$, its burden is exactly the objective in (S4.4), and Theorem S4.1 identifies its feasible vectors with the exact innovation-safe sublibraries. Zero-frontier rows are redundant after $x_{s_0} = 1$, but retaining them makes the representation a literal tagged copy of every declared belief and avoids a hidden positive-frontier convention.

Why arbitrary closure is different.

For a nonidentity closure operator, equality of closed unions need not be equivalent to retaining a raw carrier for each member of the source closure. Capabilities may be generated jointly. A correct general-closure model must enforce the set-valued equality

$$\text{cl}\left(\bigcup_{s: x_s=1} \text{mods}(s)\right) = C_L$$

or use a separately proved valid extended formulation. The independent module rows in (S4.6) have no automatic validity in that setting.

Exact boundary examples.

The following finite examples use integer profiles and exact finite sets.

1. *Frontier rows alone are insufficient.* Take one belief, one module m , identity closure, and $L = \{s_0, s_m\}$, with both profiles zero, $\text{mods}(s_0) = \emptyset$, and $\text{mods}(s_m) = \{m\}$. The sublibrary $\{s_0\}$ covers the only frontier row and preserves the zero frontier, but its closure is empty rather than $\{m\}$.
2. *Module rows alone are insufficient.* Take one belief, identity closure, and $L = \{s_0, s_1\}$, with $j_{s_0} = 0$, $j_{s_1} = 1$, and both module sets empty. The source module row family is empty, so $\{s_0\}$ satisfies it vacuously, but its frontier is zero rather than one.
3. *Independent raw-module rows can fail under closure complementarity.* Let $M = \{a, b, c\}$ and

$$\text{cl}(A) = \begin{cases} A \cup \{c\}, & \{a, b\} \subseteq A, \\ A, & \text{otherwise.} \end{cases}$$

This map is extensive, monotone, and idempotent. Let the zero-profile source be $L = \{s_0, s_a, s_b\}$, where s_a carries only a and s_b carries only b . Then $C_L = \{a, b, c\}$, and the source is of course safe-feasible relative to itself. Nevertheless a raw-carrier row for $c \in C_L$ has no covering source strategy. The naive independent-row model therefore rejects even the source library.

The deterministic machine-readable realization is `journal/aor/fixtures/tagged_cover_equivalence_v1.json`. It records exact finite computation in Julia,

including all sublibraries of its declared small-library grid; it is neither an exhaustive proof over arbitrary finite instances nor a solver optimality certificate. The universal core biconditional and exact burden identity are separately checked in the Lean source `TaggedSafeCompression.lean` under `formal/StrategyInnovation/Optimization/`. Those declarations formalize neither a complexity result nor an approximation guarantee.

S4.2 Computational complexity of exact compression

We use an explicit finite encoding. A source instance lists a finite strategy library $L = \{s_0, s_1, \dots, s_n\}$, a finite nonempty belief set $\mathcal{B}$, exact binary-rational profiles $j_s(b)$, a finite module universe $\mathcal{M}$, and the incidence rows $\text{mods}(s) \subseteq \mathcal{M}$. Strategy s_0 is the mandatory inactive strategy, has profile zero at every belief, carries no modules, and has zero burden. Every active strategy has a positive binary-integer weight w_i , and the threshold B is a nonnegative binary integer. Closure is identity closure,

$$C(A) = \bigcup_{s \in A} \text{mods}(s).$$

The source frontier is $F_L(b) = \max_{s \in L} j_s(b)$, and the burden of a sublibrary is $W(A) = \sum_{s_i \in A, i \geq 1} w_i$. The decision problem **IDENTITY-CLOSURE SAFE COMPRESSION (ISC-ID)** asks whether there is an $A \subseteq L$ such that $s_0 \in A$, $W(A) \leq B$, $F_A = F_L$ at every listed belief, and $C(A) = C(L)$.

The hardness source below is a deliberately normalized version of weighted set cover. In **FULL-UNION WEIGHTED SET COVER (FU-WSC)**, the input is a nonempty finite universe U , an indexed finite family $(A_i)_{i=1}^n$ whose union is U , positive binary-integer costs c_i , and a nonnegative binary-integer budget B . The question is whether some index set $I \subseteq \{1, \dots, n\}$ satisfies $\bigcup_{i \in I} A_i = U$ and $\sum_{i \in I} c_i \leq B$. Encodings that fail these stated syntactic restrictions are rejected, so FU-WSC is a language rather than an unverified promise problem.

Lemma S4.2 (Normalized weighted set cover). *FU-WSC is NP-complete.*

Proof Input validity is checkable in polynomial time. A selected-index bit vector is a polynomial certificate: unioning its listed incidence rows and adding its binary-integer costs verifies feasibility in polynomial time. Thus FU-WSC is in NP.

For hardness, reduce the standard **SET COVER** decision problem, one of Karp's NP-complete covering problems (Karp 1972). Write an input as $(U, (A_i)_{i=1}^n, k)$, with nonnegative cardinality bound k . If $U = \emptyset$, output the fixed yes-instance consisting of universe $\{u\}$, the one set $\{u\}$, cost one, and budget one. If $U \neq \emptyset$ but $\bigcup_i A_i \neq U$, output the fixed no-instance with universe $\{u\}$, the one set $\{u\}$, cost two, and budget one. Otherwise retain the universe and family, assign every set unit cost, and set the budget to k . Every output has a nonempty universe, a full-union family, positive integer costs, and a nonnegative integer budget. The first branch preserves the fact that the empty universe is coverable; the second preserves the fact that an uncovered element makes the source instance a no-instance; and in the third branch cost is exactly selected-set cardinality. The case split, union check, and copied incidence representation have polynomial size and construction time. Hence the mapping preserves yes- and no-instances and proves NP-hardness. $\square$

Theorem S4.3 (Identity-closure safe-compression complexity). *ISC-ID is NP-complete. More specifically, it remains NP-complete on the subclass with exactly one belief, zero profiles for every active strategy, an inactive strategy with zero profile, zero burden, and no modules, and a source library whose active module rows have full union. The result holds even when every active weight is one. Consequently, minimum-burden identity-closure safe compression is NP-hard. Any broader safe-compression model class containing this identity-closure subclass is NP-hard; no membership claim for an arbitrary nonidentity closure operator follows without a polynomial closure-equality verifier in its declared encoding.*

Proof Membership in NP. Encode a proposed retained library by one bit per source strategy. Inactive retention and source inclusion are direct bit checks. Binary-integer addition checks $W(A) \leq B$. For every explicitly listed belief, exact rational comparison finds the source and candidate maxima and checks their equality. Under identity closure, bitwise union of the listed module-incidence rows checks $C(A) = C(L)$. Integer and rational operations have bit complexity polynomial in the explicit input length. Thus ISC-ID is in NP.

Reduction algorithm. Given a valid FU-WSC instance $(U, (A_i)_{i=1}^n, (c_i)_{i=1}^n, B)$, create one module m_u for each $u \in U$, one active strategy s_i for each set A_i , and one mandatory inactive strategy s_0 . Use one belief b_0 . Set

$$j_{s_i}(b_0) = j_{s_0}(b_0) = 0, \quad \text{mods}(s_i) = \{m_u : u \in A_i\}, \quad \text{mods}(s_0) = \emptyset.$$

Give s_i weight c_i , give s_0 weight zero, take the complete constructed library as the source, use identity closure, and copy the budget without alteration.

Polynomial construction size. The construction adds one module per universe element, one active strategy per set, one inactive strategy, and one belief. Its module-incidence table is the set-cover incidence table; the remaining profiles are $n + 1$ zero entries, and costs and budget are copied. A dense encoding has size $O(n|U|)$ plus the copied identifier and integer bit lengths, while a sparse encoding is linear in the copied incidence list plus this overhead. Either encoding is polynomial in the corresponding explicit source encoding.

Forward feasibility. Suppose I is a set cover with $\sum_{i \in I} c_i \leq B$. Retain $A = \{s_0\} \cup \{s_i : i \in I\}$. Both source and retained frontiers equal zero at b_0 , with s_0 itself attaining that value. Identity closure gives

$$C(A) = \bigcup_{i \in I} \text{mods}(s_i) = \{m_u : u \in U\} = \bigcup_{i=1}^n \text{mods}(s_i) = C(L),$$

where the last equality uses the full-union condition. The inactive strategy is retained, and $W(A) = \sum_{i \in I} c_i \leq B$, so A is a yes-certificate for the constructed ISC-ID instance.

Reverse feasibility. Suppose A is a safe constructed sublibrary within threshold, and let $I = \{i : s_i \in A\}$. Exact closure preservation and the empty inactive module row imply

$$\bigcup_{i \in I} \{m_u : u \in A_i\} = C(A) = C(L) = \{m_u : u \in U\}.$$

The module labels are in one-to-one correspondence with U , so $\bigcup_{i \in I} A_i = U$. Thus I is a set cover. Frontier preservation adds no hidden selection requirement because the mandatory inactive strategy already attains the common zero frontier.

Threshold preservation. The inactive strategy has zero burden, active weights are copied exactly, and the threshold is copied exactly. Therefore $W(A) = \sum_{i \in I} c_i$ for every selected index

set I , not merely for optimal selections. Hence $W(A) \leq B$ if and only if the corresponding set cover has cost at most B . The reduction is a polynomial many-one reduction. Together with membership in NP and Lemma S4.2, this proves NP-completeness of the stated restricted subclass and therefore of ISC-ID. The normalization proof uses unit costs on its nontrivial branch, so the restricted subclass remains NP-complete with unit active weights. Solving the associated minimum-burden problem would decide whether its optimum is at most B , which proves NP-hardness of the optimization problem. $\square$

Reduction audit

The proof depends on each of the following points; none is supplied by the executable examples alone.

1. The hardness source is the decision problem with an at-most budget, not the optimization problem and not maximum coverage.
2. The source family has full union. The preceding lemma proves that this normalization preserves NP-completeness instead of assuming it silently. Invalid encodings are rejected, so the restricted source is a language rather than a promise supplied by an oracle.
3. The universe is nonempty. Empty-universe source inputs are mapped to a fixed valid yes-instance before the compression reduction.
4. Costs are positive binary integers and the budget is a nonnegative binary integer. No pseudo-polynomial expansion of either number occurs.
5. Sets are indexed. Duplicate or empty sets are allowed and become distinct active strategies; neither feature affects the two directions.
6. The mandatory inactive strategy is part of every candidate, carries no modules, has zero profile, and has zero burden. If its burden were positive, the copied-threshold argument would require an explicit offset.
7. All active profiles are exactly zero at the single belief. Therefore the inactive strategy preserves the source frontier without selecting any active strategy, including when a set row is empty.
8. Module names are distinct and correspond one-to-one with universe elements. Identity closure is exactly row union; no module derivation, complementarity, multiplicity, provenance, or parent-conditioned production is present in the restricted subclass.
9. Safety means exact source-relative equality of both frontier and closure, source inclusion, mandatory inactive retention, and burden at most the threshold. No tolerance or held-out outcome enters the decision.
10. The source library contains every constructed active strategy. Its closure is the entire module universe precisely because the source family has full union.
11. The explicit encoding lists profiles and module incidences. The fixture's 62-bit masks are an implementation limit, not an assumption of the complexity theorem.
12. NP membership for a broader nonidentity-closure model is not inherited from this reduction. It requires a polynomial-time verifier for exact closure equality under that model's own encoding.

This theorem and proof are human-readable mathematical evidence. The exact Julia fixture separately enumerates five representative budgeted reductions, and the

focused test exhausts 5,565 full-union three-set/three-element instances and all 44,520 corresponding selected masks. Those exact finite computations check the implemented feasibility, weight, and threshold correspondences; they are not a Lean kernel proof, an exhaustive proof over arbitrary inputs, a solver-status claim, an approximation result, or a runtime result.

S5 Exact Preprocessing Propositions

Fix an identity-closure tagged covering instance after all declared mandatory columns have been selected. Let Q be the remaining requirement rows, A the available nonfixed columns, $R_s^Q = R_s \cap Q$, and let each optional weight be a strictly positive exact rational. There are no identity-sensitive side constraints, groups, exact-cardinality constraints, or mutual exclusions: this is the pure covering model. Every transformation records original identifiers, forced selections, fixed burden, equal-weight substitutions, and merged-row memberships.

Proposition S5.1 (Mandatory carrier and forced propagation). *If $u \in Q$ has exactly one carrier $s \in A$, every feasible completion selects s . Fixing s , adding w_s to the objective offset, deleting the rows it satisfies, and recomputing carriers preserves feasibility, the exact optimum value, every optimum identity, and reconstruction.*

Proof Any feasible completion must cover u , and s is its only available carrier, so s occurs in every completion. Removing already satisfied rows does not change the constraints on the remaining columns. Splitting every feasible objective into the fixed term w_s and its residual term is a bijection between feasible residual selections and feasible original selections containing the already forced columns. Repeating the argument after every selection proves propagation. $\square$

Proposition S5.2 (Coverage dominance). *Suppose $s, t \in A$, $s \neq t$, $R_s^Q \subseteq R_t^Q$, and $w_t \leq w_s$. If s is not mandatory, deleting s preserves feasibility, the optimum value, and at least one optimum. If $w_t < w_s$, no optimum contains s , so all optimizer identities are preserved. If $w_t = w_s$, all optimizer identities need not be preserved; a reconstruction record must retain the admissible substitution $s \leftrightarrow t$ where its surrounding selection remains feasible.*

Proof Take a feasible selection containing s . If it also contains t , then deleting s preserves coverage and strictly lowers burden because $w_s > 0$. If it does not contain t , replace s by t . Superset coverage preserves feasibility and the burden weakly falls. Thus every feasible solution has a no-worse representative without s , which proves feasibility and optimum-value preservation and supplies at least one reduced optimum. Under strict inequality, either case strictly improves any selection containing s , so no original optimum uses it. Under equality, replacement can exchange two distinct optimizer identities without changing burden. $\square$

Proposition S5.3 (Duplicate columns and empty contribution). *For columns with identical residual coverage, retaining a stable minimum-weight representative and deleting the others preserves feasibility, the exact optimum value, and at least one optimum. Strictly heavier duplicates cannot occur in an optimum. Equal-weight duplicates require an explicit substitution class to reconstruct all optimizer identities. A nonmandatory positive-weight column with empty residual coverage may be deleted; it occurs in no optimum.*

Proof Identical-coverage replacement leaves every row status unchanged and weakly reduces burden, with strict reduction for a heavier duplicate. Stable choice only fixes which representative remains in the reduced model. Recording the equal-weight class recovers the alternative identities consistent with the rest of a lifted selection. An empty-contribution column changes no residual constraint, and deleting it strictly lowers any selection that contains it. $\square$

Proposition S5.4 (Duplicate requirement rows). *If two residual requirement rows have identical carrier sets, replacing them by one representative preserves feasibility, optimum value, at least one optimum, and all optimizer identities. Recording both original row identifiers preserves the feasibility certificate.*

Proof A selection intersects one carrier set if and only if it intersects the identical other carrier set. The two inequalities therefore define the same feasible selections and do not enter the objective. Merging them changes no optimizer; the row-membership record merely expands the final certificate. $\square$

The implementation applies the rules to a stable fixed point and restarts its ordered scan after every transformation. Each nontrivial step removes a row or column or fixes a previously available column, so finiteness proves termination. The complete machine audit distinguishes forced columns, dominated columns, duplicate columns, empty columns, satisfied rows, and duplicate rows. These are exact combinatorial propositions. Julia randomized exhaustive tests and Lean finite-mask identities are separate evidence layers; neither turns a downstream solver status into a formal optimality proof.

S6 Algorithm Proofs, Certificates, and Pseudocode

The following proofs concern validated tagged identity-closure instances. Every returned library is reconstructed to original identifiers and then checked against the original source frontier, source closure, mandatory set, and exact burden. Approximation statements do not extend to arbitrary closure.

S6.1 Independent exact algorithms

The tagged representation gives two independent exact finite methods. They share only validated input, mandatory-selection preprocessing, and the final source-library certificate; their search procedures are different. Complete enumeration is the small-library oracle. Requirement-mask dynamic programming is parameterized by the number of residual tagged requirements.

Fix every mandatory strategy first. Let W_0 be their exact burden, delete every requirement they satisfy, and let the residual instance have strategies $1, \dots, n$ and requirements $1, \dots, r$. Write $R_i \subseteq [r]$ for the residual tagged coverage of strategy i and $w_i > 0$ for its exact weight. Mandatory preprocessing alone removes no optional strategy, so it preserves all optimizer identities. If a serialized instance already contains a valid stronger preprocessing map, the algorithms use its residual model, exact offset, and declared reconstruction map; complete-tie output is rejected when that map says that some equal-cost identities are unreconstructable.

S6.1.1 Feasibility-safe preprocessing core

Proposition S6.1 (Finite covering identities used in preprocessing). *For a finite residual pure covering model: every feasible selection contains the unique carrier of a required row; replacing a selected column by a column whose residual coverage contains it preserves feasibility; a nonmandatory column covering no residual requirement may be erased; and one of two requirement rows with identical carrier predicates may be erased without changing the feasible selections.*

Proof The unique-carrier statement follows directly from the covering inequality. For dominance, redirect any row witnessed by the replaced column to the coverage superset. An empty-contribution column witnesses no residual row, so erasing it preserves every existing witness. Identical carrier predicates make the two covering inequalities logically identical. These arguments are finite-set feasibility identities. Weight inequalities, objective offsets, optimizer identities, reconstruction maps, and fixed-point termination require the additional propositions and proofs in the preprocessing report. $\square$

The four feasibility clauses above are Lean verified. They are supporting lemmas, not a Lean proof of the complete preprocessing pipeline: optimum-value preservation, tie reconstruction, propagation, and fixed-point termination remain human proofs with exhaustive exact Julia tests. The shared finite incidence fixture has three required rows and cover sets $R_0 = \{0\}$, $R_1 = \{1, 2\}$, and $R_2 = \{2\}$; its Lean declaration checks that row zero has unique carrier zero and hence every complete cover selects strategy zero. This is one exact instance of the first clause, not a replacement for the universal proposition.

S6.1.2 Complete enumeration

The enumeration method visits every one of the 2^n residual binary selections. It tests tagged coverage directly, sums burden in exact rational arithmetic, and retains either every minimum or the declared stable representative. It then lifts forced selections and rechecks mandatory retention, the original operating frontier, the original identity closure, and the exact burden independently of the search test.

Proposition S6.2 (Correctness of complete enumeration). *On a valid finite tagged identity-closure instance, complete enumeration returns a minimum-burden safe source sublibrary. When complete-tie retention is requested and the preprocessing map preserves all optimizer identities, it returns every minimum-burden source selection.*

Proof Under mandatory-only preprocessing, every admissible source sublibrary is the forced set together with exactly one of the 2^n residual subsets. Enumeration evaluates all of them. Its feasibility test is therefore true for precisely the feasible residual subsets, and its exact minimum is the residual optimum. Adding the selection-independent burden W_0 preserves objective order, and lifting is a feasibility- and burden-preserving bijection.

If the input already carries stronger preprocessing, the preprocessing propositions establish optimum-value preservation and a valid canonical lift; the solver does not reassert a bijection for eliminated columns. When the map also declares all optimizer identities reconstructable, its equal-coverage classes enumerate the additional optimal identities. Finally, the independent original-library checks exclude a corrupted lift or incidence-only certificate. The tagged-cover equivalence theorem identifies every checked lift with a safe source sublibrary. $\square$

S6.1.3 Requirement-mask dynamic program

Encode a requirement subset by an r -bit mask; union is bitwise OR. The implemented forward recurrence is the following.

1. Set $D_0[0] = 0$ and leave every other mask unreachable.
2. For $i = 1, \dots, n$ and every mask q , carry the skip transition $(q, D_{i-1}[q])$ and the take transition $(q \text{ OR } R_i, D_{i-1}[q] + w_i)$ into layer i , keeping the lower exact burden.
3. At equal burden, keep the first predecessor under the declared stable traversal, or retain every tied predecessor when complete-tie mode is requested.
4. Return the entry for the all-ones mask, add W_0 , reconstruct the original selection, and run the independent source-library certificate.

The implementation scans masks in increasing integer order and strategies in canonical identifier order, emitting the skip transition before the take transition. In representative mode it keeps the first exact predecessor; in complete-tie mode it stores the complete optimal predecessor DAG and expands it only during reconstruction. Thus iteration and representative tie selection are deterministic; hash-table iteration order does not determine output.

The shared finite fixture checks the take-transition identity $[1, 0, 1] \text{ OR } [0, 1, 0] = [1, 1, 1]$ in both Julia and Lean. That calculation validates one mask union only; it is not a Lean proof of the inductive optimum invariant, reconstruction, complexity, or FPT statement.

Theorem S6.3 (Exact requirement-mask dynamic programming). *For a valid finite tagged identity-closure instance, the recurrence above returns a minimum-burden safe source sublibrary and reconstructs at least one optimizer. In representative mode it uses $O(n2^r)$ exact-arithmetic and mask operations and $O(2^r)$ cost states plus $O(n2^r)$ reconstruction-aware predecessor records. If only optimum value were required, the two cost layers would use $O(2^r)$ memory. Complete-tie retention is output-sensitive: materializing T tied reconstruction vectors adds $O(Tn)$ bits and can be exponential in n .*

Consequently, the representative algorithm is fixed-parameter tractable in r : its bit-complexity is 2^r times a polynomial in the explicit input length. It is not polynomial in the full input size when r is unrestricted.

Proof For $i \in \{0, \dots, n\}$ and mask q , let $D_i[q]$ be the minimum exact burden of a subset of the first i residual strategies whose coverage union is exactly q , with $D_i[q] = +\infty$ if no such subset exists. We prove the stored invariant by induction on i . At $i = 0$, only the empty subset exists, so $D_0[0] = 0$ and all other states are unreachable.

Assume the invariant at $i - 1$. Every subset of the first i strategies either omits strategy i , in which case it supplies the skip candidate already characterized by D_{i-1} , or contains strategy i . Removing i from the latter subset leaves some predecessor mask q with optimal predecessor cost $D_{i-1}[q]$; reinserting it produces mask $q \text{ OR } R_i$ and cost $D_{i-1}[q] + w_i$. Conversely, every skip or take transition constructs a subset of the first i strategies with the stated mask and burden. Taking the minimum over these exhaustive and disjoint cases proves the invariant for layer i . Stored predecessor records follow the same transitions, so every finite state has a valid path to the base layer. Equal-cost merging preserves the first predecessor under the declared deterministic traversal; retaining every equal-cost predecessor preserves the complete optimal predecessor DAG. Expanding that DAG preserves every tied residual identity, subject to the explicitly enforced output cap.

At $i = n$, a residual selection is feasible exactly when its union is the all-ones mask. Hence $D_n[2^r - 1]$ is the residual optimum. Adding the fixed mandatory offset and lifting preserve objective value directly under mandatory-only preprocessing; for a supplied stronger map, those properties are the content of the preprocessing propositions. Applying the tagged-cover equivalence proves safe-compression optimality. The independent exact certificate rechecks the source frontier, source closure, mandatory strategies, and burden rather than using the DP mask as its sole evidence.

There are 2^r masks and two relaxations per strategy-mask pair, giving $O(n2^r)$ recurrence operations in the unit-cost mask/arithmetic model. Two layers contain at most 2^r exact costs. The implementation additionally stores optimal predecessor records across the n layers, giving the stated reconstruction-aware bound. Exact rational sums and comparisons have bit cost polynomial in the explicit input bit length: sums of at most n encoded weights have polynomially bounded numerator and denominator length. Thus the bit-time bound is $2^r \text{ poly}(|I|)$, which is fixed-parameter tractable in r and remains exponential when r is part of unrestricted input. $\square$

S6.1.4 Evidence boundary and counters

Both methods are exact finite computations, not Lean kernel verification and not mixed-integer solver evidence. The supporting mask-union declaration does not change that status. The standard result object records the exact objective and checked selections separately from wall-clock measurements. For enumeration, candidate count is the number of binary subsets tested. For the DP, a visited state is a reachable mask at one strategy layer and a transition is one evaluated skip or take relaxation; the final reachable-state count is reported separately. Runtime fields are single-run measurements split into preprocessing, search, and reconstruction/certification scopes. They are instrumentation, not an algorithmic scaling finding; scaling evidence requires the separately locked benchmark design.

S6.2 Approximation-guaranteed greedy construction

The identity-closure reduction also permits a fast constructive method with a classical performance guarantee. Fix all mandatory strategies, including the inactive strategy, and remove the tagged requirements they cover. Let U^0 be the residual requirement

universe, let $W_0 \geq 0$ be the fixed mandatory burden, and let each residual strategy s have positive exact weight w_s and residual coverage $R_s \subseteq U^0$. For nonempty U^0 , write

$$d = \max_s |R_s|, \quad H(d) = \sum_{i=1}^d \frac{1}{i}.$$

The weighted rule is the standard set-cover greedy algorithm of Chvátal (1979), applied only after the semantic reduction:

1. initialize the selected set with every mandatory strategy and set $U \leftarrow U^0$;
2. while $U \neq \emptyset$, choose a residual strategy minimizing

$$\frac{w_s}{|R_s \cap U|}$$

among strategies with nonzero new coverage, breaking an exact tie by the lowest original strategy index in canonical identifier order;

3. add the chosen strategy and replace $U \leftarrow U \setminus R_s$;
4. optionally examine the chosen nonmandatory strategies in reverse selection order and delete one exactly when complete tagged coverage remains.

All scores and comparisons in the implementation use `Rational{BigInt}`. A separate cardinality variant replaces w_s in the score by one, equivalently choosing the largest number of newly covered requirements. That variant has the burden guarantee below only when all residual optional weights are equal; with unequal weights, no approximation guarantee for the actual retention burden is asserted.

Theorem S6.4 (Transferred weighted-greedy guarantee). *Suppose the source library is finite, closure is identity closure, every residual tagged requirement has a carrier, mandatory weights are nonnegative, and every nonmandatory weight is positive. If $U^0 \neq \emptyset$, the weighted greedy construction returns an innovation-safe source sublibrary of burden W_G satisfying*

$$W_G \leq H(d) \text{ OPT},$$

where OPT is the minimum total burden of any safe source sublibrary and $d = \max_s |R_s|$. Deterministic reverse deletion returns burden $W_{GR} \leq W_G$ and hence satisfies the same bound. If U^0 is empty, the mandatory library is optimal and the factor is one.

Proof First suppose U^0 is nonempty. The residual instance is an ordinary positive-cost weighted set-cover instance. Chvátal’s theorem applies to the rule minimizing cost per newly covered element and gives

$$G \leq H(d) \text{ OPT}_R,$$

where G and OPT_R are, respectively, the greedy and optimal residual costs. The theorem permits any choice among exact minimum-ratio ties, so the canonical deterministic tie rule does not change the bound.

Every admissible source selection contains the mandatory strategies, and mandatory-only preprocessing removes no optional strategy. Thus $W_G = W_0 + G$ and $\text{OPT} = W_0 + \text{OPT}_R$. Since $W_0 \geq 0$ and $H(d) \geq 1$,

$$W_G = W_0 + G \leq W_0 + H(d) \text{OPT}_R \leq H(d)(W_0 + \text{OPT}_R) = H(d) \text{OPT}.$$

The tagged-cover equivalence then converts residual complete coverage plus mandatory retention into exact preservation of the source operating frontier and source identity closure.

Reverse deletion accepts a removal only after complete tagged coverage is rechecked. It never removes a mandatory strategy, and every removed optional strategy has positive weight. Therefore each accepted deletion strictly lowers burden, every rejected deletion leaves burden unchanged, and $W_{GR} \leq W_G$. Moreover, one reverse pass is enough for inclusion-wise irreducibility: if a strategy cannot be removed at its examination, later deletions cannot restore any missing coverage. The same $H(d)$ bound follows immediately. If U^0 is empty, mandatory retention is already feasible and every optional positive-weight strategy is unnecessary, so the mandatory library is the exact optimum. $\square$

A primary-source near-tight fixture.

Chvátal's example has m requirements, singleton set $\{j\}$ of cost $1/j$ for each $j = 1, \dots, m$, and a set covering the entire universe with cost $1 + \epsilon$, where $\epsilon > 0$ and $1 + \epsilon < H(m)$. Greedy selects the singletons in order $m, m-1, \dots, 1$: when q requirements remain, singleton q has score $1/q$, whereas the universe set has score $(1 + \epsilon)/q > 1/q$. Its burden is $H(m)$, while the universe set alone is optimal with burden $1 + \epsilon$. The ratio $H(m)/(1 + \epsilon)$ tends to $H(m)$ as $\epsilon \downarrow 0$. The journal fixture realizes each requirement as an identity-closure module tag and lets the mandatory zero-weight inactive strategy cover the sole zero-frontier tag. Reverse deletion removes no singleton from the greedy endpoint.

Example S6.5 (Shared three-row greedy arithmetic). For $m = 3$ and $\epsilon = 1/12$, the singleton burden is $1 + 1/2 + 1/3 = 11/6$, the bundle burden is $1 + 1/12 = 13/12$, and the exact ratio is $22/13$. The Julia–Lean shared fixture checks only this arithmetic; it does not formalize Chvátal's theorem or the transfer to safe compression.

Limitations and evidence boundary.

The guarantee is inherited set-cover theory; the contribution here is its proved transfer through the source-relative frontier-closure representation and exact postchecked implementation. The theorem does not apply to arbitrary nonidentity closure, because independent module rows need not represent closure complementarities. It does not give the backward heaviest-safe-first deletion heuristic an approximation guarantee, and it does not make cardinality greedy a weighted-burden approximation when weights differ. The returned certificate is exact finite computation: it rechecks mandatory retention, tagged coverage, the original frontier, the original identity closure, and burden. It is neither exhaustive optimality evidence, mixed-integer solver evidence, nor Lean kernel verification.

S6.3 Certified rechecked-deletion portfolio

Forward greedy constructs a cover from the mandatory library. Rechecked deletion instead starts from the source and removes one entry at a time. The common implementation performs a complete fresh scan after every accepted deletion. For each retained nonmandatory strategy, it constructs the trial library and recomputes exact source-frontier equality and exact source-closure equality from the original profiles and module memberships. A deletion is accepted only after both equalities, mandatory retention, tagged coverage, and burden reconciliation pass. All other certificates are then discarded.

The deterministic portfolio contains heaviest-safe-first, lightest-safe-first, maximum immediate burden release, minimum remaining unique-carrier exposure, and declared source order. Under additive burden, the release from deleting one strategy is exactly its weight, so maximum immediate release and heaviest-safe-first are two names for the same ordering on this model; they are not treated as independent evidence.

For current retained set S , safe deletion candidate s , fixed tagged universe U_L , and tagged coverage R_t , define

$$E(s \mid S) = \left| \left\{ u \in U_L : \sum_{t \in S \setminus \{s\}} \mathbf{1}\{u \in R_t\} = 1 \right\} \right|.$$

The exposure rule minimizes $E(s \mid S)$, then maximizes exact immediate burden release, then uses the lowest canonical original strategy index. It uses only the source-defined incidence structure; it does not use dynamic value, held-out financial quality, realized return, or descendant outcomes. Heaviest and lightest rules break exact weight ties by lowest canonical original index. Declared-order deletion uses a complete caller-supplied permutation, or canonical original order by default.

The stochastic portfolio first draws a stable seeded permutation and then uses its earliest currently safe strategy at every fresh scan. Multi-start random deletion returns the lowest-burden certified endpoint across the declared starts; equal burdens are resolved by the lexicographically smallest selected-index tuple and then earliest start. The master seed, each derived seed, every random order, and every endpoint summary are retained.

Proposition S6.6 (Certified deletion endpoint). *Every algorithm in the rechecked-deletion portfolio returns a safe source sublibrary. Its reported endpoint is one-deletion irreducible after a complete exact final scan.*

Proof The source is safe relative to itself. At every accepted step, the trial library has been recomputed and certified to have the source frontier and source closure while retaining all mandatory entries. It is therefore safe, independently of the score used to choose it. Induction over the finite deletion trace gives safety of the endpoint. Every accepted deletion removes one retained nonmandatory strategy, so the trace terminates.

The final scan examines every retained nonmandatory strategy and records that its deletion fails exact source-relative feasibility. Hence no safe one-strategy deletion remains. Random

ordering changes only the choice among already certified candidates and does not alter this argument. $\square$

The generic finite-trace safety implication in the first paragraph of the proof is Lean verified by `recheckedSafeDeletionEndpoint_feasible`. The declaration assumes a trace whose current-library redundancy certificates are already supplied. It does not verify the Julia scan, scoring, randomization, termination counters, or the reported final irreducibility scan; those clauses remain human proof plus exact Julia tests.

Safety and one-deletion irreducibility do not imply global resource efficiency. The singleton–bundle family in Theorem S7.1 remains a regression test: heaviest-safe-first and maximum immediate release return burden k while the global optimum has burden $1 + \epsilon$. Exact enumeration and mask dynamic programming, rather than any deletion status, supply small-instance optimum comparisons. Runtime counters are instrumentation for the separately registered computational study, not scaling evidence in this draft.

S7 Worst-Case Local–Global Distinction

This section proves the parameterized counterexample used to separate exact safety certification from optimization quality. It concerns the named heaviest-safe-first rule, not every deletion order.

S7.1 Worst-case gap for heaviest-safe-first deletion

The worked $(2, 2, 3)$ example establishes a strict local–global separation. The following family shows that the separation for one specific deletion rule is not bounded by any constant. Define HSF to recheck exact source-relative safety at the current library, delete a safely deletable active strategy of greatest weight, and repeat until none remains. Ties may be resolved by any fixed rule; the construction below has a unique greatest-weight choice at its decisive step. For a minimization instance with positive optimum, its output ratio is the retained burden divided by the global minimum safe burden.

Theorem S7.1 (Unbounded gap for heaviest-safe-first deletion). *Let $k \geq 2$ be an integer and let $\epsilon \in \mathbb{Q}$ satisfy $0 < \epsilon < k - 1$. Consider one belief, identity module closure, modules $m_1, \dots, m_k$, and a mandatory inactive strategy of zero weight, zero profile, and no modules. For each i , let s_i have zero profile, weight one, and module set $\{m_i\}$. Let the bundle s_B have zero profile, weight $1 + \epsilon$, and all k modules. At the full source library:*

1. *every active strategy is safely deletable, and s_B is the unique heaviest safely deletable strategy;*
2. *HSF deletes s_B and stops at the library of all k singletons, which is inclusion-wise irreducible and has burden k ;*
3. *the bundle-only active library is the unique global minimum safe representation and has burden $1 + \epsilon$.*

Consequently, the output ratio of HSF is

$$\frac{k}{1 + \epsilon}.$$

The rule HSF therefore has no constant approximation guarantee over finite identity-closure innovation-safe compression instances. Moreover, one-deletion irreducibility alone has no constant worst-case burden guarantee relative to the global safe optimum.

Proof All active profiles and the mandatory inactive profile are zero. The source frontier is therefore zero and remains zero in every admissible sublibrary. Under identity closure, exact safety in this family reduces to covering all of $m_1, \dots, m_k$. A retained active set covers those modules if and only if it contains s_B , or it contains every singleton $s_1, \dots, s_k$.

At the full source, removing any singleton leaves s_B , so every singleton is safely deletable. Removing s_B leaves all singletons, so the bundle is safely deletable as well. Because $\epsilon > 0$, the bundle weight $1 + \epsilon$ is strictly greater than each singleton weight one. Thus it is the unique heaviest currently safe strategy, and HSF deletes it first.

After that deletion, singleton s_i is the only retained carrier of m_i . Deleting it would change the closure, so no further deletion is safe. More strongly, every proper sublibrary of the singleton endpoint omits some s_i and hence loses m_i ; the endpoint is inclusion-wise irreducible. Additivity of burden gives endpoint burden k .

The bundle-only active library is safe and has burden $1 + \epsilon$. Any safe library without the bundle must retain all k singletons and has burden $k > 1 + \epsilon$. Any safe library with the bundle and at least one singleton has burden strictly above $1 + \epsilon$, since active weights are positive. The bundle-only library is therefore the unique global optimum, and division by its positive burden gives the stated ratio.

Finally, fix the rational value $\epsilon = 1/2$. It satisfies the family condition for every $k \geq 2$, while the ratio is $2k/3$, which exceeds any proposed finite constant for a sufficiently large integer k . This proves the two negative guarantees: the named rule reaches these endpoints, and the endpoints themselves are one-deletion irreducible. It does not say that every deletion rule reaches them. $\square$

Example S7.2 (Shared finite gap arithmetic). At $k = 4$ and $\epsilon = 1/2$, the heaviest-safe-first endpoint has burden 4, the bundle optimum has burden $3/2$, and their exact ratio is $8/3$. The common Julia–Lean fixture checks this displayed arithmetic and the Julia generator separately checks the corresponding finite safety trace. The Lean declaration does not formalize the universal safety characterization, the named algorithm, or the unboundedness argument.

The deletion-order boundary is exact. If a rechecked maximal deletion sequence removes the bundle first, each remaining singleton is essential and the sequence stops at burden k . If it removes any singleton first, the bundle becomes essential for that singleton’s module; every remaining singleton stays redundant while the bundle is retained, so maximal deletion ends at the bundle-only optimum. Thus the theorem applies to the specified heaviest-safe-first rule and supplies bad one-deletion-irreducible endpoints, but it does not apply to every deletion order. In particular, it is not a lower bound for forward weighted-cover construction or for all certified deletion heuristics.

The earlier $(2, 2, 3)$ worked example is the case $k = 2$, $\epsilon = 1/2$, after multiplying every active burden by two. It is retained because its complete five-library table is a compact intuitive instance of the parameterized result. The universal statement above has a human-readable proof. The accompanying committed Julia artifact uses exact `Rational{BigInt}` arithmetic on several finite fixtures and an exhaustive small grid; it is finite computation, not Lean kernel verification and not a substitute for the proof. The separate shared Lean fixture above is likewise only exact finite arithmetic.

S8 Frontier–Closure Interaction

The main paper’s compressed state has two economically different margins. A higher frontier improves current operation and makes further operational improvement harder. A richer closure expands the menu of policies that research can generate. This section gives the complete finite interaction theory, its primitive sufficient conditions, and the exact mechanisms that make the sign ambiguous outside those conditions.

The results use the main paper’s finite-horizon unified action value, including cost, positive duration, operation during research, the joint completion law, compressed updating, and lower-horizon optimized continuation. They add no path–outcome independence assumption.

S8.1 Closure increments and relative saturation

Definition S8.1 (Closure increment and interaction). For $F_0 \leq F_1$ and $C_0 \subseteq C_1$, define

$$\Delta_C V_h(F; C_1, C_0) := V_h(F, C_1) - V_h(F, C_0)$$

and

$$J_h := \Delta_C V_h(F_1; C_1, C_0) - \Delta_C V_h(F_0; C_1, C_0).$$

The two margins are substitutes on the rectangle when $J_h \leq 0$ and complements when $J_h \geq 0$.

Frontier independence holds project availability, initiation cost, duration, operation during research, and the complete joint completion law fixed across the two frontiers. Closure expansion makes every closure-poor action available under the rich closure. These restrictions make each fixed opportunity saturate, but they do not compare saturation across projects. The theorem therefore needs a relative condition at the action level.

Theorem S8.2 (Frontier–closure substitution under relative saturation). *Consider the four realizable compressed states*

$$(F_0, C_0), \quad (F_0, C_1), \quad (F_1, C_0), \quad (F_1, C_1), \quad F_0 \leq F_1, \quad C_0 \subseteq C_1.$$

Suppose research primitives are frontier independent and the closure-rich menu contains the closure-poor menu. Suppose also that, at every finite Bellman node, every action

a_1 feasible with C_1 and every action a_0 feasible with C_0 satisfy

$$Q_h(F_1, C_1, a_1) - Q_h(F_1, C_0, a_0) \leq Q_h(F_0, C_1, a_1) - Q_h(F_0, C_0, a_0). \quad (\text{S8.1})$$

Here Q_h is the unified action value, including cost, positive duration, operation during research, the joint completion law, compressed updating, and lower-horizon optimized continuation. Then, for every finite horizon and belief,

$$\Delta_C V_h(F_1; C_1, C_0) \leq \Delta_C V_h(F_0; C_1, C_0), \quad \text{equivalently } J_h \leq 0.$$

At a fixed finite horizon and belief, for the four realizable corners choose a_{11} maximizing at (F_1, C_1) and a_{00} maximizing at (F_0, C_0) . Finiteness of the action menus supplies these maximizers. Frontier independence transports them to the comparison corners; no probabilistic independence is asserted. Relative action saturation and finite maximum attainment give

$$\begin{aligned} V_h(F_1, C_1) - V_h(F_1, C_0) &\leq Q_h(F_1, C_1, a_{11}) - Q_h(F_1, C_0, a_{00}) \\ &\leq Q_h(F_0, C_1, a_{11}) - Q_h(F_0, C_0, a_{00}) \\ &\leq V_h(F_0, C_1) - V_h(F_0, C_0). \end{aligned}$$

This is exactly $J_h \leq 0$. Neither detectability nor contraction is used. Without the middle inequality, individual project saturation does not control optimizer switching.

S8.2 Primitive common-gap sufficient conditions

Proposition S8.3 (Primitive common-gap frontier saturation). *Maintain the realizable rectangle, frontier-independent primitives, and menu inclusion of Theorem S8.2. Suppose that at every finite Bellman node and belief the four action-value tables admit*

$$\begin{aligned} Q_h(F_i, C_1, a) &= B_{h,i} + \eta_{h,a}^1 + \lambda_{h,a}^1 g_{h,i}, \\ Q_h(F_i, C_0, a) &= B_{h,i} + \eta_{h,a}^0 + \lambda_{h,a}^0 g_{h,i}, \quad i \in \{0, 1\}, \end{aligned} \quad (\text{S8.2})$$

where $g_{h,1} \leq g_{h,0}$, every rich-menu exposure satisfies $\lambda_{h,a}^1 \geq 0$, and every feasible poor-menu action satisfies $\lambda_{h,a}^0 = 0$. Then relative action saturation (S8.1) holds. Consequently

$$\Delta_C V_h(F_1; C_1, C_0) \leq \Delta_C V_h(F_0; C_1, C_0)$$

for every finite horizon and belief.

The subclass is economically narrow but primitive. A fixed descendant profile supplies the common positive-part gap g ; frontier-independent cost, duration, operation, candidate, admission, and fixed-continuation terms enter the intercepts and nonnegative exposures. The poor closure is Continue-only in the leading case; more generally its feasible actions may have fixed payoffs but no exposure to the descendant gap. The shared decomposition across frontiers is the required fixed-continuation factorization. It

is an assumption defining this subclass, not a claim that arbitrary unified continuation preserves the form.

There is a transition-level sufficient condition for the displayed gap order. Let the low- and high-frontier gaps share the process belief kernel P and discount $\beta \geq 0$, and let

$$\begin{aligned} g_{0,i}(b) &= \bar{g}_i(b), \\ g_{h+1,i}(b) &= s_i(b) + \beta \sum_{b'} P(b, b') g_{h,i}(b'), \quad i \in \{0, 1\}. \end{aligned}$$

If $\bar{g}_1 \leq \bar{g}_0$ and $s_1 \leq s_0$ pointwise, positivity of the fixed transition operator gives $g_{h,1} \leq g_{h,0}$ by finite-horizon induction. Thus the common-gap proposition is recursion stable when the descendant continuation uses this fixed kernel and remains outside the action-dependent optimizer.

The induction makes this preservation step explicit. Assume $\beta \geq 0$, $P(b, b') \geq 0$, $\bar{g}_1(b) \leq \bar{g}_0(b)$, and $s_1(b) \leq s_0(b)$ pointwise. The terminal order is the induction base. If $g_{h,1}(b') \leq g_{h,0}(b')$ for every b' , then

$$\begin{aligned} g_{h+1,1}(b) - g_{h+1,0}(b) &= s_1(b) - s_0(b) \\ &\quad + \beta \sum_{b'} P(b, b') (g_{h,1}(b') - g_{h,0}(b')) \leq 0. \end{aligned}$$

Hence $g_{h,1} \leq g_{h,0}$ for every finite h . Row mass one gives the kernel its probability interpretation; positivity, rather than normalization, is the sign-bearing property in the induction.

At each node, fixed project and continuation primitives give

$$Q_h(F_i, C_j, a) = B_{h,i} + \eta_{h,a}^j + \lambda_{h,a}^j g_{h,i}.$$

Take a rich action a_1 and a feasible poor action a_0 . The frontier-specific base cancels within their relative payoff. Because every poor action has $\lambda_{h,a_0}^0 = 0$, its change from the low to the high frontier is

$$\begin{aligned} &[Q_h(F_1, C_1, a_1) - Q_h(F_1, C_0, a_0)] \\ &\quad - [Q_h(F_0, C_1, a_1) - Q_h(F_0, C_0, a_0)] = \lambda_{h,a_1}^1 (g_{h,1} - g_{h,0}) \leq 0, \end{aligned}$$

using $\lambda_{h,a_1}^1 \geq 0$. This is precisely relative action saturation. Frontier independence transports the high-rich and low-poor maximizers to the other corners, and the finite-max comparison above then yields $J_h \leq 0$.

S8.3 Complementarity and sign-ambiguous cases

The zero-exposure restriction cannot be replaced by merely ordering the added project's exposure above the incumbent project's. Because Continue is always available and has zero exposure, its advantage over a positive-exposure poor project rises when the common gap shrinks. An exact 2,430-row rational search finds 648 failures of

all-pairs relative saturation under the broader exposure order and none among the 810 zero-poor-exposure rows. This is finite falsification and validation, not proof; the proposition is separately kernel checked.

The substitution theorem is sharp. Exact rational finite examples attain all three interaction signs:

construction	$\Delta_C V(F_0)$	$\Delta_C V(F_1)$	J
fixed opportunity with saturation	2	1	-1
frontier-dependent success $0 \rightarrow 1$	0	1/2	1/2
separable closure bonus	3/2	3/2	0

Proposition S8.4 (Complementarity from project switching). *Frontier independence and individual project saturation do not imply substitution. Let $\beta = 1/2$, the descendant payoff be 10, and $(F_0, F_1) = (0, 8)$. The old project has fixed $(\pi, \kappa) = (1, 2)$, while closure adds a project with fixed $(\pi, \kappa) = (1/2, 0)$. Their premia fall respectively*

$$3 \rightarrow 0 \quad \text{and} \quad 5/2 \rightarrow 1/2.$$

Nevertheless, optimization gives

$$\Delta_C V(F_0) = 0, \quad \Delta_C V(F_1) = 1/2, \quad J = 1/2.$$

The added project’s advantage over the old project rises from $-1/2$ to $1/2$, violating relative action saturation. This produces strict complementarity even though both project premia decline individually. The example remains outside Proposition S8.3: its poor-menu project has positive common-gap exposure. Frontier-dependent success provides a second exact complementarity mechanism; the fixed-opportunity and separable examples show that substitution and zero interaction are also attainable. Hence neither closure enrichment nor frontier improvement has an unconditional interaction sign.

The numerical interaction response remains evidence separate from the theory. It preserves the registered 2,430-row frontier–project boundary grid and five canonical four-corner fixtures: primitive strict substitution, the saturation boundary, optimizer-switching complementarity, frontier-dependent-success complementarity, and nonconstant separability. Every row reports the two closure increments, their cross difference, the four selected projects, four realizability flags, and the primitive sufficient-condition certificate. Only jointly realizable four-corner rows enter any sign frequency; nonrealizable diagnostic rows remain recorded but are not aggregated.

S9 Extended Comparative Statics and Counterexamples

These results are supporting interpretations of the finite dynamic model. They are not required by the tagged-cover reduction and are not used to select financial retention decisions.

S9.1 Additional comparative statics and counterexamples

This section records the supporting one-way comparative statics and the exact counterexamples that delimit the coverage results in Section S3. Every comparison holds conditional on the displayed current state and fixed action or on the stated one-shot problem. Unnamed primitives are held fixed. These results do not promote a one-shot cost-covering region to an optimized Bellman research region.

S9.2 Cost, admission, survival, and delay

The supporting order comparisons are compact because every nondisplayed primitive is held fixed:

Change	Direction	Required boundary
higher initiation cost	optimized value weakly falls	the same project actions change; Continue and other primitives are fixed
higher admission or survival	project value weakly rises	successful completion weakly dominates failure
longer duration	project value weakly falls	every added operating date satisfies $F_t \leq (1 - \beta)W$; suspended operation is $F_t = 0$
higher incumbent frontier	fixed-candidate operating gain weakly falls	candidate profile and research opportunities are fixed

Proposition S9.1 (One-shot cutoff comparative statics). *Holding all other one-shot primitives fixed, higher cost shrinks $\mathcal{C}$; higher nonnegative survival or admission expands it; and a higher operating frontier shrinks it for a fixed candidate. When both compared nonempty regions have upper-threshold representations, higher cost or frontier raises the cutoff, while higher survival or admission lowers it.*

These set inclusions concern the one-shot cost-covering region in Definition S3.9. Cost and survival also order an exact finite weak-research region for a fixed research action versus Continue when all other unified-model returns are unchanged. Neither result, by itself, orders the full optimal research region when opportunity menus, continuation laws, or competing actions also change.

The cutoff directions are pointwise set comparisons. Raising κ removes weakly covered beliefs. Raising a nonnegative survival or admission factor raises G and adds beliefs. Raising the frontier lowers the positive-part gap and removes beliefs. If two nonempty regions are $\text{Ici}(b_0^*)$ and $\text{Ici}(b_1^*)$, set inclusion reverses the cutoff order, yielding the directions in Proposition S9.1.

At a fixed finite horizon and conditional on the current decision state, holding every nondisplayed primitive fixed, a higher initiation cost subtracts the same nonnegative increment from each affected project action and leaves Continue unchanged. Backward induction and maximization over the finite action menu therefore make optimized value antitone in the cost schedule.

For a binary completion law, condition on the fixed current state and chosen project, with success continuation W^+ and failure continuation W^- . The binary specialization assumes the conditional success mass $m = \pi\rho^d$; it is not inferred from unrelated marginals. The completion component is

$$mW^+ + (1 - m)W^-.$$

For two admission–survival configurations its exact difference is

$$[m_1W^+ + (1 - m_1)W^-] - [m_0W^+ + (1 - m_0)W^-] = (m_1 - m_0)(W^+ - W^-).$$

Thus greater admission or survival raises project value when $W^+ \geq W^-$, and reverses sign if success is harmful.

For continued operation, keep the initial state, belief path, and terminal continuation W fixed while comparing durations. Write the elapsed return at duration d as

$$R_d = -\kappa + \sum_{t < d} \beta^t F_t + \beta^d W.$$

The one-date increment is

$$R_{d+1} - R_d = \beta^d [F_d - (1 - \beta)W].$$

Consequently, for $d_0 < d_1$,

$$R_{d_1} - R_{d_0} = \sum_{t=d_0}^{d_1-1} \beta^t [F_t - (1 - \beta)W].$$

Longer delay is therefore harmful when every added date satisfies $F_t \leq (1 - \beta)W$. Suspended operation is the special case $F_t = 0$. Nonnegativity alone is insufficient: with $\beta = 1/2$ and $F_t = W = 1$, $R_1 = 3/2 < R_2 = 7/4$.

S9.3 Threshold and multi-gap counterexamples

The assumptions of the positive threshold theorem have exact finite boundaries. The following examples preserve the original rational values and isolate the failed condition in each case.

Counterexample S9.2 (A nonmonotone kernel defeats the upper threshold). On $B = \{0, 1, 2\}$, take deterministic destinations $(1, 0, 1)$. The resulting row-stochastic kernel maps the increasing nonnegative gap $\Delta = (0, 1, 2)$ to $P\Delta = (1, 0, 1)$. With unit survival and antitone constant unit cost, the weak cost-covering set is $\{0, 2\}$, which is neither upper nor order-connected. Thus the stochastic-monotonicity assumption is independently necessary. The same kernel also maps the single-peaked gap $(0, 1, 0)$ to $(1, 0, 1)$, retaining the earlier boundary against unrestricted single-peaked preservation.

Counterexample S9.3 (Non-antitone cost disconnects a monotone covering set). The increasing potential $(1, 2, 3)$ and cost $(0, 3, 0)$ produce the cost-covering set $\{1, 3\}$. Antitonicity of research cost cannot be dropped from Theorem S3.10.

Counterexample S9.4 (One-project multi-gap disconnection). Let $B = \{0, 1, 2, 3, 4\}$. One project supplies candidates with certified gaps

$$\Delta^L = (4, 0, 0, 0, 0), \quad \Delta^R = (0, 0, 0, 0, 4)$$

over the zero frontier. Under the degree-four Bernstein/binomial kernel

$$P_{ij} = \binom{4}{j} (i/4)^j (1 - i/4)^{4-j}, \quad i, j \in B,$$

their aggregate one-shot coverage potential is $P(\Delta^L + \Delta^R) = (4, 41/32, 1/2, 41/32, 4)$. Constant strict cost one therefore gives the one-shot cost-covering set $\{0, 1\} \cup \{3, 4\}$, which is not order-connected.

Counterexample S9.5 (Arbitrary cost defeats a component bound). With the same five-state kernel, constant gap and potential $(2, 2, 2, 2, 2)$, and cost $(0, 3, 0, 3, 0)$, the strict cost-covering set is $\{0, 2, 4\}$. A kernel-only variation-diminishing condition cannot control net-region components under unrestricted belief-dependent cost.

The multi-gap example shows that the upper-threshold conclusion does not extend to separated candidate gaps. Separately, the first two examples show that neither monotone gaps alone nor a well-behaved gross-value profile alone ensures connected cost covering. The arbitrary-cost example rules out a kernel-only component bound when belief-dependent cost is unrestricted.

S9.4 Auxiliary finite-model illustration

The following exact construction isolates the distinction between operating profiles and generative modules in the main paper's finite model. It is an illustration rather than an additional assumption or theorem.

Example S9.6 (A two-module generator). Let $X = \{\text{down}, \text{up}\}$, let $\mu_{b-} = (3/4, 1/4)$ and $\mu_{b+} = (1/4, 3/4)$, and take identity closure. Strategies s_T and s_H carry modules m_T and m_H . A project q requires both. When $\{m_T, m_H\} \subseteq C$, set

$$G(q, b, C)(\text{some}(c)) = \frac{2}{3}, \quad G(q, b, C)(\text{none}) = \frac{1}{3}, \quad \nu(q, b, C, c) = \frac{3}{4};$$

otherwise set $G(q, b, C) = \delta_{\text{none}}$. Retaining both modules therefore admits c with probability $1/2$, while deleting the sole carrier of either module makes admission impossible. If $u_c(\text{down}) = -1$ and $u_c(\text{up}) = 2$, then $j_c(b^-) = -1/4$ and $j_c(b^+) = 5/4$: current operating payoff depends on the filtered state, while candidate availability

depends on retained closure. For a positive-duration instance, take $d_q = 2$, initiation cost $1/10$, and operation flag $o_q = 1$.

S9.5 Conditional replacement under a capacity bound

The supporting question is what an incumbent should release when an outer-certified candidate $c \notin L$ becomes available for retention and capacity binds. This formulation is conditional on outer certification and is separate from the raw generation, verification, and admission law in the main paper.

Let the incumbent satisfy $W(L) \leq B$, let $w_c \leq B$, and permit deletion of active incumbents $D \subseteq L \setminus \{s_0\}$. Additivity makes the required release

$$r_c(B) := [W(L) + w_c - B]_+,$$

so the feasible deletion family is

$$\mathcal{D}_c(B) := \{D \subseteq L \setminus \{s_0\} : W(D) \geq r_c(B)\}.$$

The conditional replacement problem is

$$\underset{D \in \mathcal{D}_c(B)}{\text{maximize}} V_\theta(b, (L \setminus D) \cup \{c\}). \quad (\text{S9.1})$$

For interpretation, define the unconstrained candidate gain and displacement loss

$$\begin{aligned} G_c(b, L) &:= V_\theta(b, L \cup \{c\}) - V_\theta(b, L), \\ \ell_c(D) &:= V_\theta(b, L \cup \{c\}) - V_\theta(b, (L \setminus D) \cup \{c\}). \end{aligned}$$

Then the best net admission value is the definition-level identity

$$A_c(b, L, B) = G_c(b, L) - \min_{D \in \mathcal{D}_c(B)} \ell_c(D).$$

Under common insertion and the productive factorization, a deletion safe before admission is sufficient for zero displacement loss, but it is not necessary: the candidate may replace a capability carried by a deleted incumbent. This supporting formulation is not part of the Lean-verified theorem set, and it makes no constrained-penalized equivalence claim.

S9.6 Secondary sensitivity results

This section retains sensitivity calculations that are useful for interpreting the economic model but are not needed for the safe-compression reduction. Each calculation holds along the named parameter path or conditional on the displayed initial belief and fixed action. Unnamed primitives are fixed. Finiteness is used for termwise differentiation and finite action maximization. No closure-detectability or stationary-contraction argument is used.

S9.7 Bridge-margin sensitivity

Write

$$A_{\text{br}} := \beta^d \rho^d \pi C, \quad M_{\text{br}} := A_{\text{br}} - \kappa, \quad m_{\text{br}} := \frac{M_{\text{br}}}{A_{\text{br}}}. \quad (\text{S9.2})$$

On the declared real extension with fixed integer d , positive gross coordinates, and $M_{\text{br}} > 0$, realized bridge value equals the margin. Its named-coordinate elasticities are

$$(\varepsilon_{\beta}^M, \varepsilon_{\rho}^M, \varepsilon_{\pi}^M, \varepsilon_C^M, \varepsilon_{\kappa}^M) = \left(\frac{d}{m_{\text{br}}}, \frac{d}{m_{\text{br}}}, \frac{1}{m_{\text{br}}}, \frac{1}{m_{\text{br}}}, -\frac{1 - m_{\text{br}}}{m_{\text{br}}} \right). \quad (\text{S9.3})$$

Threshold fragility is governed by the normalized positive margin m_{br} . The magnitudes diverge as $m_{\text{br}} \downarrow 0$, but not merely because $M_{\text{br}} \downarrow 0$: if cost is zero and gross scale vanishes, then $m_{\text{br}} = 1$. At $M_{\text{br}} = 0$, bridge-value elasticity is undefined. At and below that threshold, the signed margin, action region, or exact finite changes in realized value replace a percentage elasticity.

For the derivation, condition on the canonical project and hold every unnamed coordinate fixed. Differentiating on the positive real extension gives

$$\beta \partial_{\beta} M_{\text{br}} = d A_{\text{br}}, \quad \rho \partial_{\rho} M_{\text{br}} = d A_{\text{br}}, \quad \pi \partial_{\pi} M_{\text{br}} = C \partial_C M_{\text{br}} = A_{\text{br}}, \quad \kappa \partial_{\kappa} M_{\text{br}} = -\kappa.$$

On $M_{\text{br}} > 0$, realized value $[M_{\text{br}}]_+$ equals the signed margin. Dividing by M_{br} and using $m_{\text{br}} = M_{\text{br}}/A_{\text{br}}$ gives (S9.3). At the threshold, the positive-part value has no ordinary percentage elasticity. The costless path $\kappa = 0$, $A_{\text{br}} \downarrow 0$ keeps $m_{\text{br}} = 1$, so a vanishing net level alone does not imply the threshold-fragility limit.

S9.8 Operational and generative elasticity contributions

Let one named positive scalar x vary the same primitive in total insertion value $I(x)$, operational value $O(x)$, and generative value $G(x)$, holding every unlisted primitive fixed and remaining on a differentiable branch. Differentiating $I = O + G$ gives

$$xI'(x) = xO'(x) + xG'(x).$$

Proposition S9.7 (Operational–generative elasticity contributions). *At $x_0 > 0$, suppose $I(x) = O(x) + G(x)$ throughout a neighborhood of x_0 , and the three channel paths are differentiable there along the same named perturbation. If $I(x_0) > 0$, define*

$$C_x^{\text{op}} := \frac{x_0 O'(x_0)}{I(x_0)}, \quad C_x^{\text{gen}} := \frac{x_0 G'(x_0)}{I(x_0)}.$$

Then

$$\varepsilon_x^I = C_x^{\text{op}} + C_x^{\text{gen}}.$$

Only when both channel levels are positive may this be written as

$$\varepsilon_x^I = \frac{O}{I} \varepsilon_x^{\text{op}} + \frac{G}{I} \varepsilon_x^{\text{gen}}.$$

The contribution form remains meaningful when one channel level is zero; the corresponding ordinary component log elasticity does not. If total value is zero, no total elasticity or total-normalized contribution is reported. The scaled derivative identity above, or the exact finite-change identity $\Delta I = \Delta O + \Delta G$, remains well defined.

Along one fixed differentiable scalar path, differentiating the level identity and applying common scaling gives

$$xI'(x) = xO'(x) + xG'(x).$$

If $I(x) > 0$, division by the total level gives the two signed contributions in Proposition S9.7. Rewriting a contribution as

$$\frac{xO'}{I} = \frac{O}{I} \frac{xO'}{O}$$

requires $O > 0$, and the analogous rewrite requires $G > 0$. If a component vanishes, its scaled derivative remains available but its log elasticity does not. If $I = 0$, neither total elasticity nor a total-normalized contribution is defined; the level derivative or exact finite-change identity is used.

Belief	Library	V	V^{op}	V^{gen}	D_β	E_β^{op}	E_β^{gen}
low	I	$\frac{115}{288}$	0	$\frac{115}{288}$	$\frac{10777}{2070}$	—	$\frac{10777}{2070}$
low	I+A	1	0	1	$\frac{56}{15}$	—	$\frac{56}{15}$
low	I+D	$\frac{14}{3}$	$\frac{14}{3}$	0	$\frac{25}{21}$	25	—
high	I	$\frac{23}{288}$	0	$\frac{23}{288}$	$\frac{14089}{2070}$	—	$\frac{14089}{2070}$
high	I+A	$\frac{7}{6}$	0	$\frac{7}{6}$	$\frac{15}{29}$	—	$\frac{15}{29}$
high	I+D	$\frac{22}{3}$	$\frac{22}{3}$	0	$\frac{29}{33}$	29	—

Table S1 Exact productive value, channel levels, and fixed-policy discount elasticity in the canonical benchmark. A dash marks an undefined ordinary component elasticity at a zero component level; it is never recorded as zero. Scaled contributions are evaluated directly.

S9.9 Fixed-exposure innovation duration

For a fixed nonnegative, nonzero exposure sequence $z_0, \dots, z_{H-1}$ and effective discount $\alpha > 0$, let

$$\Psi_H(\alpha) := \sum_{t=0}^{H-1} \alpha^t z_t, \quad \omega_t := \frac{\alpha^t z_t}{\Psi_H(\alpha)}, \quad D_\Psi := \sum_t t \omega_t, \quad C_\Psi := \sum_t \omega_t (t - D_\Psi)^2. \quad (\text{S9.4})$$

Because the sequence is nonzero, $\Psi_H(\alpha) > 0$. The exact finite scaled-derivative identities are

$$\alpha \Psi'_H(\alpha) = D_\Psi \Psi_H(\alpha), \quad \alpha D'_\Psi(\alpha) = C_\Psi \geq 0. \quad (\text{S9.5})$$

Equivalently, D_Ψ is the derivative of $\log \Psi_H$ with respect to $\log \alpha$, and C_Ψ is the corresponding derivative of D_Ψ . These are interpretations of the displayed scaled identities. Innovation duration is the contribution-weighted date at which fixed uncovered value arrives, and innovation convexity is its timing variance. A higher effective discount shifts weight toward later exposures and weakly raises duration. This is not primitive project delay, time to admission, or a convexity theorem for an optimized Bellman value whose exposure sequence or selected action changes. If exposure is zero, duration and log elasticity are undefined; level contributions or finite differences are used instead.

For the derivation, introduce

$$N_1(\alpha) := \sum_t t \alpha^t z_t, \quad N_2(\alpha) := \sum_t t^2 \alpha^t z_t.$$

Termwise finite differentiation gives

$$\alpha \Psi'_H(\alpha) = N_1(\alpha), \quad \alpha N'_1(\alpha) = N_2(\alpha).$$

Since $\Psi_H > 0$, $D_\Psi = N_1/\Psi_H$, and the quotient rule yields

$$\alpha D'_\Psi = \frac{N_2 \Psi_H - N_1^2}{\Psi_H^2} = \sum_t \omega_t (t - D_\Psi)^2 = C_\Psi \geq 0.$$

The nonnegative curvature is curvature of $\log \Psi_H$ in $\log \alpha$ for a fixed exposure sequence. Changing project duration, belief dynamics, completion laws, or the selected action changes the sequence and lies outside this derivative.

S9.10 Finite switch and one-sided-change conventions

Within a strictly dominant price branch, the penalized envelope inherits the branch slope $-W(L')$. A pairwise crossing is only a candidate switch; it becomes a resource-price breakpoint only when both branches are globally active. At an unequal-burden tie the optimizer is set-valued, so one-sided slopes or cross-breakpoint finite changes replace a derivative selected by an unstated tie breaker.

For a finite library family, record the best-versus-second-best objective margin; for a Bellman node, record the corresponding action-value margin. A positive margin measures distance from a tie in value space. A zero margin flags a possible switch, but distinct libraries or actions may generate identical branches, so it is not sufficient for a change of optimizer. At an actual switch with positive value, one-sided or arc elasticities may be reported; at zero value, only level slopes and finite differences are defined.

S9.11 Finite capacity and resource-price geometry

Let $\mathcal{L}_\theta$ be a nonempty finite family of outer-certified libraries, let $W(L') \geq 0$ be retention burden, and let $V_\theta(b, L')$ be productive value under the unchanged inner

process. Define

$$V_\theta^*(b; B) := \max_{L' \in \mathfrak{L}_\theta: W(L') \leq B} V_\theta(b, L').$$

Theorem S9.8 (Finite hard-capacity retention). *Suppose $\mathfrak{L}_\theta$ is finite and contains the zero-burden inactive-only library. For every $B \geq 0$, the maximum is attained. The function $B \mapsto V_\theta^*(b; B)$ is nondecreasing, is constant between consecutive attainable burdens, and can change only at an attainable burden. Every exact forward capacity increment is nonnegative. Neither uniqueness nor inclusion nesting of maximizing libraries follows.*

Proof For fixed B , the feasible family is a nonempty subset of a finite set, so the maximum is attained. If capacity rises, feasible families are nested, which proves monotonicity. Between consecutive values in the finite set $\{W(L') : L' \in \mathfrak{L}_\theta\}$, the feasible family does not change. The hypotheses impose neither strict value comparisons nor inclusion relations among maximizers. $\square$

Discreteness matters. With two unit-burden capabilities that produce no value separately and value one jointly, successive attainable values are $(0, 0, 1)$. The capacity function is not generally concave, marginal capacity increments need not diminish, and a finite nonconcave capacity optimum need not be supported by a scalar price.

For $\lambda \geq 0$, define the penalized problem

$$J_\theta^*(b; \lambda) := \max_{L' \in \mathfrak{L}_\theta} \{V_\theta(b, L') - \lambda W(L')\}.$$

Theorem S9.9 (Finite resource-price envelope). *For a fixed nonempty finite eligible family, the maximum is attained. The optimized value is continuous, nonincreasing, convex, and piecewise affine in the real price $\lambda \geq 0$, with switching candidates among finitely many pairwise branch intersections. If $0 \leq \lambda_1 < \lambda_2$, every optimizer at λ_2 has weakly lower burden than every optimizer at λ_1 . Wherever one library is strictly dominant, the local slope is $-W(L')$.*

Proof Each eligible library supplies the affine branch $V_\theta(b, L') - \lambda W(L')$. The maximum of finitely many continuous affine functions is continuous, convex, and piecewise affine; nonnegative burdens make every branch nonincreasing. Let L_i optimize at λ_i , $i = 1, 2$. Adding the two optimality inequalities and using $\lambda_2 > \lambda_1$ gives $W(L_2) \leq W(L_1)$. On an interval with a unique active branch, differentiation gives slope $-W(L')$. $\square$

For two unequal-burden libraries, their value difference divided by their burden difference is only a candidate crossing price; it is an actual breakpoint only if both branches reach the upper envelope. At a tie the optimizer is set-valued. The burden ordering across prices does not imply that raw strategy identities are nested.

S10 Complete Canonical Numerical Records

This section preserves the full canonical numerical records from the preprint program and supports the journal article’s condensed economic section. The tables below are typeset directly from the committed CSV files; compilation does not rerun either numerical solver. Exact rational fields retain their machine-readable `numerator//denominator` form. The compact stationary table is included to make the record self-contained.

S10.1 Convergence history and stationary solution

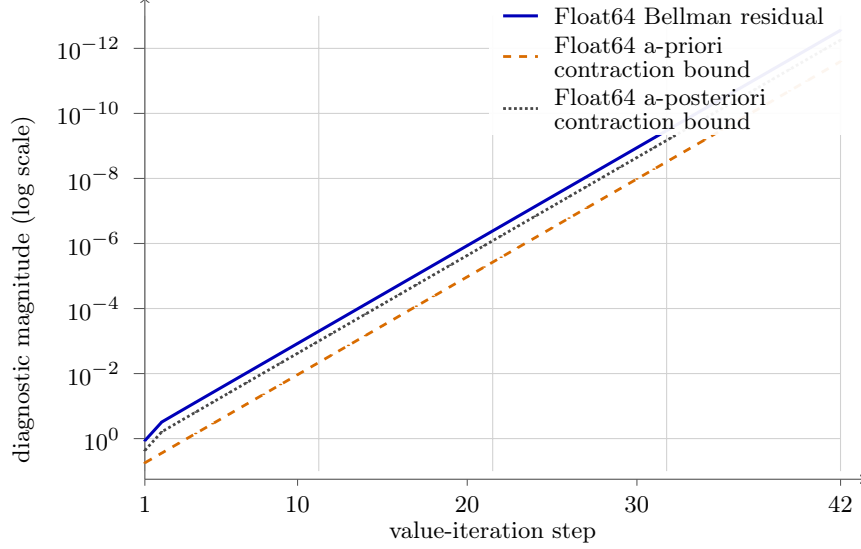


Fig. S1 Complete Float64 convergence history for the unified six-state benchmark. Markers are the 42 committed iterations; joining segments are visual guides. The dotted series is the Float64 a-posteriori contraction bound. Exact rational policy iteration is a separate calculation with Bellman residual zero.

The final convergence-row bound is $r_{42}^{64}/(1 - \beta_{64}) = 5.54223 \times 10^{-13}$, computed in Float64. The canonical summary separately converts the binary iterate exactly to `Rational{BigInt}`, records maximum exact error $937/1688849860263936 \approx 5.5482 \times 10^{-13}$, and evaluates the exact rational Bellman residual to obtain the exactly re-evaluated residual-based certificate $2499/4503599627370496 \approx 5.5489 \times 10^{-13}$. The latter is not the residual-bound column in the convergence table.

Table S2: Full 42-row Float64 convergence record.

Iter.	Float64 increment	Float64 Bellman residual	Float64 a-priori contraction bound	Float64 a-posteriori contraction bound
1	5.5625	1.15625	5.5625	2.3125
2	1.15625	0.30859375	2.78125	0.6171875
3	0.30859375	0.1533203125	1.390625	0.306640625
4	0.1533203125	0.076416015625	0.6953125	0.15283203125
5	0.076416015625	0.03814697265625	0.34765625	0.0762939453125
6	0.03814697265625	0.0190582275390625	0.173828125	0.038116455078125
7	0.0190582275390625	0.009525299072265625	0.0869140625	0.01905059814453125
8	0.009525299072265625	0.0047616958618164062	0.04345703125	0.0095233917236328125
9	0.0047616958618164062	0.0023806095123291016	0.021728515625	0.0047612190246582031
10	0.0023806095123291016	0.0011902451515197754	0.0108642578125	0.0023804903030395508
11	0.0011902451515197754	0.00059510767459869385	0.00543212890625	0.0011902153491973877
12	0.00059510767459869385	0.00029755011200904846	0.002716064453125	0.00059510022401809692
13	0.00029755011200904846	0.00014877412468194962	0.0013580322265625	0.00029754824936389923
14	0.00014877412468194962	7.4386829510331154e-05	0.00067901611328125	0.00014877365902066231
15	7.4386829510331154e-05	3.7193356547504663e-05	0.000339508056640625	7.4386713095009327e-05
16	3.7193356547504663e-05	1.8596663721837103e-05	0.0001697540283203125	3.7193327443674207e-05
17	1.8596663721837103e-05	9.2983282229397446e-06	8.487701416015625e-05	1.8596656445879489e-05
18	9.2983282229397446e-06	4.6491632019751705e-06	4.2438507080078125e-05	9.298326403950341e-06
19	4.6491632019751705e-06	2.3245813736139098e-06	2.1219253540039062e-05	4.6491627472278196e-06
20	2.3245813736139098e-06	1.162290629963536e-06	1.0609626770019531e-05	2.3245812599270721e-06
21	1.162290629963536e-06	5.8114530077091331e-07	5.3048133850097656e-06	1.1622906015418266e-06
22	5.8114530077091331e-07	2.9057264683274298e-07	2.6524066925048828e-06	5.8114529366548595e-07
23	2.9057264683274298e-07	1.4528632252819307e-07	1.3262033462524414e-06	2.9057264505638614e-07
24	1.4528632252819307e-07	7.2643161708185744e-08	6.631016731262207e-07	1.4528632341637149e-07
25	7.2643161708185744e-08	3.6321580410003662e-08	3.3155083656311035e-07	7.2643160820007324e-08
26	3.6321580410003662e-08	1.8160790205001831e-08	1.6577541828155518e-07	3.6321580410003662e-08
27	1.8160790205001831e-08	9.0803951025009155e-09	8.2887709140777588e-08	1.8160790205001831e-08
28	9.0803951025009155e-09	4.5401975512504578e-09	4.1443854570388794e-08	9.0803951025009155e-09
29	4.5401975512504578e-09	2.2700987756252289e-09	2.0721927285194397e-08	4.5401975512504578e-09
30	2.2700987756252289e-09	1.1350493878126144e-09	1.0360963642597198e-08	2.2700987756252289e-09
31	1.1350493878126144e-09	5.6752469390630722e-10	5.1804818212985992e-09	1.1350493878126144e-09
32	5.6752469390630722e-10	2.8376234695315361e-10	2.5902409106492996e-09	5.6752469390630722e-10
33	2.8376234695315361e-10	1.4188117347657681e-10	1.2951204553246498e-09	2.8376234695315361e-10
34	1.4188117347657681e-10	7.0940586738288403e-11	6.4756022766232491e-10	1.4188117347657681e-10
35	7.0940586738288403e-11	3.5470293369144201e-11	3.2378011383116245e-10	7.0940586738288403e-11

Iter.	Float64 increment	Float64 Bellman residual	Float64 a-priori contraction bound	Float64 a-posteriori contraction bound
36	3.5470293369144201e-11	1.7735146684572101e-11	1.6189005691558123e-10	3.5470293369144201e-11
37	1.7735146684572101e-11	8.8675733422860503e-12	8.0945028457790613e-11	1.7735146684572101e-11
38	8.8675733422860503e-12	4.4337866711430252e-12	4.0472514228895307e-11	8.8675733422860503e-12
39	4.4337866711430252e-12	2.2168933355715126e-12	2.0236257114447653e-11	4.4337866711430252e-12
40	2.2168933355715126e-12	1.1084466677857563e-12	1.0118128557223827e-11	2.2168933355715126e-12
41	1.1084466677857563e-12	5.5422333389287814e-13	5.0590642786119133e-12	1.1084466677857563e-12
42	5.5422333389287814e-13	2.7711166694643907e-13	2.5295321393059567e-12	5.5422333389287814e-13

State Belief V^* (payoff units) π^*		Winning margin (payoff units)	
K_0	ℓ	$\frac{115}{288}$ Discover	$\frac{23}{96}$
K_0	h	$\frac{23}{288}$ Continue	$\frac{245}{384}$
K_1	ℓ	1 Scale	$\frac{23}{48}$
K_1	h	$\frac{7}{6}$ Scale	$\frac{29}{48}$
K_2	ℓ	$\frac{14}{3}$ Continue	$\frac{3}{16}$
K_2	h	$\frac{22}{3}$ Continue	$\frac{3}{16}$

Table S3: Exact rational stationary solution, duplicated from the main paper.

Table S4: Complete stationary action-value record.

State	Belief	Value	Chosen	Second	Continue (Q)	Research	Research (Q)	Margin	Duration	Reward	block	Discounted	continuation
K0	low	115//288	research:discover	continue	23//144	research:discover	115//288	23//96	1	-1//32		31//72	
K0	high	23//288	continue	research:discover	23//288	research:discover	-643//1152	245//384	1	-1//1		509//1152	
K1	low	1//1	research:scale	continue	25//48	research:scale	1//1	23//48	2	-3//16		19//16	
K1	high	7//6	research:scale	continue	9//16	research:scale	7//6	29//48	2	-3//16		65//48	
K2	low	14//3	continue	research:scale	14//3	research:scale	215//48	3//16	2	49//16		17//12	
K2	high	22//3	continue	research:scale	22//3	research:scale	343//48	3//16	2	89//16		19//12	

S10.2 Horizon-by-horizon values and actions

The value record includes horizons zero through eight and all four stationary solution/evaluation outputs. The policy record includes every positive finite horizon and both stationary policy outputs. This retains the exact and floating-point rows together rather than presenting a selectively rounded comparison.

Table S5: Complete horizon-by-horizon and stationary value record.

Solution	Horizon	Arithmetic	State	Belief	Value
finite_horizon_exact	0	Rational{BigInt}	K0	low	0//1
finite_horizon_exact	0	Rational{BigInt}	K0	high	0//1
finite_horizon_exact	0	Rational{BigInt}	K1	low	0//1
finite_horizon_exact	0	Rational{BigInt}	K1	high	0//1
finite_horizon_exact	0	Rational{BigInt}	K2	low	0//1
finite_horizon_exact	0	Rational{BigInt}	K2	high	0//1
finite_horizon_exact	1	Rational{BigInt}	K0	low	0//1
finite_horizon_exact	1	Rational{BigInt}	K0	high	0//1
finite_horizon_exact	1	Rational{BigInt}	K1	low	0//1
finite_horizon_exact	1	Rational{BigInt}	K1	high	0//1
finite_horizon_exact	1	Rational{BigInt}	K2	low	2//1
finite_horizon_exact	1	Rational{BigInt}	K2	high	4//1
finite_horizon_exact	2	Rational{BigInt}	K0	low	0//1
finite_horizon_exact	2	Rational{BigInt}	K0	high	0//1
finite_horizon_exact	2	Rational{BigInt}	K1	low	0//1
finite_horizon_exact	2	Rational{BigInt}	K1	high	0//1
finite_horizon_exact	2	Rational{BigInt}	K2	low	13//4
finite_horizon_exact	2	Rational{BigInt}	K2	high	23//4
finite_horizon_exact	3	Rational{BigInt}	K0	low	0//1
finite_horizon_exact	3	Rational{BigInt}	K0	high	0//1
finite_horizon_exact	3	Rational{BigInt}	K1	low	3//8
finite_horizon_exact	3	Rational{BigInt}	K1	high	1//2
finite_horizon_exact	3	Rational{BigInt}	K2	low	63//16
finite_horizon_exact	3	Rational{BigInt}	K2	high	105//16
finite_horizon_exact	4	Rational{BigInt}	K0	low	31//256
finite_horizon_exact	4	Rational{BigInt}	K0	high	0//1
finite_horizon_exact	4	Rational{BigInt}	K1	low	21//32
finite_horizon_exact	4	Rational{BigInt}	K1	high	13//16
finite_horizon_exact	4	Rational{BigInt}	K2	low	275//64

Solution	Horizon	Arithmetic	State	Belief	Value
finite_horizon_exact	4	Rational{BigInt}	K2	high	445//64
finite_horizon_exact	5	Rational{BigInt}	K0	low	1973//8192
finite_horizon_exact	5	Rational{BigInt}	K0	high	31//2048
finite_horizon_exact	5	Rational{BigInt}	K1	low	105//128
finite_horizon_exact	5	Rational{BigInt}	K1	high	63//64
finite_horizon_exact	5	Rational{BigInt}	K2	low	1147//256
finite_horizon_exact	5	Rational{BigInt}	K2	high	1829//256
finite_horizon_exact	6	Rational{BigInt}	K0	low	82523//262144
finite_horizon_exact	6	Rational{BigInt}	K0	high	2345//65536
finite_horizon_exact	6	Rational{BigInt}	K1	low	465//512
finite_horizon_exact	6	Rational{BigInt}	K1	high	275//256
finite_horizon_exact	6	Rational{BigInt}	K2	low	4683//1024
finite_horizon_exact	6	Rational{BigInt}	K2	high	7413//1024
finite_horizon_exact	7	Rational{BigInt}	K0	low	2982325//8388608
finite_horizon_exact	7	Rational{BigInt}	K0	high	110663//2097152
finite_horizon_exact	7	Rational{BigInt}	K1	low	1953//2048
finite_horizon_exact	7	Rational{BigInt}	K1	high	1147//1024
finite_horizon_exact	7	Rational{BigInt}	K2	low	18923//4096
finite_horizon_exact	7	Rational{BigInt}	K2	high	29845//4096
finite_horizon_exact	8	Rational{BigInt}	K0	low	101185083//268435456
finite_horizon_exact	8	Rational{BigInt}	K0	high	4310281//67108864
finite_horizon_exact	8	Rational{BigInt}	K1	low	8001//8192
finite_horizon_exact	8	Rational{BigInt}	K1	high	4683//4096
finite_horizon_exact	8	Rational{BigInt}	K2	low	76075//16384
finite_horizon_exact	8	Rational{BigInt}	K2	high	119765//16384
stationary_exact_policy_iteration		Rational{BigInt}	K0	low	115//288
stationary_exact_policy_evaluation		Rational{BigInt}	K0	low	115//288
stationary_float64_value_iteration		Float64	K0	low	0.39930555555531799
stationary_float64_policy_evaluation		Float64	K0	low	0.3993055555555552
stationary_exact_policy_iteration		Rational{BigInt}	K0	high	23//288
stationary_exact_policy_evaluation		Rational{BigInt}	K0	high	23//288
stationary_float64_value_iteration		Float64	K0	high	0.079861111110873587

Solution	Horizon	Arithmetic	State	Belief	Value
stationary_float64_policy_evaluation		Float64	K0	high	0.079861111111111105
stationary_exact_policy_iteration		Rational{BigInt}	K1	low	1//1
stationary_exact_policy_evaluation		Rational{BigInt}	K1	low	1//1
stationary_float64_value_iteration		Float64	K1	low	0.9999999999999976241
stationary_float64_policy_evaluation		Float64	K1	low	1
stationary_exact_policy_iteration		Rational{BigInt}	K1	high	7//6
stationary_exact_policy_evaluation		Rational{BigInt}	K1	high	7//6
stationary_float64_value_iteration		Float64	K1	high	1.166666666666664289
stationary_float64_policy_evaluation		Float64	K1	high	1.1666666666666667
stationary_exact_policy_iteration		Rational{BigInt}	K2	low	14//3
stationary_exact_policy_evaluation		Rational{BigInt}	K2	low	14//3
stationary_float64_value_iteration		Float64	K2	low	4.666666666666661119
stationary_float64_policy_evaluation		Float64	K2	low	4.666666666666667
stationary_exact_policy_iteration		Rational{BigInt}	K2	high	22//3
stationary_exact_policy_evaluation		Rational{BigInt}	K2	high	22//3
stationary_float64_value_iteration		Float64	K2	high	7.33333333333327788
stationary_float64_policy_evaluation		Float64	K2	high	7.333333333333339

Table S6: Complete horizon-by-horizon and stationary action record.

Solution	Horizon	State	Belief	Action
finite_horizon_exact	1	K0	low	continue
finite_horizon_exact	1	K0	high	continue
finite_horizon_exact	1	K1	low	continue
finite_horizon_exact	1	K1	high	continue
finite_horizon_exact	1	K2	low	continue
finite_horizon_exact	1	K2	high	continue
finite_horizon_exact	2	K0	low	continue
finite_horizon_exact	2	K0	high	continue
finite_horizon_exact	2	K1	low	continue
finite_horizon_exact	2	K1	high	continue
finite_horizon_exact	2	K2	low	continue
finite_horizon_exact	2	K2	high	continue
finite_horizon_exact	3	K0	low	continue
finite_horizon_exact	3	K0	high	continue
finite_horizon_exact	3	K1	low	research:scale
finite_horizon_exact	3	K1	high	research:scale
finite_horizon_exact	3	K2	low	continue
finite_horizon_exact	3	K2	high	continue
finite_horizon_exact	4	K0	low	research:discover
finite_horizon_exact	4	K0	high	continue
finite_horizon_exact	4	K1	low	research:scale
finite_horizon_exact	4	K1	high	research:scale
finite_horizon_exact	4	K2	low	continue
finite_horizon_exact	4	K2	high	continue
finite_horizon_exact	5	K0	low	research:discover
finite_horizon_exact	5	K0	high	continue
finite_horizon_exact	5	K1	low	research:scale
finite_horizon_exact	5	K1	high	research:scale
finite_horizon_exact	5	K2	low	continue
finite_horizon_exact	5	K2	high	continue
finite_horizon_exact	6	K0	low	research:discover
finite_horizon_exact	6	K0	high	continue
finite_horizon_exact	6	K1	low	research:scale
finite_horizon_exact	6	K1	high	research:scale
finite_horizon_exact	6	K2	low	continue
finite_horizon_exact	6	K2	high	continue
finite_horizon_exact	7	K0	low	research:discover

Solution	Horizon	State	Belief	Action
finite_horizon_exact	7	K0	high	continue
finite_horizon_exact	7	K1	low	research:scale
finite_horizon_exact	7	K1	high	research:scale
finite_horizon_exact	7	K2	low	continue
finite_horizon_exact	7	K2	high	continue
finite_horizon_exact	8	K0	low	research:discover
finite_horizon_exact	8	K0	high	continue
finite_horizon_exact	8	K1	low	research:scale
finite_horizon_exact	8	K1	high	research:scale
finite_horizon_exact	8	K2	low	continue
finite_horizon_exact	8	K2	high	continue
stationary_exact_policy_iteration		K0	low	research:discover
stationary_float64_value_iteration		K0	low	research:discover
stationary_exact_policy_iteration		K0	high	continue
stationary_float64_value_iteration		K0	high	continue
stationary_exact_policy_iteration		K1	low	research:scale
stationary_float64_value_iteration		K1	low	research:scale
stationary_exact_policy_iteration		K1	high	research:scale
stationary_float64_value_iteration		K1	high	research:scale
stationary_exact_policy_iteration		K2	low	continue
stationary_float64_value_iteration		K2	low	continue
stationary_exact_policy_iteration		K2	high	continue
stationary_float64_value_iteration		K2	high	continue

S10.3 Path, admission, and operating-reward atoms

Table S7: Complete positive-duration path and admission-atom record.

Project	Start state	Start belief	Duration	Path	Terminal belief	Admitted	Joint mass	Path mass	Conditional admission
discover	K0	low	1	low>low	low	failure	3//16	3//4	1//4
discover	K0	low	1	low>low	low	carrier_a	3//8	3//4	1//2
discover	K0	low	1	low>low	low	carrier_b	3//16	3//4	1//4
discover	K0	low	1	low>high	high	failure	1//16	1//4	1//4
discover	K0	low	1	low>high	high	carrier_a	1//8	1//4	1//2
discover	K0	low	1	low>high	high	carrier_b	1//16	1//4	1//4
discover	K0	high	1	high>low	low	failure	1//16	1//4	1//4
discover	K0	high	1	high>low	low	carrier_a	1//8	1//4	1//2
discover	K0	high	1	high>low	low	carrier_b	1//16	1//4	1//4
discover	K0	high	1	high>high	high	failure	3//16	3//4	1//4
discover	K0	high	1	high>high	high	carrier_a	3//8	3//4	1//2
discover	K0	high	1	high>high	high	carrier_b	3//16	3//4	1//4
scale	K1	low	2	low>low>low	low	failure	9//40	9//16	2//5
scale	K1	low	2	low>low>low	low	descendant	27//80	9//16	3//5
scale	K1	low	2	low>low>high	high	descendant	3//16	3//16	1//1
scale	K1	low	2	low>high>low	low	failure	1//40	1//16	2//5
scale	K1	low	2	low>high>low	low	descendant	3//80	1//16	3//5
scale	K1	low	2	low>high>high	high	descendant	3//16	3//16	1//1
scale	K1	high	2	high>low>low	low	failure	1//8	3//16	2//3
scale	K1	high	2	high>low>low	low	descendant	1//16	3//16	1//3
scale	K1	high	2	high>low>high	high	descendant	1//16	1//16	1//1
scale	K1	high	2	high>high>low	low	failure	1//8	3//16	2//3
scale	K1	high	2	high>high>low	low	descendant	1//16	3//16	1//3
scale	K1	high	2	high>high>high	high	descendant	9//16	9//16	1//1
scale	K2	low	2	low>low>low	low	failure	9//40	9//16	2//5
scale	K2	low	2	low>low>low	low	descendant	27//80	9//16	3//5
scale	K2	low	2	low>low>high	high	descendant	3//16	3//16	1//1
scale	K2	low	2	low>high>low	low	failure	1//40	1//16	2//5
scale	K2	low	2	low>high>low	low	descendant	3//80	1//16	3//5
scale	K2	low	2	low>high>high	high	descendant	3//16	3//16	1//1
scale	K2	high	2	high>low>low	low	failure	1//8	3//16	2//3
scale	K2	high	2	high>low>low	low	descendant	1//16	3//16	1//3
scale	K2	high	2	high>low>high	high	descendant	1//16	1//16	1//1
scale	K2	high	2	high>high>low	low	failure	1//8	3//16	2//3
scale	K2	high	2	high>high>low	low	descendant	1//16	3//16	1//3
scale	K2	high	2	high>high>high	high	descendant	9//16	9//16	1//1

Table S8: Complete operating-reward block record.

State	Belief	Action	Duration	Operates	Initiation cost	Incumbent reward	Net reward block
K0	low	continue	1	false	0//1	0//1	0//1
K0	low	discover	1	true	1//32	0//1	-1//32
K0	high	continue	1	false	0//1	0//1	0//1
K0	high	discover	1	true	1//1	0//1	-1//1
K1	low	continue	1	false	0//1	0//1	0//1
K1	low	scale	2	true	3//16	0//1	-3//16
K1	high	continue	1	false	0//1	0//1	0//1

State	Belief	Action	Duration	Operates	Initiation cost	Incumbent reward	Net reward block
K1	high	scale	2	true	3//16	0//1	-3//16
K2	low	continue	1	false	0//1	2//1	2//1
K2	low	scale	2	true	3//16	13//4	49//16
K2	high	continue	1	false	0//1	4//1	4//1
K2	high	scale	2	true	3//16	23//4	89//16

S10.4 Safe compression and capacity correspondences

The following tables retain every source and every optimizer tie. Masks use the coordinate order (I, A, B, D) ; semicolon-separated entries are tied solutions, not a sequence of selections. The primary burden keeps $w_I = w_{s_0} = 0$. The committed canonical resource artifacts use the separately displayed total $\widetilde{W}(L) = 1 + W(L)$, so their machine fields `source_burden`, `capacity`, `optimal_burdens`, and `selected_burden` are display burdens. Consequently the capacity objective is $\widetilde{V}^*(b; \widetilde{B})$ and the penalized net-value field is $\widetilde{J}^*(b; \lambda) = J^*(b; \lambda) - \lambda$. Absolute burden reductions and all switching prices are invariant to this common-unit translation. The capacity theorem and Lean formalization use $\widetilde{W}$, not W .

Table S9: Complete safe-compression optimizer correspondence.

Schedule	Source mask	Source library	Display source		Minimum display		Optimal masks	Optimal libraries	Burden reduction		Frontier	Closure	Ties complete
			burden $\widetilde{W}$	Feasible masks	burden $\widetilde{W}$				$\Delta \widetilde{W} = \Delta W$				
equal_active	0	I	1//1	0	1//1	0	I	I	0//1	0//1;0//1			true
equal_active	1	I+A	2//1	1	2//1	1	I+A	I+A	0//1	0//1;0//1	1		true
equal_active	2	I+B	2//1	2	2//1	2	I+B	I+B	0//1	0//1;0//1	1		true
equal_active	3	I+A+B	3//1	1;2;3	2//1	1;2	I+A;I+B	I+A;I+B	1//1	0//1;0//1	1		true
equal_active	4	I+D	2//1	4	2//1	4	I+D	I+D	0//1	2//1;4//1	1		true
equal_active	5	I+A+D	3//1	4;5	2//1	4	I+D	I+D	1//1	2//1;4//1	1		true
equal_active	6	I+B+D	3//1	4;6	2//1	4	I+D	I+D	1//1	2//1;4//1	1		true
equal_active	7	I+A+B+D	4//1	4;5;6;7	2//1	4	I+D	I+D	2//1	2//1;4//1	1		true
carrier_heavy	0	I	1//1	0	1//1	0	I	I	0//1	0//1;0//1			true
carrier_heavy	1	I+A	3//1	1	3//1	1	I+A	I+A	0//1	0//1;0//1	1		true
carrier_heavy	2	I+B	3//1	2	3//1	2	I+B	I+B	0//1	0//1;0//1	1		true
carrier_heavy	3	I+A+B	5//1	1;2;3	3//1	1;2	I+A;I+B	I+A;I+B	2//1	0//1;0//1	1		true
carrier_heavy	4	I+D	2//1	4	2//1	4	I+D	I+D	0//1	2//1;4//1	1		true
carrier_heavy	5	I+A+D	4//1	4;5	2//1	4	I+D	I+D	2//1	2//1;4//1	1		true
carrier_heavy	6	I+B+D	4//1	4;6	2//1	4	I+D	I+D	2//1	2//1;4//1	1		true
carrier_heavy	7	I+A+B+D	6//1	4;5;6;7	2//1	4	I+D	I+D	4//1	2//1;4//1	1		true
descendant_heavy	0	I	1//1	0	1//1	0	I	I	0//1	0//1;0//1			true
descendant_heavy	1	I+A	2//1	1	2//1	1	I+A	I+A	0//1	0//1;0//1	1		true
descendant_heavy	2	I+B	2//1	2	2//1	2	I+B	I+B	0//1	0//1;0//1	1		true
descendant_heavy	3	I+A+B	3//1	1;2;3	2//1	1;2	I+A;I+B	I+A;I+B	1//1	0//1;0//1	1		true
descendant_heavy	4	I+D	3//1	4	3//1	4	I+D	I+D	0//1	2//1;4//1	1		true
descendant_heavy	5	I+A+D	4//1	4;5	3//1	4	I+D	I+D	1//1	2//1;4//1	1		true
descendant_heavy	6	I+B+D	4//1	4;6	3//1	4	I+D	I+D	1//1	2//1;4//1	1		true
descendant_heavy	7	I+A+B+D	5//1	4;5;6;7	3//1	4	I+D	I+D	2//1	2//1;4//1	1		true

Table S10: Complete capacity paths and value certificates.

Schedule	Belief	Display capacity		Optimal masks	Optimal libraries	Optimal display		Selected library	Op. value	Gen. value	Total value	Ties complete
		$\bar{B}$	$\tilde{V}^*$			burdens $\tilde{W}$						
equal_active	low	1//1	115//288	0	I	1//1	I	I	0//1	115//288	115//288	true
equal_active	low	2//1	14//3	4	I+D	2//1	I+D	I+D	14//3	0//1	14//3	true
equal_active	low	3//1	14//3	4;5;6	I+D; I+A+D; I+B+D	2//1;3//1;3//1	I+D	I+D	14//3	0//1	14//3	true
equal_active	low	4//1	14//3	4;5;6;7	I+D; I+A+D; I+B+D; I+A+B+D	2//1;3//1;3//1;4//1	I+D	I+D	14//3	0//1	14//3	true
equal_active	high	1//1	23//288	0	I	1//1	I	I	0//1	23//288	23//288	true
equal_active	high	2//1	22//3	4	I+D	2//1	I+D	I+D	22//3	0//1	22//3	true
equal_active	high	3//1	22//3	4;5;6	I+D; I+A+D; I+B+D	2//1;3//1;3//1	I+D	I+D	22//3	0//1	22//3	true
equal_active	high	4//1	22//3	4;5;6;7	I+D; I+A+D; I+B+D; I+A+B+D	2//1;3//1;3//1;4//1	I+D	I+D	22//3	0//1	22//3	true
carrier_heavy	low	1//1	115//288	0	I	1//1	I	I	0//1	115//288	115//288	true
carrier_heavy	low	2//1	14//3	4	I+D	2//1	I+D	I+D	14//3	0//1	14//3	true
carrier_heavy	low	3//1	14//3	4	I+D	2//1	I+D	I+D	14//3	0//1	14//3	true
carrier_heavy	low	4//1	14//3	4;5;6	I+D; I+A+D; I+B+D	2//1;4//1;4//1	I+D	I+D	14//3	0//1	14//3	true
carrier_heavy	low	5//1	14//3	4;5;6	I+D; I+A+D; I+B+D	2//1;4//1;4//1	I+D	I+D	14//3	0//1	14//3	true
carrier_heavy	low	6//1	14//3	4;5;6;7	I+D; I+A+D; I+B+D; I+A+B+D	2//1;4//1;4//1;6//1	I+D	I+D	14//3	0//1	14//3	true
carrier_heavy	high	1//1	23//288	0	I	1//1	I	I	0//1	23//288	23//288	true
carrier_heavy	high	2//1	22//3	4	I+D	2//1	I+D	I+D	22//3	0//1	22//3	true
carrier_heavy	high	3//1	22//3	4	I+D	2//1	I+D	I+D	22//3	0//1	22//3	true
carrier_heavy	high	4//1	22//3	4;5;6	I+D; I+A+D; I+B+D	2//1;4//1;4//1	I+D	I+D	22//3	0//1	22//3	true
carrier_heavy	high	5//1	22//3	4;5;6	I+D; I+A+D; I+B+D	2//1;4//1;4//1	I+D	I+D	22//3	0//1	22//3	true
carrier_heavy	high	6//1	22//3	4;5;6;7	I+D; I+A+D; I+B+D; I+A+B+D	2//1;4//1;4//1;6//1	I+D	I+D	22//3	0//1	22//3	true
descendant_heavy	low	1//1	115//288	0	I	1//1	I	I	0//1	115//288	115//288	true
descendant_heavy	low	2//1	1//1	1;2	I+A; I+B	2//1;2//1	I+A	I+A	0//1	1//1	1//1	true
descendant_heavy	low	3//1	14//3	4	I+D	3//1	I+D	I+D	14//3	0//1	14//3	true
descendant_heavy	low	4//1	14//3	4;5;6	I+D; I+A+D; I+B+D	3//1;4//1;4//1	I+D	I+D	14//3	0//1	14//3	true
descendant_heavy	low	5//1	14//3	4;5;6;7	I+D; I+A+D; I+B+D; I+A+B+D	3//1;4//1;4//1;5//1	I+D	I+D	14//3	0//1	14//3	true
descendant_heavy	high	1//1	23//288	0	I	1//1	I	I	0//1	23//288	23//288	true
descendant_heavy	high	2//1	7//6	1;2	I+A; I+B	2//1;2//1	I+A	I+A	0//1	7//6	7//6	true
descendant_heavy	high	3//1	22//3	4	I+D	3//1	I+D	I+D	22//3	0//1	22//3	true
descendant_heavy	high	4//1	22//3	4;5;6	I+D; I+A+D; I+B+D	3//1;4//1;4//1	I+D	I+D	22//3	0//1	22//3	true
descendant_heavy	high	5//1	22//3	4;5;6;7	I+D; I+A+D; I+B+D; I+A+B+D	3//1;4//1;4//1;5//1	I+D	I+D	22//3	0//1	22//3	true

S10.5 Penalized intervals and pairwise crossings

The interval record retains breakpoint ties and the open and terminal cells of each finite affine envelope. The crossing record then lists every pair of branches, including coincident and parallel pairs, nonnegative intersections that never reach the upper envelope, and the globally active breakpoints.

Table S11: Complete penalized-envelope interval and breakpoint record.

Optimal display													
Schedule	Belief Cell		Lower	Upper	Low incl.	Up incl.	Probe	Optimal masks	Optimal libraries	burdens $\bar{W}$	Selected $\tilde{J}^*$	Ties complete	
equal_active	low	breakpoint	0//1	0//1	true	true	0//1	4;5;6;7	I+D;I+A+D;I+B+D;I+A+B+D	2//1;3//1;3//1;4//1	I+D	14//3	true
equal_active	low	open_interval	0//1	1229//288	false	false	1229//576	4	I+D	2//1	I+D	115//288	true
equal_active	low	breakpoint	1229//288	1229//288	true	true	1229//288	0;4	I;I+D	1//1;2//1	I	-557//144	true
equal_active	low	terminal_interval	1229//288		false	false	1517//288	0	I	1//1	I	-701//144	true
equal_active	high	breakpoint	0//1	0//1	true	true	0//1	4;5;6;7	I+D;I+A+D;I+B+D;I+A+B+D	2//1;3//1;3//1;4//1	I+D	22//3	true
equal_active	high	open_interval	0//1	2089//288	false	false	2089//576	4	I+D	2//1	I+D	23//288	true
equal_active	high	breakpoint	2089//288	2089//288	true	true	2089//288	0;4	I;I+D	1//1;2//1	I	-1033//144	true
equal_active	high	terminal_interval	2089//288		false	false	2377//288	0	I	1//1	I	-1177//144	true
carrier_heavy	low	breakpoint	0//1	0//1	true	true	0//1	4;5;6;7	I+D;I+A+D;I+B+D;I+A+B+D	2//1;4//1;4//1;6//1	I+D	14//3	true
carrier_heavy	low	open_interval	0//1	1229//288	false	false	1229//576	4	I+D	2//1	I+D	115//288	true
carrier_heavy	low	breakpoint	1229//288	1229//288	true	true	1229//288	0;4	I;I+D	1//1;2//1	I	-557//144	true
carrier_heavy	low	terminal_interval	1229//288		false	false	1517//288	0	I	1//1	I	-701//144	true
carrier_heavy	high	breakpoint	0//1	0//1	true	true	0//1	4;5;6;7	I+D;I+A+D;I+B+D;I+A+B+D	2//1;4//1;4//1;6//1	I+D	22//3	true
carrier_heavy	high	open_interval	0//1	2089//288	false	false	2089//576	4	I+D	2//1	I+D	23//288	true
carrier_heavy	high	breakpoint	2089//288	2089//288	true	true	2089//288	0;4	I;I+D	1//1;2//1	I	-1033//144	true
carrier_heavy	high	terminal_interval	2089//288		false	false	2377//288	0	I	1//1	I	-1177//144	true
descendant_heavy	low	breakpoint	0//1	0//1	true	true	0//1	4;5;6;7	I+D;I+A+D;I+B+D;I+A+B+D	3//1;4//1;4//1;5//1	I+D	14//3	true
descendant_heavy	low	open_interval	0//1	1229//576	false	false	1229//1152	4	I+D	3//1	I+D	563//384	true
descendant_heavy	low	breakpoint	1229//576	1229//576	true	true	1229//576	0;4	I;I+D	1//1;3//1	I	-111//64	true
descendant_heavy	low	terminal_interval	1229//576		false	false	1805//576	0	I	1//1	I	-175//64	true
descendant_heavy	high	breakpoint	0//1	0//1	true	true	0//1	4;5;6;7	I+D;I+A+D;I+B+D;I+A+B+D	3//1;4//1;4//1;5//1	I+D	22//3	true
descendant_heavy	high	open_interval	0//1	2089//576	false	false	2089//1152	4	I+D	3//1	I+D	727//384	true
descendant_heavy	high	breakpoint	2089//576	2089//576	true	true	2089//576	0;4	I;I+D	1//1;3//1	I	-227//64	true
descendant_heavy	high	terminal_interval	2089//576		false	false	2665//576	0	I	1//1	I	-291//64	true

Table S12: All pairwise resource-price relationships and crossing candidates.

Schedule	Belief	Left library	Left value	Left display burden W	Right library	Right value	Right display burden W	Relation	Price	Nonneg.	Globally active
equal_active	low	I	115//2 88	1//1	I+A	1//1	2//1	intersection	173//2 88	true	false
equal_active	low	I	115//2 88	1//1	I+B	1//1	2//1	intersection	173//2 88	true	false
equal_active	low	I	115//2 88	1//1	I+A+B	1//1	3//1	intersection	173//5 76	true	false
equal_active	low	I	115//2 88	1//1	I+D	14//3	2//1	intersection	1229// 288	true	true
equal_active	low	I	115//2 88	1//1	I+A+D	14//3	3//1	intersection	1229// 576	true	false
equal_active	low	I	115//2 88	1//1	I+B+D	14//3	3//1	intersection	1229// 576	true	false
equal_active	low	I	115//2 88	1//1	I+A+B+D	14//3	4//1	intersection	1229// 864	true	false
equal_active	low	I+A	1//1	2//1	I+B	1//1	2//1	coincident		false	false
equal_active	low	I+A	1//1	2//1	I+A+B	1//1	3//1	intersection	0//1	true	false
equal_active	low	I+A	1//1	2//1	I+D	14//3	2//1	parallel		false	false
equal_active	low	I+A	1//1	2//1	I+A+D	14//3	3//1	intersection	11//3	true	false
equal_active	low	I+A	1//1	2//1	I+B+D	14//3	3//1	intersection	11//3	true	false
equal_active	low	I+A	1//1	2//1	I+A+B+D	14//3	4//1	intersection	11//6	true	false
equal_active	low	I+B	1//1	2//1	I+A+B	1//1	3//1	intersection	0//1	true	false
equal_active	low	I+B	1//1	2//1	I+D	14//3	2//1	parallel		false	false
equal_active	low	I+B	1//1	2//1	I+A+D	14//3	3//1	intersection	11//3	true	false

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Schedule	Belief	Left library	Left value	Left display burden W	Right library	Right value	Right display burden W	Relation	Price	Nonneg.	Globally active
equal_active	low	I+B	1//1	2//1	I+B+D	14//3	3//1	intersection	11//3	true	false
equal_active	low	I+B	1//1	2//1	I+A+B+D	14//3	4//1	intersection	11//6	true	false
equal_active	low	I+A+B	1//1	3//1	I+D	14//3	2//1	intersection	-11//3	false	false
equal_active	low	I+A+B	1//1	3//1	I+A+D	14//3	3//1	parallel		false	false
equal_active	low	I+A+B	1//1	3//1	I+B+D	14//3	3//1	parallel		false	false
equal_active	low	I+A+B	1//1	3//1	I+A+B+D	14//3	4//1	intersection	11//3	true	false
equal_active	low	I+D	14//3	2//1	I+A+D	14//3	3//1	intersection	0//1	true	true
equal_active	low	I+D	14//3	2//1	I+B+D	14//3	3//1	intersection	0//1	true	true
equal_active	low	I+D	14//3	2//1	I+A+B+D	14//3	4//1	intersection	0//1	true	true
equal_active	low	I+A+D	14//3	3//1	I+B+D	14//3	3//1	coincident		false	false
equal_active	low	I+A+D	14//3	3//1	I+A+B+D	14//3	4//1	intersection	0//1	true	true
equal_active	low	I+B+D	14//3	3//1	I+A+B+D	14//3	4//1	intersection	0//1	true	true
equal_active	high	I	23//288	1//1	I+A	7//6	2//1	intersection	313//288	true	false
equal_active	high	I	23//288	1//1	I+B	7//6	2//1	intersection	313//288	true	false
equal_active	high	I	23//288	1//1	I+A+B	7//6	3//1	intersection	313//76	true	false
equal_active	high	I	23//288	1//1	I+D	22//3	2//1	intersection	2089//288	true	true
equal_active	high	I	23//288	1//1	I+A+D	22//3	3//1	intersection	2089//576	true	false

Schedule	Belief	Left library	Left value	Left display burden W	Right library	Right value	Right display burden W	Relation	Price	Nonneg.	Globally active
equal_active	high	I	23//288	1//1	I+B+D	22//3	3//1	intersection	2089//576	true	false
equal_active	high	I	23//288	1//1	I+A+B+D	22//3	4//1	intersection	2089//864	true	false
equal_active	high	I+A	7//6	2//1	I+B	7//6	2//1	coincident		false	false
equal_active	high	I+A	7//6	2//1	I+A+B	7//6	3//1	intersection	0//1	true	false
equal_active	high	I+A	7//6	2//1	I+D	22//3	2//1	parallel		false	false
equal_active	high	I+A	7//6	2//1	I+A+D	22//3	3//1	intersection	37//6	true	false
equal_active	high	I+A	7//6	2//1	I+B+D	22//3	3//1	intersection	37//6	true	false
equal_active	high	I+A	7//6	2//1	I+A+B+D	22//3	4//1	intersection	37//12	true	false
equal_active	high	I+B	7//6	2//1	I+A+B	7//6	3//1	intersection	0//1	true	false
equal_active	high	I+B	7//6	2//1	I+D	22//3	2//1	parallel		false	false
equal_active	high	I+B	7//6	2//1	I+A+D	22//3	3//1	intersection	37//6	true	false
equal_active	high	I+B	7//6	2//1	I+B+D	22//3	3//1	intersection	37//6	true	false
equal_active	high	I+B	7//6	2//1	I+A+B+D	22//3	4//1	intersection	37//12	true	false
equal_active	high	I+A+B	7//6	3//1	I+D	22//3	2//1	intersection	-37//6	false	false
equal_active	high	I+A+B	7//6	3//1	I+A+D	22//3	3//1	parallel		false	false
equal_active	high	I+A+B	7//6	3//1	I+B+D	22//3	3//1	parallel		false	false
equal_active	high	I+A+B	7//6	3//1	I+A+B+D	22//3	4//1	intersection	37//6	true	false

Schedule	Belief	Left library	Left value	Left display burden $\widetilde{W}$	Right library	Right value	Right display burden $\widetilde{W}$	Relation	Price	Nonneg.	Globally active
carrie_r_heavy	high	I+B	7//6	3//1	I+A+B+D	22//3	6//1	intersection	37//18	true	false
carrie_r_heavy	high	I+A+B	7//6	5//1	I+D	22//3	2//1	intersection	-37//18	false	false
carrie_r_heavy	high	I+A+B	7//6	5//1	I+A+D	22//3	4//1	intersection	-37//6	false	false
carrie_r_heavy	high	I+A+B	7//6	5//1	I+B+D	22//3	4//1	intersection	-37//6	false	false
carrie_r_heavy	high	I+A+B	7//6	5//1	I+A+B+D	22//3	6//1	intersection	37//6	true	false
carrie_r_heavy	high	I+D	22//3	2//1	I+A+D	22//3	4//1	intersection	0//1	true	true
carrie_r_heavy	high	I+D	22//3	2//1	I+B+D	22//3	4//1	intersection	0//1	true	true
carrie_r_heavy	high	I+D	22//3	2//1	I+A+B+D	22//3	6//1	intersection	0//1	true	true
carrie_r_heavy	high	I+A+D	22//3	4//1	I+B+D	22//3	4//1	coincident		false	false
carrie_r_heavy	high	I+A+D	22//3	4//1	I+A+B+D	22//3	6//1	intersection	0//1	true	true
carrie_r_heavy	high	I+B+D	22//3	4//1	I+A+B+D	22//3	6//1	intersection	0//1	true	true
descendant_heavy	low	I	115//288	1//1	I+A	1//1	2//1	intersection	173//288	true	false
descendant_heavy	low	I	115//288	1//1	I+B	1//1	2//1	intersection	173//288	true	false
descendant_heavy	low	I	115//288	1//1	I+A+B	1//1	3//1	intersection	173//576	true	false
descendant_heavy	low	I	115//288	1//1	I+D	14//3	3//1	intersection	1229//576	true	true

Schedule	Belief	Left library	Left value	Left display burden W	Right library	Right value	Right display burden W	Relation	Price	Nonneg.	Globally active
descen dant_h eavy	low	I	115//2 88	1//1	I+A+D	14//3	4//1	intersection	1229// 864	true	false
descen dant_h eavy	low	I	115//2 88	1//1	I+B+D	14//3	4//1	intersection	1229// 864	true	false
descen dant_h eavy	low	I	115//2 88	1//1	I+A+B+D	14//3	5//1	intersection	1229// 1152	true	false
descen dant_h eavy	low	I+A	1//1	2//1	I+B	1//1	2//1	coincident		false	false
descen dant_h eavy	low	I+A	1//1	2//1	I+A+B	1//1	3//1	intersection	0//1	true	false
descen dant_h eavy	low	I+A	1//1	2//1	I+D	14//3	3//1	intersection	11//3	true	false
descen dant_h eavy	low	I+A	1//1	2//1	I+A+D	14//3	4//1	intersection	11//6	true	false
descen dant_h eavy	low	I+A	1//1	2//1	I+B+D	14//3	4//1	intersection	11//6	true	false
descen dant_h eavy	low	I+A	1//1	2//1	I+A+B+D	14//3	5//1	intersection	11//9	true	false
descen dant_h eavy	low	I+B	1//1	2//1	I+A+B	1//1	3//1	intersection	0//1	true	false
descen dant_h eavy	low	I+B	1//1	2//1	I+D	14//3	3//1	intersection	11//3	true	false

Schedule	Belief	Left library	Left value	Left display burden $\bar{W}$	Right library	Right value	Right display burden $\bar{W}$	Relation	Price	Nonneg.	Globally active
descen dant_h eavy	low	I+B	1//1	2//1	I+A+D	14//3	4//1	intersection	11//6	true	false
descen dant_h eavy	low	I+B	1//1	2//1	I+B+D	14//3	4//1	intersection	11//6	true	false
descen dant_h eavy	low	I+B	1//1	2//1	I+A+B+D	14//3	5//1	intersection	11//9	true	false
descen dant_h eavy	low	I+A+B	1//1	3//1	I+D	14//3	3//1	parallel		false	false
descen dant_h eavy	low	I+A+B	1//1	3//1	I+A+D	14//3	4//1	intersection	11//3	true	false
descen dant_h eavy	low	I+A+B	1//1	3//1	I+B+D	14//3	4//1	intersection	11//3	true	false
descen dant_h eavy	low	I+A+B	1//1	3//1	I+A+B+D	14//3	5//1	intersection	11//6	true	false
descen dant_h eavy	low	I+D	14//3	3//1	I+A+D	14//3	4//1	intersection	0//1	true	true
descen dant_h eavy	low	I+D	14//3	3//1	I+B+D	14//3	4//1	intersection	0//1	true	true
descen dant_h eavy	low	I+D	14//3	3//1	I+A+B+D	14//3	5//1	intersection	0//1	true	true
descen dant_h eavy	low	I+A+D	14//3	4//1	I+B+D	14//3	4//1	coincident		false	false

Schedule	Belief	Left library	Left value	Left display burden W	Right library	Right value	Right display burden W	Relation	Price	Nonneg.	Globally active
descen dant_h eavy	low	I+A+D	14//3	4//1	I+A+B+D	14//3	5//1	intersection	0//1	true	true
descen dant_h eavy	low	I+B+D	14//3	4//1	I+A+B+D	14//3	5//1	intersection	0//1	true	true
descen dant_h eavy	high	I	23//288	1//1	I+A	7//6	2//1	intersection	313//288	true	false
descen dant_h eavy	high	I	23//288	1//1	I+B	7//6	2//1	intersection	313//288	true	false
descen dant_h eavy	high	I	23//288	1//1	I+A+B	7//6	3//1	intersection	313//576	true	false
descen dant_h eavy	high	I	23//288	1//1	I+D	22//3	3//1	intersection	2089//576	true	true
descen dant_h eavy	high	I	23//288	1//1	I+A+D	22//3	4//1	intersection	2089//864	true	false
descen dant_h eavy	high	I	23//288	1//1	I+B+D	22//3	4//1	intersection	2089//864	true	false
descen dant_h eavy	high	I	23//288	1//1	I+A+B+D	22//3	5//1	intersection	2089//1152	true	false
descen dant_h eavy	high	I+A	7//6	2//1	I+B	7//6	2//1	coincident		false	false
descen dant_h eavy	high	I+A	7//6	2//1	I+A+B	7//6	3//1	intersection	0//1	true	false

Schedule	Belief	Left library	Left value	Left display burden $\bar{W}$	Right library	Right value	Right display burden $\bar{W}$	Relation	Price	Nonneg.	Globally active
descen dant_h eavy	high	I+A	7//6	2//1	I+D	22//3	3//1	intersection	37//6	true	false
descen dant_h eavy	high	I+A	7//6	2//1	I+A+D	22//3	4//1	intersection	37//12	true	false
descen dant_h eavy	high	I+A	7//6	2//1	I+B+D	22//3	4//1	intersection	37//12	true	false
descen dant_h eavy	high	I+A	7//6	2//1	I+A+B+D	22//3	5//1	intersection	37//18	true	false
descen dant_h eavy	high	I+B	7//6	2//1	I+A+B	7//6	3//1	intersection	0//1	true	false
descen dant_h eavy	high	I+B	7//6	2//1	I+D	22//3	3//1	intersection	37//6	true	false
descen dant_h eavy	high	I+B	7//6	2//1	I+A+D	22//3	4//1	intersection	37//12	true	false
descen dant_h eavy	high	I+B	7//6	2//1	I+B+D	22//3	4//1	intersection	37//12	true	false
descen dant_h eavy	high	I+B	7//6	2//1	I+A+B+D	22//3	5//1	intersection	37//18	true	false
descen dant_h eavy	high	I+A+B	7//6	3//1	I+D	22//3	3//1	parallel		false	false
descen dant_h eavy	high	I+A+B	7//6	3//1	I+A+D	22//3	4//1	intersection	37//6	true	false

Schedule	Belief	Left library	Left value	Left display burden W	Right library	Right value	Right display burden W	Relation	Price	Nonneg.	Globally active
descen dant_h eavy	high	I+A+B	7//6	3//1	I+B+D	22//3	4//1	intersection	37//6	true	false
descen dant_h eavy	high	I+A+B	7//6	3//1	I+A+B+D	22//3	5//1	intersection	37//12	true	false
descen dant_h eavy	high	I+D	22//3	3//1	I+A+D	22//3	4//1	intersection	0//1	true	true
descen dant_h eavy	high	I+D	22//3	3//1	I+B+D	22//3	4//1	intersection	0//1	true	true
descen dant_h eavy	high	I+D	22//3	3//1	I+A+B+D	22//3	5//1	intersection	0//1	true	true
descen dant_h eavy	high	I+A+D	22//3	4//1	I+B+D	22//3	4//1	coincident		false	false
descen dant_h eavy	high	I+A+D	22//3	4//1	I+A+B+D	22//3	5//1	intersection	0//1	true	true
descen dant_h eavy	high	I+B+D	22//3	4//1	I+A+B+D	22//3	5//1	intersection	0//1	true	true

The committed CSV files retain additional machine-oriented counts and certificate fields not repeated as display columns above. In particular, they record feasible and enumerated counts, all-optima-feasible flags, productive-value preservation, capacity shadow increments, and optimizer fragility diagnostics. These fields remain part of the reproducibility record; the tables here preserve every numerical row and every optimizer or crossing candidate needed to reconstruct the reported correspondences.

S11 Frozen Registered Randomized Library Study (N=1024)

Historical evidence boundary. This is the original frozen $N = 1024$ randomized-library study inherited unchanged from the preprint program. Its registry, seeds, design locks, amendments, and committed results predate and are distinct from the Registered Algorithmic Compression Benchmark v2 in Section S12.

This section preserves the detailed estimands, raw-realizability gates, interaction diagnostics, sequential-prefix records, and frozen-pilot comparison behind the historical randomized results reported in the preprint program. The study is a synthetic structural stress test of a fixed finite generator, not a population sample or a population-frequency estimate.

S11.1 Synthetic estimands and exact fixture bridge

For fixture f , horizon H , and declared input set X_f , define

$$\epsilon_{f,H}^{\text{proj}} := \max_{(b,L) \in X_f} |V_H^{\text{raw}}(b, L) - V_H^K(b, \phi(L))|.$$

For deletion $L \rightarrow L^-$, frontier-only pruning $L \rightarrow L^F$, and insertion, the remaining exact-fixture estimands are

$$\begin{aligned} \epsilon_F &:= \|F_L - F_{L^-}\|_\infty, \quad \epsilon_C := \mathbf{1}\{C_L \neq C_{L^-}\}, \quad \epsilon_V := \max_b |V_H(b, L) - V_H(b, L^-)|, \\ \text{Loss}_H^F &:= V_H(b, L) - V_H(b, L^F), \quad \lambda^F := \frac{\text{Loss}_H^F}{\beta^d \rho^d \pi C - \kappa}, \quad \epsilon^{\text{dec}} := \Delta^{\text{tot}} - \Delta^{\text{op}} - \Delta^{\text{gen}}. \end{aligned}$$

The ratio λ^F is used only when its denominator is positive. For a four-corner value fixture and a one-shot cost-covering set, define

$$J := [V(F_1, C_1) - V(F_1, C_0)] - [V(F_0, C_1) - V(F_0, C_0)], \quad \mathcal{R} := \{b : \Psi(b) - \kappa(b) \geq 0\},$$

and let $c(\mathcal{R})$ be the number of connected components on the ordered belief grid.

The exact fixture outcomes are retained here as a bridge between those estimands and the randomized design.

The scaled frontier-only family obtains loss $R/2$ at each $R \in \{0, 2, 4, 10, 20, 40\}$. The unit-cap fixture is the normalized result; the bad-kernel and separated-gap fixtures retain their disconnection diagnostics.

S11.2 Registered design and randomized estimands

The randomized layer uses master seed 6075990691714899803 and all $N = 1024$ registered horizon-four trials. The complete 2^7 factorial crosses sparse/dense frontiers, low/high module overlap, weak/strong module complementarity, low/high project cost,

Mechanism	Registered exact input	Computed diagnostic
Safe compression	Four-strategy, two-belief library	Minimum and greedy sizes both 3; frontier, closure, and value loss all zero
Frontier-only loss	Future rewards 0, 2, 4, 10, 20, 40	Loss $R/2$ in all six cases; largest displayed loss 20
Value decomposition	Operational-only, generative-only, and joint insertions	$(\Delta^{\text{op}}, \Delta^{\text{gen}}) = (3, 0), (0, 4), (3, 1)$
Monotone single gap	Potential $(0, 1/2, 3/2, 3)$	Upper-grid research set; one component
Single peak, bad kernel	Gap $(0, 1, 0)$, potential $(1, 0, 1)$	Preservation assumption fails; two components
Separated multi-gap	Potential $(4, 41/32, 1/2, 41/32, 4)$	Constant-cost research set has two components

Table S13 Selected-instance outcomes from the exact mechanism fixtures.

short/long duration, low/high admission, and low/high persistence. Each of the 128 cells has eight replicates. The complete registry records 1,024 trial, catalog, project, and deletion seeds; all 4,096 are distinct. The initial design, sequential-precision amendment, and executable-protocol amendment were locked before outcomes.

For trial t , source library L_t , pruning method m , and pruned library L_{tm} , let

$$P_t(L) := \text{passive value}, \quad \Omega_t(L) := V_t(L) - P_t(L),$$

$$\begin{aligned} L_{tm}^{\text{op}} &:= P_t(L_t) - P_t(L_{tm}), & L_{tm}^{\text{gen}} &:= \Omega_t(L_t) - \Omega_t(L_{tm}), \\ L_{tm}^{\text{tot}} &:= V_t(L_t) - V_t(L_{tm}) = L_{tm}^{\text{op}} + L_{tm}^{\text{gen}}, & r_{tm} &:= 1 - |L_{tm}|/|L_t|. \end{aligned}$$

For frontier-only pruning F , define

$$\begin{aligned} \hat{p}_F &:= N^{-1} \sum_t \mathbf{1}\{L_{tF}^{\text{tot}} > 0\}, & \bar{L}_F^{+|+} &:= \frac{\sum_t [L_{tF}^{\text{tot}}]_+}{\sum_t \mathbf{1}\{L_{tF}^{\text{tot}} > 0\}}, \\ \tilde{L}_F^{\text{max}} &:= \max_t \frac{[L_{tF}^{\text{tot}}]_+}{V_t(L_t)}. \end{aligned}$$

Strict positivity of every source value is a construction gate. An operationally silent generative asset $s \in L_t$ satisfies

$$F_{L_t} = F_{L_t \setminus \{s\}}, \quad C_{L_t} \neq C_{L_t \setminus \{s\}}, \quad V_t(L_t) - V_t(L_t \setminus \{s\}) > 0.$$

Its asset-level frequency uses every noninactive source asset; a separate library-level count records whether any such asset is present.

Every interaction observation starts from four actual raw libraries:

$$\begin{aligned} L_{00} &= L_{\text{base}}, & L_{10} &= L_{\text{base}} \cup A_F, \\ L_{01} &= L_{\text{base}} \cup A_C, & L_{11} &= L_{\text{base}} \cup A_F \cup A_C, \end{aligned}$$

where A_F is closure-silent and frontier-improving, A_C is frontier-silent and closure-improving, and the two additions commute. All four libraries use the same catalog and module closure. Their frontiers, closures, project laws, menus, transitions, and values are recomputed from raw primitives. With J_t defined by the four-corner difference above, the sign frequencies are

$$\hat{p}_{\text{sub}} := \frac{\#\{t : J_t < 0\}}{1024}, \quad \hat{p}_{\text{comp}} := \frac{\#\{t : J_t > 0\}}{1024},$$

with the analogous $\hat{p}_0$ for $J_t = 0$. No synthetic compressed rectangle enters these denominators.

All profiles, module rows, closure tables, kernels, project requirements, admitted laws, raw updates, passive values, and Bellman values use `Rational{BigInt}`. Before writing any artifact, hard gates require all 1,024 trials, all 4,096 raw corners, and every raw library to be valid; every compressed state to have an enumerated raw witness; all raw/compressed values to agree at every belief and remaining horizon $0, \dots, 4$; frontier-only passive loss to be zero; innovation-safe frontier, closure, operational, generative, and total loss to be exactly zero; every signed decomposition to close; and every displayed theorem-condition flag to be computed from exact laws and action tables. The completed audit records 47,458 raw-witness rows and 1,920 raw/compressed value comparisons per trial.

S11.3 Registered outcomes and factor slices

Registered estimand	Exact count	Exact estimate
Positive frontier-only dynamic loss	398/1024	199/512
Conditional mean positive loss	398 positive trials	124909129/78249984
Maximum normalized loss	1024 trials	84963/222163
Positive innovation-safe loss	0/1024	0
Mean frontier-only compression ratio	1024 trials	5/12
Mean innovation-safe compression ratio	1024 trials	1/8
Silent generative asset	388/5120	97/1280
Substitution, $J_t < 0$	256/1024	1/4
Complementarity, $J_t > 0$	141/1024	141/1024
Zero interaction, $J_t = 0$	627/1024	627/1024

Table S14 Registered finite-generator outcomes. Ratios are structural stress-test frequencies, not estimates of real-population prevalence.

Frontier-only pruning has positive loss in 398 trials; conditional on positive loss, its mean magnitude is $124909129/78249984 \approx 1.5963$, and the largest normalized loss is $84963/222163 \approx 0.3824$. Innovation-safe pruning has exact zero frontier, closure, operational, generative, and total loss in every trial. Its smaller mean compression ratio reflects the additional closure- preservation constraint. Silent generative carriers occur in 388 of 5,120 noninactive source-asset observations.

The primitive common-gap predicate is the mechanical conjunction of frontier-independent project primitives and menus, poor-to-rich menu inclusion, an antitone recursive descendant gap, nonnegative rich exposure, zero poor exposure, and exact action-value identities. It holds in 512 trials. Those rows contain 255 substitutions, 257 zeros, and no complementarity. The other 512 rows are registered boundary constructions: the 256 frontier-dependent-generator rows contain all 141 complementarities, one substitution, and 114 zeros; the 256 positive-poor-exposure rows are all zero. Hence every positive J_t occurs where frontier-independent generation and completion fail. Marginally, complementarity is more frequent under low cost than high cost (97 versus 44), short than long duration (107 versus 34), and high than low admission (95 versus 46). These balanced factor slices locate generated cases; they are neither causal comparisons nor a theorem that complementarity is rare.

The companion exact interaction response preserves the registered 2,430-row frontier-project boundary grid and five canonical four-corner fixtures: primitive strict substitution, the saturation boundary, optimizer-switching complementarity, frontier-dependent-success complementarity, and nonconstant separability. It crosses strict low/high frontier pairs, closure richness, project cost, admission probability, descendant payoff, incumbent operating reward, project duration, and frontier dependence of generator quality. Each row records the two closure increments, their cross difference, four selected projects, four realizability flags, and the primitive sufficient-condition certificate. Nonrealizable rows remain diagnostic records but are never aggregated into interaction-sign frequencies; Section S8 reports the corresponding interaction-theory interpretation.

S11.4 Sequential prefixes and descriptive uncertainty

The sequential amendment fixed cumulative views at $N = 50, 100, 200, 300, 500, 750, 1000, 1024$ before outcomes and required the final estimate to use all 1,024 trials. The early prefixes are not claimed to be factor-balanced.

At $N = 1024$, the descriptive Wilson intervals are $[0.3593, 0.4189]$ for frontier loss, $[0, 0.00374]$ for innovation-safe loss, $[0.06885, 0.08335]$ for silent assets, $[0.2244, 0.2774]$ for substitution, and $[0.1179, 0.1602]$ for complementarity. The corresponding Monte Carlo standard errors are 0.08416 for conditional mean positive loss, 0.00261 for mean frontier compression, and 0.00432 for mean safe compression. Innovation-safe loss has zero events at every prefix and therefore retains the registered sparse-event warning; complementarity has zero events through $N = 100$, illustrating why the registered maximum cannot be selected from an early sign.

The complete 64-row cumulative table, 896-row factor-stratified table, exact sample variances, interval values, warnings, and stabilization figure remain in the registered artifacts indexed in Section S16.

N	Frontier loss	Safe loss	Silent asset	Substitution	Complementarity
50	25/50	0/50	25/250	27/50	0/50
100	45/100	0/100	45/500	47/100	0/100
200	101/200	0/200	101/1000	64/200	39/200
300	153/300	0/300	153/1500	87/300	69/300
500	198/500	0/500	192/2500	127/500	69/500
750	321/750	0/750	315/3750	192/750	130/750
1000	396/1000	0/1000	388/5000	256/1000	141/1000
1024	398/1024	0/1024	388/5120	256/1024	141/1024

Table S15 Cumulative registered event counts. The silent-asset denominator is five noninactive source assets per trial. Prefixes diagnose simulation precision and never select the final sample.

N	Positive support	Mean positive loss	Frontier compression	Safe compression
50	25	1818477/1638400	31/75	9/100
100	45	1623911/1474560	5/12	3/25
200	101	5624367/3309568	5/12	51/400
300	153	55259345/30081024	187/450	227/1800
500	198	63735497/38928384	1249/3000	47/375
750	321	107121079/63111168	1873/4500	557/4500
1000	396	124655857/77856768	5/12	47/375
1024	398	124909129/78249984	5/12	1/8

Table S16 Cumulative mean diagnostics, all computed exactly.

S11.5 Frozen pilot and policy-map diagnostics

The earlier $N = 90$ study remains a frozen pilot and is not pooled with the registered v2 estimates. The two generators differ materially, so the comparison below records provenance rather than a trend.

The separately locked resource extension reuses only the frozen v2 trial registry and does not pool its outcomes with the parent study. Every trial enumerates all 32 inactive-containing sublibraries. Complete optimizer correspondences, capacity staircases, active pairwise breakpoints, price paths, and channel arcs are indexed in Section S16.

Quantity	Frozen pilot, $N = 90$	Registered v2, $N = 1024$
Design	twelve marginally balanced three-level factors	complete 2^7 factorial, eight replicates per cell
Positive frontier-only loss	8/90	398/1024
Conditional mean positive loss	3705/16384	124909129/78249984
Mean frontier compression	22/45	5/12
Mean innovation-safe compression	2009/5400	1/8
Positive innovation-safe loss	0/90	0/1024
Silent generative assets	3/360	388/5120
Interaction comparison	omitted: no four-row-witness gate	256/141/627 negative/positive/zero

Table S17 Frozen-pilot and v2 results are reported separately. The pilot interaction statistic is not compared because its synthetic compressed rectangle did not require four raw witnesses.

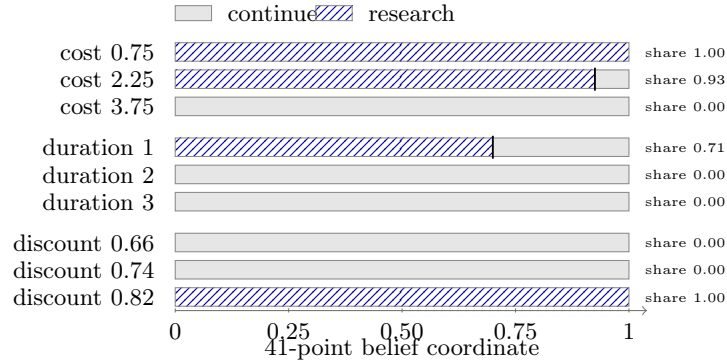


Fig. S2 Policy-map slices from the unified positive-duration response experiment. The estimand in each row is the share of 41 enumerated belief points selecting research in a converged raw-derived Float64 solve; light unhatched spans denote Continue and diagonal hatching denotes research. Filled spans summarize the connected finite-grid action set and the boundary line is a grid cutoff, not a continuous root or smooth interpolation. All duration rows are strictly positive, use P_B^d , and apply the active-operation flag.

S12 Registered Algorithmic Compression Benchmark v2

New journal-program evidence. This prospectively registered algorithmic benchmark is distinct from the frozen preprint-era $N = 1024$ randomized study in Section S11. No seed, registry row, design lock, or result is shared or relabeled across the two programs.

The predecessor v1 final execution was stopped at the author’s request on 2026-08-27 after 1634 of 3907 registered run records had become terminal and 68 of 173 final instances had been generated. The visible information was incomplete infrastructure progress; no result-level burden, runtime, solver, or algorithm comparison was used to alter a v1 row, seed, limit, method, or analysis rule. V1 remains an incomplete historical record. V2 used new identifiers, seeds, configuration, result root, and lock; it changed execution infrastructure by parallelizing independent one-thread run units and did not pool v1 outcomes. This chronology is recorded in the committed v1 incomplete-execution notice.

The locked v2 analysis population contains 173 instances: 24 small exact, 128 structured, and 21 adversarial. Its 3907 registered run units retain every terminal outcome; 3526 returned libraries passed independent exact checks and 381 records returned no accepted library. These counts are generated from the committed audit summary through `tables/evidence_macros.tex`.

The design, analysis plan, reporting rules, seed and instance registries, amendments, active design lock, execution-start lock, environment record, raw run matrix, and independent audit are under `experiments/algorithmic_compression_v2/`. The final runner used 8 concurrent one-thread subprocesses. Runtime is therefore an eight-worker contention estimand on an Apple M1 Pro, not a claim about serial algorithm complexity or another machine.

S12.1 Realized registered design

The committed realized-design table records requested and measured structure; invalid generator outcomes remain terminal failures rather than silently repaired instances.

Table S18: Registered design and realized structural diagnostics. Generation failures retain requested fields and unavailable realized fields. Part 1 of 2; join by the repeated identifying columns.

S74	instance id	family	size	generation status	coverage density level
	V2-A-N007-R006-X01	small_exact	N7-R6	GENERATED	one_eighth
	V2-A-N007-R006-X02	small_exact	N7-R6	GENERATED	one_eighth
	V2-A-N007-R006-X03	small_exact	N7-R6	GENERATED	three_eighths
	V2-A-N007-R006-X04	small_exact	N7-R6	GENERATED	three_eighths
	V2-A-N007-R010-X01	small_exact	N7-R10	GENERATED	one_eighth
	V2-A-N007-R010-X02	small_exact	N7-R10	GENERATED	one_eighth
	V2-A-N007-R010-X03	small_exact	N7-R10	GENERATED	three_eighths
	V2-A-N007-R010-X04	small_exact	N7-R10	GENERATED	three_eighths
	V2-A-N011-R006-X01	small_exact	N11-R6	GENERATED	one_eighth
	V2-A-N011-R006-X02	small_exact	N11-R6	GENERATED	one_eighth
	V2-A-N011-R006-X03	small_exact	N11-R6	GENERATED	three_eighths
	V2-A-N011-R006-X04	small_exact	N11-R6	GENERATED	three_eighths
	V2-A-N011-R010-X01	small_exact	N11-R10	GENERATED	one_eighth
	V2-A-N011-R010-X02	small_exact	N11-R10	GENERATED	one_eighth
	V2-A-N011-R010-X03	small_exact	N11-R10	GENERATED	three_eighths
	V2-A-N011-R010-X04	small_exact	N11-R10	GENERATED	three_eighths
	V2-A-N015-R006-X01	small_exact	N15-R6	GENERATED	one_eighth
	V2-A-N015-R006-X02	small_exact	N15-R6	GENERATED	one_eighth
	V2-A-N015-R006-X03	small_exact	N15-R6	GENERATED	three_eighths
	V2-A-N015-R006-X04	small_exact	N15-R6	GENERATED	three_eighths
	V2-A-N015-R010-X01	small_exact	N15-R10	GENERATED	one_eighth
	V2-A-N015-R010-X02	small_exact	N15-R10	GENERATED	one_eighth
	V2-A-N015-R010-X03	small_exact	N15-R10	GENERATED	three_eighths

instance id	family	size	generation status	coverage density level
V2-A-N015-R010-X04	small_exact	N15-R10	GENERATED	three_eighths
V2-B-N064-R032-C01-X01	structured	N64-R32	GENERATED	one_eighth
V2-B-N064-R032-C01-X02	structured	N64-R32	GENERATED	one_eighth
V2-B-N064-R032-C02-X01	structured	N64-R32	GENERATED	one_eighth
V2-B-N064-R032-C02-X02	structured	N64-R32	GENERATED	one_eighth
V2-B-N064-R032-C03-X01	structured	N64-R32	GENERATED	three_eighths
V2-B-N064-R032-C03-X02	structured	N64-R32	GENERATED	three_eighths
V2-B-N064-R032-C04-X01	structured	N64-R32	GENERATED	three_eighths
V2-B-N064-R032-C04-X02	structured	N64-R32	GENERATED	three_eighths
V2-B-N064-R032-C05-X01	structured	N64-R32	GENERATED	one_eighth
V2-B-N064-R032-C05-X02	structured	N64-R32	GENERATED	one_eighth
V2-B-N064-R032-C06-X01	structured	N64-R32	GENERATED	one_eighth
V2-B-N064-R032-C06-X02	structured	N64-R32	GENERATED	one_eighth
V2-B-N064-R032-C07-X01	structured	N64-R32	GENERATED	three_eighths
V2-B-N064-R032-C07-X02	structured	N64-R32	GENERATED	three_eighths
V2-B-N064-R032-C08-X01	structured	N64-R32	GENERATED	three_eighths
V2-B-N064-R032-C08-X02	structured	N64-R32	GENERATED	three_eighths
V2-B-N064-R032-C09-X01	structured	N64-R32	GENERATION_FAILED	one_eighth
V2-B-N064-R032-C09-X02	structured	N64-R32	GENERATION_FAILED	one_eighth
V2-B-N064-R032-C10-X01	structured	N64-R32	GENERATED	one_eighth
V2-B-N064-R032-C10-X02	structured	N64-R32	GENERATED	one_eighth
V2-B-N064-R032-C11-X01	structured	N64-R32	GENERATED	three_eighths
V2-B-N064-R032-C11-X02	structured	N64-R32	GENERATED	three_eighths
V2-B-N064-R032-C12-X01	structured	N64-R32	GENERATED	three_eighths
V2-B-N064-R032-C12-X02	structured	N64-R32	GENERATED	three_eighths
V2-B-N064-R032-C13-X01	structured	N64-R32	GENERATION_FAILED	one_eighth

instance id	family	size	generation status	coverage density level
V2-B-N064-R032-C13-X02	structured	N64-R32	GENERATION_FAILED	one_eighth
V2-B-N064-R032-C14-X01	structured	N64-R32	GENERATED	one_eighth
V2-B-N064-R032-C14-X02	structured	N64-R32	GENERATED	one_eighth
V2-B-N064-R032-C15-X01	structured	N64-R32	GENERATED	three_eighths
V2-B-N064-R032-C15-X02	structured	N64-R32	GENERATED	three_eighths
V2-B-N064-R032-C16-X01	structured	N64-R32	GENERATED	three_eighths
V2-B-N064-R032-C16-X02	structured	N64-R32	GENERATED	three_eighths
V2-B-N064-R064-C01-X01	structured	N64-R64	GENERATED	one_eighth
V2-B-N064-R064-C01-X02	structured	N64-R64	GENERATED	one_eighth
V2-B-N064-R064-C02-X01	structured	N64-R64	GENERATED	one_eighth
V2-B-N064-R064-C02-X02	structured	N64-R64	GENERATED	one_eighth
V2-B-N064-R064-C03-X01	structured	N64-R64	GENERATED	three_eighths
V2-B-N064-R064-C03-X02	structured	N64-R64	GENERATED	three_eighths
V2-B-N064-R064-C04-X01	structured	N64-R64	GENERATED	three_eighths
V2-B-N064-R064-C04-X02	structured	N64-R64	GENERATED	three_eighths
V2-B-N064-R064-C05-X01	structured	N64-R64	GENERATED	one_eighth
V2-B-N064-R064-C05-X02	structured	N64-R64	GENERATED	one_eighth
V2-B-N064-R064-C06-X01	structured	N64-R64	GENERATED	one_eighth
V2-B-N064-R064-C06-X02	structured	N64-R64	GENERATED	one_eighth
V2-B-N064-R064-C07-X01	structured	N64-R64	GENERATED	three_eighths
V2-B-N064-R064-C07-X02	structured	N64-R64	GENERATED	three_eighths
V2-B-N064-R064-C08-X01	structured	N64-R64	GENERATED	three_eighths
V2-B-N064-R064-C08-X02	structured	N64-R64	GENERATED	three_eighths
V2-B-N064-R064-C09-X01	structured	N64-R64	GENERATION_FAILED	one_eighth
V2-B-N064-R064-C09-X02	structured	N64-R64	GENERATION_FAILED	one_eighth
V2-B-N064-R064-C10-X01	structured	N64-R64	GENERATED	one_eighth

instance id	family	size	generation status	coverage density level
V2-B-N064-R064-C10-X02	structured	N64-R64	GENERATED	one_eighth
V2-B-N064-R064-C11-X01	structured	N64-R64	GENERATED	three_eighths
V2-B-N064-R064-C11-X02	structured	N64-R64	GENERATED	three_eighths
V2-B-N064-R064-C12-X01	structured	N64-R64	GENERATED	three_eighths
V2-B-N064-R064-C12-X02	structured	N64-R64	GENERATED	three_eighths
V2-B-N064-R064-C13-X01	structured	N64-R64	GENERATION_FAILED	one_eighth
V2-B-N064-R064-C13-X02	structured	N64-R64	GENERATION_FAILED	one_eighth
V2-B-N064-R064-C14-X01	structured	N64-R64	GENERATED	one_eighth
V2-B-N064-R064-C14-X02	structured	N64-R64	GENERATED	one_eighth
V2-B-N064-R064-C15-X01	structured	N64-R64	GENERATED	three_eighths
V2-B-N064-R064-C15-X02	structured	N64-R64	GENERATED	three_eighths
V2-B-N064-R064-C16-X01	structured	N64-R64	GENERATED	three_eighths
V2-B-N064-R064-C16-X02	structured	N64-R64	GENERATED	three_eighths
V2-B-N192-R032-C01-X01	structured	N192-R32	GENERATED	one_eighth
V2-B-N192-R032-C01-X02	structured	N192-R32	GENERATED	one_eighth
V2-B-N192-R032-C02-X01	structured	N192-R32	GENERATED	one_eighth
V2-B-N192-R032-C02-X02	structured	N192-R32	GENERATED	one_eighth
V2-B-N192-R032-C03-X01	structured	N192-R32	GENERATED	three_eighths
V2-B-N192-R032-C03-X02	structured	N192-R32	GENERATED	three_eighths
V2-B-N192-R032-C04-X01	structured	N192-R32	GENERATED	three_eighths
V2-B-N192-R032-C04-X02	structured	N192-R32	GENERATED	three_eighths
V2-B-N192-R032-C05-X01	structured	N192-R32	GENERATED	one_eighth
V2-B-N192-R032-C05-X02	structured	N192-R32	GENERATED	one_eighth
V2-B-N192-R032-C06-X01	structured	N192-R32	GENERATED	one_eighth
V2-B-N192-R032-C06-X02	structured	N192-R32	GENERATED	one_eighth
V2-B-N192-R032-C07-X01	structured	N192-R32	GENERATED	three_eighths

instance id	family	size	generation status	coverage density level
V2-B-N192-R032-C07-X02	structured	N192-R32	GENERATED	three_eighths
V2-B-N192-R032-C08-X01	structured	N192-R32	GENERATED	three_eighths
V2-B-N192-R032-C08-X02	structured	N192-R32	GENERATED	three_eighths
V2-B-N192-R032-C09-X01	structured	N192-R32	GENERATION_FAILED	one_eighth
V2-B-N192-R032-C09-X02	structured	N192-R32	GENERATION_FAILED	one_eighth
V2-B-N192-R032-C10-X01	structured	N192-R32	GENERATED	one_eighth
V2-B-N192-R032-C10-X02	structured	N192-R32	GENERATED	one_eighth
V2-B-N192-R032-C11-X01	structured	N192-R32	GENERATED	three_eighths
V2-B-N192-R032-C11-X02	structured	N192-R32	GENERATED	three_eighths
V2-B-N192-R032-C12-X01	structured	N192-R32	GENERATED	three_eighths
V2-B-N192-R032-C12-X02	structured	N192-R32	GENERATED	three_eighths
V2-B-N192-R032-C13-X01	structured	N192-R32	GENERATION_FAILED	one_eighth
V2-B-N192-R032-C13-X02	structured	N192-R32	GENERATION_FAILED	one_eighth
V2-B-N192-R032-C14-X01	structured	N192-R32	GENERATED	one_eighth
V2-B-N192-R032-C14-X02	structured	N192-R32	GENERATED	one_eighth
V2-B-N192-R032-C15-X01	structured	N192-R32	GENERATED	three_eighths
V2-B-N192-R032-C15-X02	structured	N192-R32	GENERATED	three_eighths
V2-B-N192-R032-C16-X01	structured	N192-R32	GENERATED	three_eighths
V2-B-N192-R032-C16-X02	structured	N192-R32	GENERATED	three_eighths
V2-B-N192-R064-C01-X01	structured	N192-R64	GENERATED	one_eighth
V2-B-N192-R064-C01-X02	structured	N192-R64	GENERATED	one_eighth
V2-B-N192-R064-C02-X01	structured	N192-R64	GENERATED	one_eighth
V2-B-N192-R064-C02-X02	structured	N192-R64	GENERATED	one_eighth
V2-B-N192-R064-C03-X01	structured	N192-R64	GENERATED	three_eighths
V2-B-N192-R064-C03-X02	structured	N192-R64	GENERATED	three_eighths
V2-B-N192-R064-C04-X01	structured	N192-R64	GENERATED	three_eighths

instance id	family	size	generation status	coverage density level
V2-B-N192-R064-C04-X02	structured	N192-R64	GENERATED	three_eighths
V2-B-N192-R064-C05-X01	structured	N192-R64	GENERATED	one_eighth
V2-B-N192-R064-C05-X02	structured	N192-R64	GENERATED	one_eighth
V2-B-N192-R064-C06-X01	structured	N192-R64	GENERATED	one_eighth
V2-B-N192-R064-C06-X02	structured	N192-R64	GENERATED	one_eighth
V2-B-N192-R064-C07-X01	structured	N192-R64	GENERATED	three_eighths
V2-B-N192-R064-C07-X02	structured	N192-R64	GENERATED	three_eighths
V2-B-N192-R064-C08-X01	structured	N192-R64	GENERATED	three_eighths
V2-B-N192-R064-C08-X02	structured	N192-R64	GENERATED	three_eighths
V2-B-N192-R064-C09-X01	structured	N192-R64	GENERATION_FAILED	one_eighth
V2-B-N192-R064-C09-X02	structured	N192-R64	GENERATION_FAILED	one_eighth
V2-B-N192-R064-C10-X01	structured	N192-R64	GENERATED	one_eighth
V2-B-N192-R064-C10-X02	structured	N192-R64	GENERATED	one_eighth
V2-B-N192-R064-C11-X01	structured	N192-R64	GENERATED	three_eighths
V2-B-N192-R064-C11-X02	structured	N192-R64	GENERATED	three_eighths
V2-B-N192-R064-C12-X01	structured	N192-R64	GENERATED	three_eighths
V2-B-N192-R064-C12-X02	structured	N192-R64	GENERATED	three_eighths
V2-B-N192-R064-C13-X01	structured	N192-R64	GENERATION_FAILED	one_eighth
V2-B-N192-R064-C13-X02	structured	N192-R64	GENERATION_FAILED	one_eighth
V2-B-N192-R064-C14-X01	structured	N192-R64	GENERATED	one_eighth
V2-B-N192-R064-C14-X02	structured	N192-R64	GENERATED	one_eighth
V2-B-N192-R064-C15-X01	structured	N192-R64	GENERATED	three_eighths
V2-B-N192-R064-C15-X02	structured	N192-R64	GENERATED	three_eighths
V2-B-N192-R064-C16-X01	structured	N192-R64	GENERATED	three_eighths
V2-B-N192-R064-C16-X02	structured	N192-R64	GENERATED	three_eighths
V2-C-M01-S01	adversarial	small	GENERATED	mechanism

instance id	family	size	generation status	coverage density level
V2-C-M01-S02	adversarial	medium	GENERATED	mechanism
V2-C-M01-S03	adversarial	large	GENERATED	mechanism
V2-C-M02-S01	adversarial	small	GENERATED	mechanism
V2-C-M02-S02	adversarial	medium	GENERATED	mechanism
V2-C-M02-S03	adversarial	large	GENERATED	mechanism
V2-C-M03-S01	adversarial	small	GENERATED	mechanism
V2-C-M03-S02	adversarial	medium	GENERATED	mechanism
V2-C-M03-S03	adversarial	large	GENERATED	mechanism
V2-C-M04-S01	adversarial	small	GENERATED	mechanism
V2-C-M04-S02	adversarial	medium	GENERATED	mechanism
V2-C-M04-S03	adversarial	large	GENERATED	mechanism
V2-C-M05-S01	adversarial	small	GENERATED	mechanism
V2-C-M05-S02	adversarial	medium	GENERATED	mechanism
V2-C-M05-S03	adversarial	large	GENERATED	mechanism
V2-C-M06-S01	adversarial	small	GENERATED	mechanism
V2-C-M06-S02	adversarial	medium	GENERATED	mechanism
V2-C-M06-S03	adversarial	large	GENERATED	mechanism
V2-C-M07-S01	adversarial	small	GENERATED	mechanism
V2-C-M07-S02	adversarial	medium	GENERATED	mechanism
V2-C-M07-S03	adversarial	large	GENERATED	mechanism

Table S19: Registered design and realized structural diagnostics. Generation failures retain requested fields and unavailable realized fields. Part 2 of 2; join by the repeated identifying columns.

instance id	coverage density realized	module overlap level	bundle prevalence level	unique carrier frequency level
V2-A-N007-R006-X01	11//36	low_clustered	zero	one_eighth
V2-A-N007-R006-X02	1//3	high_clustered	one_eighth	zero
V2-A-N007-R006-X03	7//18	low_clustered	zero	zero
V2-A-N007-R006-X04	7//18	high_clustered	one_eighth	one_eighth
V2-A-N007-R010-X01	19//60	low_clustered	zero	one_eighth
V2-A-N007-R010-X02	1//3	high_clustered	one_eighth	zero
V2-A-N007-R010-X03	23//60	low_clustered	zero	zero
V2-A-N007-R010-X04	23//60	high_clustered	one_eighth	one_eighth
V2-A-N011-R006-X01	11//60	low_clustered	zero	one_eighth
V2-A-N011-R006-X02	1//5	high_clustered	one_eighth	zero
V2-A-N011-R006-X03	23//60	low_clustered	zero	zero
V2-A-N011-R006-X04	23//60	high_clustered	one_eighth	one_eighth
V2-A-N011-R010-X01	19//100	low_clustered	zero	one_eighth
V2-A-N011-R010-X02	1//5	high_clustered	one_eighth	zero
V2-A-N011-R010-X03	19//50	low_clustered	zero	zero
V2-A-N011-R010-X04	19//50	high_clustered	one_eighth	one_eighth
V2-A-N015-R006-X01	11//84	low_clustered	zero	one_eighth
V2-A-N015-R006-X02	1//7	high_clustered	one_eighth	zero
V2-A-N015-R006-X03	8//21	low_clustered	zero	zero
V2-A-N015-R006-X04	8//21	high_clustered	one_eighth	one_eighth
V2-A-N015-R010-X01	19//140	low_clustered	zero	one_eighth
V2-A-N015-R010-X02	1//7	high_clustered	one_eighth	zero
V2-A-N015-R010-X03	53//140	low_clustered	zero	zero

instance id	coverage density realized	module overlap level	bundle prevalence level	unique carrier frequency level
V2-A-N015-R010-X04	53//140	high_clustered	one_eighth	one_eighth
V2-B-N064-R032-C01-X01	1//8	low_clustered	zero	zero
V2-B-N064-R032-C01-X02	1//8	low_clustered	zero	zero
V2-B-N064-R032-C02-X01	1//8	low_clustered	zero	one_eighth
V2-B-N064-R032-C02-X02	1//8	low_clustered	zero	one_eighth
V2-B-N064-R032-C03-X01	3//8	low_clustered	zero	one_eighth
V2-B-N064-R032-C03-X02	3//8	low_clustered	zero	one_eighth
V2-B-N064-R032-C04-X01	3//8	low_clustered	zero	zero
V2-B-N064-R032-C04-X02	3//8	low_clustered	zero	zero
V2-B-N064-R032-C05-X01	1//8	high_clustered	zero	zero
V2-B-N064-R032-C05-X02	1//8	high_clustered	zero	zero
V2-B-N064-R032-C06-X01	1//8	high_clustered	zero	one_eighth
V2-B-N064-R032-C06-X02	1//8	high_clustered	zero	one_eighth
V2-B-N064-R032-C07-X01	3//8	high_clustered	zero	one_eighth
V2-B-N064-R032-C07-X02	3//8	high_clustered	zero	one_eighth
V2-B-N064-R032-C08-X01	3//8	high_clustered	zero	zero
V2-B-N064-R032-C08-X02	3//8	high_clustered	zero	zero
V2-B-N064-R032-C09-X01	–	low_clustered	one_eighth	zero
V2-B-N064-R032-C09-X02	–	low_clustered	one_eighth	zero
V2-B-N064-R032-C10-X01	1//8	low_clustered	one_eighth	one_eighth
V2-B-N064-R032-C10-X02	1//8	low_clustered	one_eighth	one_eighth
V2-B-N064-R032-C11-X01	3//8	low_clustered	one_eighth	one_eighth
V2-B-N064-R032-C11-X02	3//8	low_clustered	one_eighth	one_eighth
V2-B-N064-R032-C12-X01	3//8	low_clustered	one_eighth	zero
V2-B-N064-R032-C12-X02	3//8	low_clustered	one_eighth	zero

instance id	coverage density realized	module overlap level	bundle prevalence level	unique carrier frequency level
V2-B-N064-R032-C13-X01	–	high_clustered	one_eighth	zero
V2-B-N064-R032-C13-X02	–	high_clustered	one_eighth	zero
V2-B-N064-R032-C14-X01	1//8	high_clustered	one_eighth	one_eighth
V2-B-N064-R032-C14-X02	1//8	high_clustered	one_eighth	one_eighth
V2-B-N064-R032-C15-X01	3//8	high_clustered	one_eighth	one_eighth
V2-B-N064-R032-C15-X02	3//8	high_clustered	one_eighth	one_eighth
V2-B-N064-R032-C16-X01	3//8	high_clustered	one_eighth	zero
V2-B-N064-R032-C16-X02	3//8	high_clustered	one_eighth	zero
V2-B-N064-R064-C01-X01	1//8	low_clustered	zero	zero
V2-B-N064-R064-C01-X02	1//8	low_clustered	zero	zero
V2-B-N064-R064-C02-X01	1//8	low_clustered	zero	one_eighth
V2-B-N064-R064-C02-X02	1//8	low_clustered	zero	one_eighth
V2-B-N064-R064-C03-X01	3//8	low_clustered	zero	one_eighth
V2-B-N064-R064-C03-X02	3//8	low_clustered	zero	one_eighth
V2-B-N064-R064-C04-X01	3//8	low_clustered	zero	zero
V2-B-N064-R064-C04-X02	3//8	low_clustered	zero	zero
V2-B-N064-R064-C05-X01	1//8	high_clustered	zero	zero
V2-B-N064-R064-C05-X02	1//8	high_clustered	zero	zero
V2-B-N064-R064-C06-X01	1//8	high_clustered	zero	one_eighth
V2-B-N064-R064-C06-X02	1//8	high_clustered	zero	one_eighth
V2-B-N064-R064-C07-X01	3//8	high_clustered	zero	one_eighth
V2-B-N064-R064-C07-X02	3//8	high_clustered	zero	one_eighth
V2-B-N064-R064-C08-X01	3//8	high_clustered	zero	zero
V2-B-N064-R064-C08-X02	3//8	high_clustered	zero	zero
V2-B-N064-R064-C09-X01	–	low_clustered	one_eighth	zero

instance id	coverage density realized	module overlap level	bundle prevalence level	unique carrier frequency level
V2-B-N064-R064-C09-X02	–	low_clustered	one_eighth	zero
V2-B-N064-R064-C10-X01	1//8	low_clustered	one_eighth	one_eighth
V2-B-N064-R064-C10-X02	1//8	low_clustered	one_eighth	one_eighth
V2-B-N064-R064-C11-X01	3//8	low_clustered	one_eighth	one_eighth
V2-B-N064-R064-C11-X02	3//8	low_clustered	one_eighth	one_eighth
V2-B-N064-R064-C12-X01	3//8	low_clustered	one_eighth	zero
V2-B-N064-R064-C12-X02	3//8	low_clustered	one_eighth	zero
V2-B-N064-R064-C13-X01	–	high_clustered	one_eighth	zero
V2-B-N064-R064-C13-X02	–	high_clustered	one_eighth	zero
V2-B-N064-R064-C14-X01	1//8	high_clustered	one_eighth	one_eighth
V2-B-N064-R064-C14-X02	1//8	high_clustered	one_eighth	one_eighth
V2-B-N064-R064-C15-X01	3//8	high_clustered	one_eighth	one_eighth
V2-B-N064-R064-C15-X02	3//8	high_clustered	one_eighth	one_eighth
V2-B-N064-R064-C16-X01	3//8	high_clustered	one_eighth	zero
V2-B-N064-R064-C16-X02	3//8	high_clustered	one_eighth	zero
V2-B-N192-R032-C01-X01	1//8	low_clustered	zero	zero
V2-B-N192-R032-C01-X02	1//8	low_clustered	zero	zero
V2-B-N192-R032-C02-X01	1//8	low_clustered	zero	one_eighth
V2-B-N192-R032-C02-X02	1//8	low_clustered	zero	one_eighth
V2-B-N192-R032-C03-X01	3//8	low_clustered	zero	one_eighth
V2-B-N192-R032-C03-X02	3//8	low_clustered	zero	one_eighth
V2-B-N192-R032-C04-X01	3//8	low_clustered	zero	zero
V2-B-N192-R032-C04-X02	3//8	low_clustered	zero	zero
V2-B-N192-R032-C05-X01	1//8	high_clustered	zero	zero
V2-B-N192-R032-C05-X02	1//8	high_clustered	zero	zero

instance id	coverage density realized	module overlap level	bundle prevalence level	unique carrier frequency level
V2-B-N192-R032-C06-X01	1//8	high_clustered	zero	one_eighth
V2-B-N192-R032-C06-X02	1//8	high_clustered	zero	one_eighth
V2-B-N192-R032-C07-X01	3//8	high_clustered	zero	one_eighth
V2-B-N192-R032-C07-X02	3//8	high_clustered	zero	one_eighth
V2-B-N192-R032-C08-X01	3//8	high_clustered	zero	zero
V2-B-N192-R032-C08-X02	3//8	high_clustered	zero	zero
V2-B-N192-R032-C09-X01	–	low_clustered	one_eighth	zero
V2-B-N192-R032-C09-X02	–	low_clustered	one_eighth	zero
V2-B-N192-R032-C10-X01	1//8	low_clustered	one_eighth	one_eighth
V2-B-N192-R032-C10-X02	1//8	low_clustered	one_eighth	one_eighth
V2-B-N192-R032-C11-X01	3//8	low_clustered	one_eighth	one_eighth
V2-B-N192-R032-C11-X02	3//8	low_clustered	one_eighth	one_eighth
V2-B-N192-R032-C12-X01	3//8	low_clustered	one_eighth	zero
V2-B-N192-R032-C12-X02	3//8	low_clustered	one_eighth	zero
V2-B-N192-R032-C13-X01	–	high_clustered	one_eighth	zero
V2-B-N192-R032-C13-X02	–	high_clustered	one_eighth	zero
V2-B-N192-R032-C14-X01	1//8	high_clustered	one_eighth	one_eighth
V2-B-N192-R032-C14-X02	1//8	high_clustered	one_eighth	one_eighth
V2-B-N192-R032-C15-X01	3//8	high_clustered	one_eighth	one_eighth
V2-B-N192-R032-C15-X02	3//8	high_clustered	one_eighth	one_eighth
V2-B-N192-R032-C16-X01	3//8	high_clustered	one_eighth	zero
V2-B-N192-R032-C16-X02	3//8	high_clustered	one_eighth	zero
V2-B-N192-R064-C01-X01	1//8	low_clustered	zero	zero
V2-B-N192-R064-C01-X02	1//8	low_clustered	zero	zero
V2-B-N192-R064-C02-X01	1//8	low_clustered	zero	one_eighth

instance id	coverage density realized	module overlap level	bundle prevalence level	unique carrier frequency level
V2-B-N192-R064-C02-X02	1//8	low_clustered	zero	one_eighth
V2-B-N192-R064-C03-X01	3//8	low_clustered	zero	one_eighth
V2-B-N192-R064-C03-X02	3//8	low_clustered	zero	one_eighth
V2-B-N192-R064-C04-X01	3//8	low_clustered	zero	zero
V2-B-N192-R064-C04-X02	3//8	low_clustered	zero	zero
V2-B-N192-R064-C05-X01	1//8	high_clustered	zero	zero
V2-B-N192-R064-C05-X02	1//8	high_clustered	zero	zero
V2-B-N192-R064-C06-X01	1//8	high_clustered	zero	one_eighth
V2-B-N192-R064-C06-X02	1//8	high_clustered	zero	one_eighth
V2-B-N192-R064-C07-X01	3//8	high_clustered	zero	one_eighth
V2-B-N192-R064-C07-X02	3//8	high_clustered	zero	one_eighth
V2-B-N192-R064-C08-X01	3//8	high_clustered	zero	zero
V2-B-N192-R064-C08-X02	3//8	high_clustered	zero	zero
V2-B-N192-R064-C09-X01	–	low_clustered	one_eighth	zero
V2-B-N192-R064-C09-X02	–	low_clustered	one_eighth	zero
V2-B-N192-R064-C10-X01	1//8	low_clustered	one_eighth	one_eighth
V2-B-N192-R064-C10-X02	1//8	low_clustered	one_eighth	one_eighth
V2-B-N192-R064-C11-X01	3//8	low_clustered	one_eighth	one_eighth
V2-B-N192-R064-C11-X02	3//8	low_clustered	one_eighth	one_eighth
V2-B-N192-R064-C12-X01	3//8	low_clustered	one_eighth	zero
V2-B-N192-R064-C12-X02	3//8	low_clustered	one_eighth	zero
V2-B-N192-R064-C13-X01	–	high_clustered	one_eighth	zero
V2-B-N192-R064-C13-X02	–	high_clustered	one_eighth	zero
V2-B-N192-R064-C14-X01	1//8	high_clustered	one_eighth	one_eighth
V2-B-N192-R064-C14-X02	1//8	high_clustered	one_eighth	one_eighth

instance id	coverage density realized	module overlap level	bundle prevalence level	unique carrier frequency level
V2-B-N192-R064-C15-X01	3//8	high_clustered	one_eighth	one_eighth
V2-B-N192-R064-C15-X02	3//8	high_clustered	one_eighth	one_eighth
V2-B-N192-R064-C16-X01	3//8	high_clustered	one_eighth	zero
V2-B-N192-R064-C16-X02	3//8	high_clustered	one_eighth	zero
V2-C-M01-S01	13//25	mechanism	mechanism	mechanism
V2-C-M01-S02	49//289	mechanism	mechanism	mechanism
V2-C-M01-S03	193//4225	mechanism	mechanism	mechanism
V2-C-M02-S01	1//1	mechanism	mechanism	mechanism
V2-C-M02-S02	1//1	mechanism	mechanism	mechanism
V2-C-M02-S03	1//1	mechanism	mechanism	mechanism
V2-C-M03-S01	1//1	mechanism	mechanism	mechanism
V2-C-M03-S02	1//1	mechanism	mechanism	mechanism
V2-C-M03-S03	1//1	mechanism	mechanism	mechanism
V2-C-M04-S01	1//2	mechanism	mechanism	mechanism
V2-C-M04-S02	5//26	mechanism	mechanism	mechanism
V2-C-M04-S03	5//66	mechanism	mechanism	mechanism
V2-C-M05-S01	13//25	mechanism	mechanism	mechanism
V2-C-M05-S02	49//289	mechanism	mechanism	mechanism
V2-C-M05-S03	193//4225	mechanism	mechanism	mechanism
V2-C-M06-S01	5//8	mechanism	mechanism	mechanism
V2-C-M06-S02	9//16	mechanism	mechanism	mechanism
V2-C-M06-S03	17//32	mechanism	mechanism	mechanism
V2-C-M07-S01	8//15	mechanism	mechanism	mechanism
V2-C-M07-S02	29//153	mechanism	mechanism	mechanism
V2-C-M07-S03	113//2145	mechanism	mechanism	mechanism

S12.2 Exact-method agreement

On 38 exact-capable instances, complete burden agreement was available on 31; 7 comparisons were unavailable after recorded failures. Equal objective burdens accompanied by different returned optimizer identities occurred in 9 cases. The table is exact finite computation plus exact semantic audit, not Lean verification.

Table S20: Exact-method agreement on registered exact-capable instances. Agreement is exact finite computation and semantic audit, not Lean verification.

instance id	family	size	enumeration burden	dp mandatory burden	mip full burden	exact agreement	optimizer identity difference
V2-A-N007-R006-X01	small_exact	N7-R6	4//1	4//1	4//1	true	false
V2-A-N007-R006-X02	small_exact	N7-R6	4//1	4//1	4//1	true	false
V2-A-N007-R006-X03	small_exact	N7-R6	4//1	4//1	4//1	true	false
V2-A-N007-R006-X04	small_exact	N7-R6	–	–	–	false	false
V2-A-N007-R010-X01	small_exact	N7-R10	4//1	4//1	4//1	true	false
V2-A-N007-R010-X02	small_exact	N7-R10	8//1	8//1	8//1	true	false
V2-A-N007-R010-X03	small_exact	N7-R10	5//1	5//1	5//1	true	false
V2-A-N007-R010-X04	small_exact	N7-R10	2//1	2//1	2//1	true	false
V2-A-N011-R006-X01	small_exact	N11-R6	5//1	5//1	5//1	true	false
V2-A-N011-R006-X02	small_exact	N11-R6	4//1	4//1	4//1	true	false
V2-A-N011-R006-X03	small_exact	N11-R6	2//1	2//1	2//1	true	true
V2-A-N011-R006-X04	small_exact	N11-R6	7//1	7//1	7//1	true	false
V2-A-N011-R010-X01	small_exact	N11-R10	5//1	5//1	5//1	true	false
V2-A-N011-R010-X02	small_exact	N11-R10	8//1	8//1	8//1	true	false
V2-A-N011-R010-X03	small_exact	N11-R10	2//1	2//1	2//1	true	false
V2-A-N011-R010-X04	small_exact	N11-R10	5//1	5//1	5//1	true	true
V2-A-N015-R006-X01	small_exact	N15-R6	3//1	3//1	3//1	true	false

instance id	family	size	enumeration burden	dp mandatory burden	mip full burden	exact agreement	optimizer identity difference
V2-A-N015-R006-X02	small_exact	N15-R6	2//1	2//1	2//1	true	false
V2-A-N015-R006-X03	small_exact	N15-R6	3//1	3//1	3//1	true	true
V2-A-N015-R006-X04	small_exact	N15-R6	4//1	4//1	4//1	true	false
V2-A-N015-R010-X01	small_exact	N15-R10	9//1	9//1	9//1	true	false
V2-A-N015-R010-X02	small_exact	N15-R10	4//1	4//1	4//1	true	false
V2-A-N015-R010-X03	small_exact	N15-R10	2//1	2//1	2//1	true	true
V2-A-N015-R010-X04	small_exact	N15-R10	3//1	3//1	3//1	true	false
V2-C-M01-S01	adversarial	small	3//2	3//2	3//2	true	false
V2-C-M01-S02	adversarial	medium	–	3//2	3//2	false	false
V2-C-M02-S01	adversarial	small	1//1	1//1	1//1	true	true
V2-C-M02-S02	adversarial	medium	1//1	1//1	1//1	true	true
V2-C-M02-S03	adversarial	large	–	1//1	1//1	false	false
V2-C-M03-S01	adversarial	small	1//1	1//1	1//1	true	true
V2-C-M04-S01	adversarial	small	2//1	2//1	2//1	true	true
V2-C-M04-S02	adversarial	medium	–	6//1	6//1	false	false
V2-C-M05-S01	adversarial	small	3//1	3//1	3//1	true	true
V2-C-M05-S02	adversarial	medium	–	9//1	9//1	false	false
V2-C-M06-S01	adversarial	small	2//1	2//1	2//1	true	false
V2-C-M06-S02	adversarial	medium	–	2//1	2//1	false	false
V2-C-M07-S01	adversarial	small	15//4	15//4	15//4	true	false
V2-C-M07-S02	adversarial	medium	–	63//4	63//4	false	false

S12.3 Complete scaling and failure records

The scaling table preserves per-family, per-size, per-algorithm solved counts, PAR-2 summaries, MIP nodes, DP states, and exact semantic-evaluation counts. Unavailable diagnostics print as dashes, never as zeros. The status table retains failed generation, implementation errors, memory limits, interruptions, nonbinary candidates, and no-primal outcomes in their registered denominators. Of the unavailable exact comparisons, 6 are preserved with the registered `IMPLEMENTATION_ERROR` status. They arose when complete enumeration crossed its declared applicability cap; the raw statuses are immutable, and no absent enumeration solution is imputed.

Table S21: Computational scaling under the fixed eight-worker v2 environment. PAR-2 retains every unsuccessful run; unavailable diagnostics are not zero. Part 1 of 2; join by the repeated identifying columns.

06S	family	size	algorithm	registered n	solved n	par2 median seconds	par2 q90 seconds
	adversarial	large	Cardinality greedy	7	7	0.555	0.9481
	adversarial	large	Enumeration	1	0	600	600
	adversarial	large	Declared source order	7	7	0.8469	1.697
	adversarial	large	Heaviest safe first	7	7	1.05	2.172
	adversarial	large	HiGHS MIP	7	7	4.558	7.15
	adversarial	large	Lightest safe first	7	7	1.093	2.104
	adversarial	large	Maximum immediate release	7	7	0.8262	2.152
	adversarial	large	Minimum unique-carrier exposure	7	7	0.9022	1.832
	adversarial	large	32-start random deletion	7	7	5.971	33.42
	adversarial	large	Random-order deletion	7	7	0.8629	1.969
	adversarial	large	Mask DP	1	1	0.6007	0.6007
	adversarial	large	Weighted greedy	7	7	0.7873	0.9742
	adversarial	large	Weighted greedy + reverse deletion	7	7	0.8394	0.9438

	family	size	algorithm	registered n	solved n	par2 median seconds	par2 q90 seconds
IS	adversarial	medium	Cardinality greedy	7	7	0.5735	0.7713
	adversarial	medium	Enumeration	6	1	600	600
	adversarial	medium	Declared source order	7	7	0.6742	1.083
	adversarial	medium	Heaviest safe first	7	7	0.7366	0.9278
	adversarial	medium	HiGHS MIP	7	7	5.228	6.448
	adversarial	medium	Lightest safe first	7	7	0.7419	1.244
	adversarial	medium	Maximum immediate release	7	7	0.8289	0.8463
	adversarial	medium	Minimum unique-carrier exposure	7	7	0.7318	1.114
	adversarial	medium	32-start random deletion	7	7	1.148	1.51
	adversarial	medium	Random-order deletion	7	7	0.9961	1.207
	adversarial	medium	Mask DP	7	7	1.351	2.122
	adversarial	medium	Weighted greedy	7	7	0.6361	0.8242
	adversarial	medium	Weighted greedy + reverse deletion	7	7	0.8367	0.9705
	adversarial	small	Cardinality greedy	7	7	0.7603	0.9167
	adversarial	small	Enumeration	7	7	0.7613	0.8595
	adversarial	small	Declared source order	7	7	0.7432	1.074
	adversarial	small	Heaviest safe first	7	7	0.8954	1.047
	adversarial	small	HiGHS MIP	7	7	4	4.925
	adversarial	small	Lightest safe first	7	7	1	1.263
	adversarial	small	Maximum immediate release	7	7	0.6173	1.005
	adversarial	small	Minimum unique-carrier exposure	7	7	0.5968	0.9179
	adversarial	small	32-start random deletion	7	7	0.5578	0.9351
	adversarial	small	Random-order deletion	7	7	0.9609	1.189

family	size	algorithm	registered n	solved n	par2 median seconds	par2 q90 seconds
adversarial	small	Mask DP	7	7	0.779	1.108
adversarial	small	Weighted greedy	7	7	0.5433	0.74
adversarial	small	Weighted greedy + reverse deletion	7	7	0.6499	0.6906
small_exact	N11-R10	Cardinality greedy	4	4	0.5693	0.5712
small_exact	N11-R10	Enumeration	4	4	0.7121	0.9128
small_exact	N11-R10	Declared source order	4	4	0.5622	0.9187
small_exact	N11-R10	Heaviest safe first	4	4	0.8631	1.268
small_exact	N11-R10	HiGHS MIP	4	4	3.068	4.186
small_exact	N11-R10	Lightest safe first	4	4	0.6314	0.7363
small_exact	N11-R10	Maximum immediate release	4	4	0.6535	0.7443
small_exact	N11-R10	Minimum unique-carrier exposure	4	4	0.7431	0.8804
small_exact	N11-R10	32-start random deletion	4	4	1.019	1.272
small_exact	N11-R10	Random-order deletion	4	4	0.6546	0.9243
small_exact	N11-R10	Mask DP	4	4	0.8333	0.8872
small_exact	N11-R10	Weighted greedy	4	4	0.6459	0.8752
small_exact	N11-R10	Weighted greedy + reverse deletion	4	4	0.7197	0.8422
small_exact	N11-R6	Cardinality greedy	4	4	0.8	1.058
small_exact	N11-R6	Enumeration	4	4	0.4245	0.6135
small_exact	N11-R6	Declared source order	4	4	0.5689	0.9913
small_exact	N11-R6	Heaviest safe first	4	4	0.6162	0.8991
small_exact	N11-R6	HiGHS MIP	4	4	3.908	4.201
small_exact	N11-R6	Lightest safe first	4	4	0.778	0.9499
small_exact	N11-R6	Maximum immediate release	4	4	0.7202	0.9603

	family	size	algorithm	registered n	solved n	par2 median	par2 q90 seconds
						seconds	
S93	small_exact	N11-R6	Minimum unique-carrier exposure	4	4	0.7653	0.812
	small_exact	N11-R6	32-start random deletion	4	4	0.7958	0.9285
	small_exact	N11-R6	Random-order deletion	4	4	0.576	0.7871
	small_exact	N11-R6	Mask DP	4	4	0.4361	0.7391
	small_exact	N11-R6	Weighted greedy	4	4	0.6334	0.7881
	small_exact	N11-R6	Weighted greedy + reverse deletion	4	4	0.5947	0.711
	small_exact	N15-R10	Cardinality greedy	4	4	0.8693	1.003
	small_exact	N15-R10	Enumeration	4	4	0.5616	0.6857
	small_exact	N15-R10	Declared source order	4	4	0.9543	0.9974
	small_exact	N15-R10	Heaviest safe first	4	4	0.8266	0.895
	small_exact	N15-R10	HiGHS MIP	4	4	4.66	5.911
	small_exact	N15-R10	Lightest safe first	4	4	0.7703	0.8397
	small_exact	N15-R10	Maximum immediate release	4	4	0.8335	1.079
	small_exact	N15-R10	Minimum unique-carrier exposure	4	4	0.6661	0.9595
	small_exact	N15-R10	32-start random deletion	4	4	0.76	1.001
	small_exact	N15-R10	Random-order deletion	4	4	0.671	1.234
	small_exact	N15-R10	Mask DP	4	4	0.68	0.8697
	small_exact	N15-R10	Weighted greedy	4	4	0.7767	0.8508
	small_exact	N15-R10	Weighted greedy + reverse deletion	4	4	0.4954	0.5664
	small_exact	N15-R6	Cardinality greedy	4	4	0.6308	0.8562
	small_exact	N15-R6	Enumeration	4	4	0.574	0.7233
	small_exact	N15-R6	Declared source order	4	4	0.8023	0.9967

family	size	algorithm	registered n	solved n	par2 median seconds	par2 q90 seconds
small_exact	N15-R6	Heaviest safe first	4	4	0.9518	1.039
small_exact	N15-R6	HiGHS MIP	4	4	4.524	5.166
small_exact	N15-R6	Lightest safe first	4	4	0.7179	0.9549
small_exact	N15-R6	Maximum immediate release	4	4	0.6032	0.834
small_exact	N15-R6	Minimum unique-carrier exposure	4	4	0.6266	0.8142
small_exact	N15-R6	32-start random deletion	4	4	0.9723	1.001
small_exact	N15-R6	Random-order deletion	4	4	0.9925	1.166
small_exact	N15-R6	Mask DP	4	4	0.7248	1.072
small_exact	N15-R6	Weighted greedy	4	4	0.8531	0.8876
small_exact	N15-R6	Weighted greedy + reverse deletion	4	4	0.5946	0.7202
small_exact	N7-R10	Cardinality greedy	4	4	0.8409	0.921
small_exact	N7-R10	Enumeration	4	4	0.8577	0.8979
small_exact	N7-R10	Declared source order	4	4	0.7111	0.993
small_exact	N7-R10	Heaviest safe first	4	4	0.7641	1.129
small_exact	N7-R10	HiGHS MIP	4	4	4.389	5.051
small_exact	N7-R10	Lightest safe first	4	4	0.7116	0.9233
small_exact	N7-R10	Maximum immediate release	4	4	0.7839	0.889
small_exact	N7-R10	Minimum unique-carrier exposure	4	4	0.5009	0.6884
small_exact	N7-R10	32-start random deletion	4	4	0.5454	0.7282
small_exact	N7-R10	Random-order deletion	4	4	0.7044	1.007
small_exact	N7-R10	Mask DP	4	4	0.6382	0.9921
small_exact	N7-R10	Weighted greedy	4	4	0.4248	0.5049

	family	size	algorithm	registered n	solved n	par2 median seconds	par2 q90 seconds
S05	small_exact	N7-R10	Weighted greedy + reverse deletion	4	4	0.6744	0.8819
	small_exact	N7-R6	Cardinality greedy	4	3	0.9573	84.29
	small_exact	N7-R6	Enumeration	4	3	0.6359	420.2
	small_exact	N7-R6	Declared source order	4	4	0.8089	1.286
	small_exact	N7-R6	Heaviest safe first	4	4	1.195	1.386
	small_exact	N7-R6	HiGHS MIP	4	3	4.244	421.5
	small_exact	N7-R6	Lightest safe first	4	4	0.7518	1.08
	small_exact	N7-R6	Maximum immediate release	4	4	0.7456	1.213
	small_exact	N7-R6	Minimum unique-carrier exposure	4	4	0.7929	1.277
	small_exact	N7-R6	32-start random deletion	4	4	1.017	1.218
	small_exact	N7-R6	Random-order deletion	4	4	0.7759	1.807
	small_exact	N7-R6	Mask DP	4	3	0.7319	420.2
	small_exact	N7-R6	Weighted greedy	4	4	0.6146	0.7898
	small_exact	N7-R6	Weighted greedy + reverse deletion	4	4	0.7385	0.9798
	structured	N192-R32	Cardinality greedy	32	28	0.6972	1080
	structured	N192-R32	Declared source order	32	28	4.648	1081
	structured	N192-R32	Heaviest safe first	32	28	4.578	1081
	structured	N192-R32	HiGHS MIP	32	27	17.91	3600
	structured	N192-R32	Lightest safe first	32	28	4.459	1081
	structured	N192-R32	Maximum immediate release	32	28	4.886	1081
	structured	N192-R32	Minimum unique-carrier exposure	32	28	4.659	1081
	structured	N192-R32	32-start random deletion	32	15	1800	1800

family	size	algorithm	registered n	solved n	par2 median seconds	par2 q90 seconds
structured	N192-R32	Random-order deletion	32	28	4.326	1621
structured	N192-R32	Weighted greedy	32	28	0.6781	1080
structured	N192-R32	Weighted greedy + reverse deletion	32	28	0.6934	1080
structured	N192-R64	Cardinality greedy	32	28	0.7642	1080
structured	N192-R64	Declared source order	32	28	6.303	1081
structured	N192-R64	Heaviest safe first	32	28	6.482	1081
structured	N192-R64	HiGHS MIP	32	28	19.47	3243
structured	N192-R64	Lightest safe first	32	28	6.27	1081
structured	N192-R64	Maximum immediate release	32	28	6.192	1081
structured	N192-R64	Minimum unique-carrier exposure	32	28	5.973	1081
structured	N192-R64	32-start random deletion	32	27	183.1	1800
structured	N192-R64	Random-order deletion	32	28	6.363	1621
structured	N192-R64	Weighted greedy	32	28	0.8326	1080
structured	N192-R64	Weighted greedy + reverse deletion	32	28	0.9031	1080
structured	N64-R32	Cardinality greedy	32	28	0.6788	540.1
structured	N64-R32	Declared source order	32	28	0.8838	540.1
structured	N64-R32	Heaviest safe first	32	28	0.8266	540.1
structured	N64-R32	HiGHS MIP	32	28	7.298	1621
structured	N64-R32	Lightest safe first	32	28	0.9161	540.1
structured	N64-R32	Maximum immediate release	32	28	0.8966	540.1
structured	N64-R32	Minimum unique-carrier exposure	32	28	0.9936	540.1
structured	N64-R32	32-start random deletion	32	28	5.899	1081

family	size	algorithm	registered n	solved n	par2 median seconds	par2 q90 seconds
structured	N64-R32	Random-order deletion	32	28	0.8489	1080
structured	N64-R32	Weighted greedy	32	28	0.7019	540.1
structured	N64-R32	Weighted greedy + reverse deletion	32	28	0.7926	540.1
structured	N64-R64	Cardinality greedy	32	28	0.674	1080
structured	N64-R64	Declared source order	32	28	0.9456	1080
structured	N64-R64	Heaviest safe first	32	28	1.084	1080
structured	N64-R64	HiGHS MIP	32	28	7.691	3241
structured	N64-R64	Lightest safe first	32	28	1.022	1080
structured	N64-R64	Maximum immediate release	32	28	0.917	1080
structured	N64-R64	Minimum unique-carrier exposure	32	28	0.9924	1080
structured	N64-R64	32-start random deletion	32	28	8.514	1621
structured	N64-R64	Random-order deletion	32	28	1.119	1620
structured	N64-R64	Weighted greedy	32	28	0.7785	1080
structured	N64-R64	Weighted greedy + reverse deletion	32	28	0.7066	1080

Table S22: Computational scaling under the fixed eight-worker v2 environment. PAR-2 retains every unsuccessful run; unavailable diagnostics are not zero. Part 2 of 2; join by the repeated identifying columns.

family	size	algorithm	mip nodes median	dp states median	frontier checks median
adversarial	large	Cardinality greedy	–	–	–
adversarial	large	Enumeration	–	–	–

family	size	algorithm	mip nodes median	dp states median	frontier checks median
adversarial	large	Declared source order	–	–	1871
adversarial	large	Heaviest safe first	–	–	496
adversarial	large	HiGHS MIP	1	–	–
adversarial	large	Lightest safe first	–	–	1585
adversarial	large	Maximum immediate release	–	–	496
adversarial	large	Minimum unique-carrier exposure	–	–	1928
adversarial	large	32-start random deletion	–	–	6.001e+04
adversarial	large	Random-order deletion	–	–	1850
adversarial	large	Mask DP	–	31	–
adversarial	large	Weighted greedy	–	–	–
adversarial	large	Weighted greedy + reverse deletion	–	–	–
adversarial	medium	Cardinality greedy	–	–	–
adversarial	medium	Enumeration	–	256	–
adversarial	medium	Declared source order	–	–	154
adversarial	medium	Heaviest safe first	–	–	101
adversarial	medium	HiGHS MIP	1	–	–
adversarial	medium	Lightest safe first	–	–	154
adversarial	medium	Maximum immediate release	–	–	101
adversarial	medium	Minimum unique-carrier exposure	–	–	154
adversarial	medium	32-start random deletion	–	–	4808
adversarial	medium	Random-order deletion	–	–	154
adversarial	medium	Mask DP	–	1.207e+04	–
adversarial	medium	Weighted greedy	–	–	–

family	size	algorithm	mip nodes median	dp states median	frontier checks median
adversarial	medium	Weighted greedy + reverse deletion	–	–	–
adversarial	small	Cardinality greedy	–	–	–
adversarial	small	Enumeration	–	64	–
adversarial	small	Declared source order	–	–	21
adversarial	small	Heaviest safe first	–	–	16
adversarial	small	HiGHS MIP	1	–	–
adversarial	small	Lightest safe first	–	–	21
adversarial	small	Maximum immediate release	–	–	16
adversarial	small	Minimum unique-carrier exposure	–	–	21
adversarial	small	32-start random deletion	–	–	657
adversarial	small	Random-order deletion	–	–	21
adversarial	small	Mask DP	–	21	–
adversarial	small	Weighted greedy	–	–	–
adversarial	small	Weighted greedy + reverse deletion	–	–	–
small_exact	N11-R10	Cardinality greedy	–	–	–
small_exact	N11-R10	Enumeration	–	1024	–
small_exact	N11-R10	Declared source order	–	–	55
small_exact	N11-R10	Heaviest safe first	–	–	55
small_exact	N11-R10	HiGHS MIP	–	–	–
small_exact	N11-R10	Lightest safe first	–	–	54
small_exact	N11-R10	Maximum immediate release	–	–	55
small_exact	N11-R10	Minimum unique-carrier exposure	–	–	55
small_exact	N11-R10	32-start random deletion	–	–	1736

	family	size	algorithm	S100		
				mip nodes median	dp states median	frontier checks median
	small_exact	N11-R10	Random-order deletion	–	–	53
	small_exact	N11-R10	Mask DP	–	91	–
	small_exact	N11-R10	Weighted greedy	–	–	–
	small_exact	N11-R10	Weighted greedy + reverse deletion	–	–	–
	small_exact	N11-R6	Cardinality greedy	–	–	–
	small_exact	N11-R6	Enumeration	–	1024	–
	small_exact	N11-R6	Declared source order	–	–	55
	small_exact	N11-R6	Heaviest safe first	–	–	54
	small_exact	N11-R6	HiGHS MIP	1	–	–
	small_exact	N11-R6	Lightest safe first	–	–	55
	small_exact	N11-R6	Maximum immediate release	–	–	54
	small_exact	N11-R6	Minimum unique-carrier exposure	–	–	55
	small_exact	N11-R6	32-start random deletion	–	–	1747
	small_exact	N11-R6	Random-order deletion	–	–	54
	small_exact	N11-R6	Mask DP	–	46.5	–
	small_exact	N11-R6	Weighted greedy	–	–	–
	small_exact	N11-R6	Weighted greedy + reverse deletion	–	–	–
	small_exact	N15-R10	Cardinality greedy	–	–	–
	small_exact	N15-R10	Enumeration	–	1.638e+04	–
	small_exact	N15-R10	Declared source order	–	–	104
	small_exact	N15-R10	Heaviest safe first	–	–	105
	small_exact	N15-R10	HiGHS MIP	1	–	–
	small_exact	N15-R10	Lightest safe first	–	–	104

family	size	algorithm	mip nodes median	dp states median	frontier checks median
small_exact	N15-R10	Maximum immediate release	–	–	105
small_exact	N15-R10	Minimum unique-carrier exposure	–	–	105
small_exact	N15-R10	32-start random deletion	–	–	3346
small_exact	N15-R10	Random-order deletion	–	–	104
small_exact	N15-R10	Mask DP	–	125	–
small_exact	N15-R10	Weighted greedy	–	–	–
small_exact	N15-R10	Weighted greedy + reverse deletion	–	–	–
small_exact	N15-R6	Cardinality greedy	–	–	–
small_exact	N15-R6	Enumeration	–	1.638e+04	–
small_exact	N15-R6	Declared source order	–	–	105
small_exact	N15-R6	Heaviest safe first	–	–	105
small_exact	N15-R6	HiGHS MIP	–	–	–
small_exact	N15-R6	Lightest safe first	–	–	104
small_exact	N15-R6	Maximum immediate release	–	–	105
small_exact	N15-R6	Minimum unique-carrier exposure	–	–	105
small_exact	N15-R6	32-start random deletion	–	–	3352
small_exact	N15-R6	Random-order deletion	–	–	105
small_exact	N15-R6	Mask DP	–	82	–
small_exact	N15-R6	Weighted greedy	–	–	–
small_exact	N15-R6	Weighted greedy + reverse deletion	–	–	–
small_exact	N7-R10	Cardinality greedy	–	–	–
small_exact	N7-R10	Enumeration	–	64	–
small_exact	N7-R10	Declared source order	–	–	20

family	size	algorithm	mip nodes median	dp states median	frontier checks median
small_exact	N7-R10	Heaviest safe first	–	–	20
small_exact	N7-R10	HiGHS MIP	1	–	–
small_exact	N7-R10	Lightest safe first	–	–	20
small_exact	N7-R10	Maximum immediate release	–	–	20
small_exact	N7-R10	Minimum unique-carrier exposure	–	–	20
small_exact	N7-R10	32-start random deletion	–	–	614
small_exact	N7-R10	Random-order deletion	–	–	19
small_exact	N7-R10	Mask DP	–	39.5	–
small_exact	N7-R10	Weighted greedy	–	–	–
small_exact	N7-R10	Weighted greedy + reverse deletion	–	–	–
small_exact	N7-R6	Cardinality greedy	–	–	–
small_exact	N7-R6	Enumeration	–	64	–
small_exact	N7-R6	Declared source order	–	–	21
small_exact	N7-R6	Heaviest safe first	–	–	20
small_exact	N7-R6	HiGHS MIP	1	–	–
small_exact	N7-R6	Lightest safe first	–	–	21
small_exact	N7-R6	Maximum immediate release	–	–	20
small_exact	N7-R6	Minimum unique-carrier exposure	–	–	21
small_exact	N7-R6	32-start random deletion	–	–	653
small_exact	N7-R6	Random-order deletion	–	–	19
small_exact	N7-R6	Mask DP	–	38	–
small_exact	N7-R6	Weighted greedy	–	–	–
small_exact	N7-R6	Weighted greedy + reverse deletion	–	–	–

family	size	algorithm	mip nodes median	dp states median	frontier checks median
structured	N192-R32	Cardinality greedy	–	–	–
structured	N192-R32	Declared source order	–	–	1.833e+04
structured	N192-R32	Heaviest safe first	–	–	1.833e+04
structured	N192-R32	HiGHS MIP	1	–	–
structured	N192-R32	Lightest safe first	–	–	1.833e+04
structured	N192-R32	Maximum immediate release	–	–	1.833e+04
structured	N192-R32	Minimum unique-carrier exposure	–	–	1.833e+04
structured	N192-R32	32-start random deletion	–	–	5.867e+05
structured	N192-R32	Random-order deletion	–	–	1.833e+04
structured	N192-R32	Weighted greedy	–	–	–
structured	N192-R32	Weighted greedy + reverse deletion	–	–	–
structured	N192-R64	Cardinality greedy	–	–	–
structured	N192-R64	Declared source order	–	–	1.831e+04
structured	N192-R64	Heaviest safe first	–	–	1.831e+04
structured	N192-R64	HiGHS MIP	1	–	–
structured	N192-R64	Lightest safe first	–	–	1.831e+04
structured	N192-R64	Maximum immediate release	–	–	1.831e+04
structured	N192-R64	Minimum unique-carrier exposure	–	–	1.831e+04
structured	N192-R64	32-start random deletion	–	–	5.859e+05
structured	N192-R64	Random-order deletion	–	–	1.831e+04
structured	N192-R64	Weighted greedy	–	–	–
structured	N192-R64	Weighted greedy + reverse deletion	–	–	–
structured	N64-R32	Cardinality greedy	–	–	–

family	size	algorithm	mip nodes median	dp states median	frontier checks median
structured	N64-R32	Declared source order	–	–	2007
structured	N64-R32	Heaviest safe first	–	–	2007
structured	N64-R32	HiGHS MIP	1	–	–
structured	N64-R32	Lightest safe first	–	–	2007
structured	N64-R32	Maximum immediate release	–	–	2007
structured	N64-R32	Minimum unique-carrier exposure	–	–	2007
structured	N64-R32	32-start random deletion	–	–	6.422e+04
structured	N64-R32	Random-order deletion	–	–	2007
structured	N64-R32	Weighted greedy	–	–	–
structured	N64-R32	Weighted greedy + reverse deletion	–	–	–
structured	N64-R64	Cardinality greedy	–	–	–
structured	N64-R64	Declared source order	–	–	1985
structured	N64-R64	Heaviest safe first	–	–	1989
structured	N64-R64	HiGHS MIP	1	–	–
structured	N64-R64	Lightest safe first	–	–	1985
structured	N64-R64	Maximum immediate release	–	–	1989
structured	N64-R64	Minimum unique-carrier exposure	–	–	1989
structured	N64-R64	32-start random deletion	–	–	6.352e+04
structured	N64-R64	Random-order deletion	–	–	1989
structured	N64-R64	Weighted greedy	–	–	–
structured	N64-R64	Weighted greedy + reverse deletion	–	–	–

Table S23: Complete terminal status counts; no failed or unsuccessful run is dropped.

family	algorithm	preprocessing variant	status	count
adversarial	Cardinality greedy	full_fixed_point_then_reconstruct	SOLVED	21
adversarial	Cardinality greedy	mandatory_only	SOLVED	21
adversarial	Enumeration	mandatory_only	IMPLEMENTATION_ERROR	6
adversarial	Enumeration	mandatory_only	SOLVED	8
adversarial	Declared source order	full_fixed_point_then_reconstruct	SOLVED	21
adversarial	Declared source order	mandatory_only	SOLVED	21
adversarial	Heaviest safe first	full_fixed_point_then_reconstruct	SOLVED	21
adversarial	Heaviest safe first	mandatory_only	SOLVED	21
adversarial	HiGHS MIP	full_fixed_point	SOLVED	15
adversarial	HiGHS MIP	full_fixed_point	exact preprocessing	6
adversarial	HiGHS MIP	mandatory_only	SOLVED	21
adversarial	Lightest safe first	full_fixed_point_then_reconstruct	SOLVED	21
adversarial	Lightest safe first	mandatory_only	SOLVED	21
adversarial	Maximum immediate release	full_fixed_point_then_reconstruct	SOLVED	21
adversarial	Maximum immediate release	mandatory_only	SOLVED	21
adversarial	Minimum unique-carrier exposure	full_fixed_point_then_reconstruct	SOLVED	21
adversarial	Minimum unique-carrier exposure	mandatory_only	SOLVED	21
adversarial	32-start random deletion	full_fixed_point_then_reconstruct	SOLVED	21

family	algorithm	preprocessing variant	status	count
adversarial	32-start random deletion	mandatory_only	SOLVED	21
adversarial	Random-order deletion	full_fixed_point_then_reconstruct	SOLVED	21
adversarial	Random-order deletion	mandatory_only	SOLVED	21
adversarial	Mask DP	mandatory_only	SOLVED	15
adversarial	Weighted greedy	full_fixed_point_then_reconstruct	SOLVED	21
adversarial	Weighted greedy	mandatory_only	SOLVED	21
adversarial	Weighted greedy + reverse deletion	full_fixed_point_then_reconstruct	SOLVED	21
adversarial	Weighted greedy + reverse deletion	mandatory_only	SOLVED	21
small_exact	Cardinality greedy	full_fixed_point_then_reconstruct	SOLVED	24
small_exact	Cardinality greedy	mandatory_only	PROCESS_INTERRUPTED	1
small_exact	Cardinality greedy	mandatory_only	SOLVED	23
small_exact	Enumeration	mandatory_only	PROCESS_INTERRUPTED	1
small_exact	Enumeration	mandatory_only	SOLVED	23
small_exact	Declared source order	full_fixed_point_then_reconstruct	SOLVED	24
small_exact	Declared source order	mandatory_only	SOLVED	24
small_exact	Heaviest safe first	full_fixed_point_then_reconstruct	SOLVED	24
small_exact	Heaviest safe first	mandatory_only	SOLVED	24
small_exact	HiGHS MIP	full_fixed_point	PROCESS_INTERRUPTED	1
small_exact	HiGHS MIP	full_fixed_point	SOLVED	5
small_exact	HiGHS MIP	full_fixed_point	exact preprocessing	18

family	algorithm	preprocessing variant	status	count
small_exact	HiGHS MIP	mandatory_only	PROCESS_INTERRUPTED	1
small_exact	HiGHS MIP	mandatory_only	SOLVED	23
small_exact	Lightest safe first	full_fixed_point_then_reconstruct	SOLVED	24
small_exact	Lightest safe first	mandatory_only	SOLVED	24
small_exact	Maximum immediate release	full_fixed_point_then_reconstruct	SOLVED	24
small_exact	Maximum immediate release	mandatory_only	SOLVED	24
small_exact	Minimum unique-carrier exposure	full_fixed_point_then_reconstruct	SOLVED	24
small_exact	Minimum unique-carrier exposure	mandatory_only	SOLVED	24
small_exact	32-start random deletion	full_fixed_point_then_reconstruct	SOLVED	24
small_exact	32-start random deletion	mandatory_only	SOLVED	24
small_exact	Random-order deletion	full_fixed_point_then_reconstruct	SOLVED	24
small_exact	Random-order deletion	mandatory_only	SOLVED	24
small_exact	Mask DP	full_fixed_point	PROCESS_INTERRUPTED	1
small_exact	Mask DP	full_fixed_point	SOLVED	23
small_exact	Mask DP	mandatory_only	PROCESS_INTERRUPTED	1
small_exact	Mask DP	mandatory_only	SOLVED	23
small_exact	Weighted greedy	full_fixed_point_then_reconstruct	PROCESS_INTERRUPTED	1
small_exact	Weighted greedy	full_fixed_point_then_reconstruct	SOLVED	23
small_exact	Weighted greedy	mandatory_only	SOLVED	24

family	algorithm	preprocessing variant	status	count
small_exact	Weighted greedy + reverse deletion	full_fixed_point_then_reconstruct	SOLVED	24
small_exact	Weighted greedy + reverse deletion	mandatory_only	SOLVED	24
structured	Cardinality greedy	full_fixed_point_then_reconstruct	GENERATION_FAILED	16
structured	Cardinality greedy	full_fixed_point_then_reconstruct	SOLVED	112
structured	Cardinality greedy	mandatory_only	GENERATION_FAILED	16
structured	Cardinality greedy	mandatory_only	SOLVED	112
structured	Declared source order	full_fixed_point_then_reconstruct	GENERATION_FAILED	16
structured	Declared source order	full_fixed_point_then_reconstruct	SOLVED	112
structured	Declared source order	mandatory_only	GENERATION_FAILED	16
structured	Declared source order	mandatory_only	SOLVED	112
structured	Heaviest safe first	full_fixed_point_then_reconstruct	GENERATION_FAILED	16
structured	Heaviest safe first	full_fixed_point_then_reconstruct	SOLVED	112
structured	Heaviest safe first	mandatory_only	GENERATION_FAILED	16
structured	Heaviest safe first	mandatory_only	SOLVED	112
structured	HiGHS MIP	full_fixed_point	GENERATION_FAILED	16
structured	HiGHS MIP	full_fixed_point	REJECTED_NONBINARY_SOLVE R_CANDIDATE	1
structured	HiGHS MIP	full_fixed_point	SOLVED	29
structured	HiGHS MIP	full_fixed_point	exact preprocessing	82

family	algorithm	preprocessing variant	status	count
structured	HiGHS MIP	mandatory_only	GENERATION_FAILED	16
structured	HiGHS MIP	mandatory_only	NO_PRIMAL_CANDIDATE	1
structured	HiGHS MIP	mandatory_only	SOLVED	111
structured	Lightest safe first	full_fixed_point_then_reconstruct	GENERATION_FAILED	16
structured	Lightest safe first	full_fixed_point_then_reconstruct	SOLVED	112
structured	Lightest safe first	mandatory_only	GENERATION_FAILED	16
structured	Lightest safe first	mandatory_only	SOLVED	112
structured	Maximum immediate release	full_fixed_point_then_reconstruct	GENERATION_FAILED	16
structured	Maximum immediate release	full_fixed_point_then_reconstruct	SOLVED	112
structured	Maximum immediate release	mandatory_only	GENERATION_FAILED	16
structured	Maximum immediate release	mandatory_only	SOLVED	112
structured	Minimum unique-carrier exposure	full_fixed_point_then_reconstruct	GENERATION_FAILED	16
structured	Minimum unique-carrier exposure	full_fixed_point_then_reconstruct	SOLVED	112
structured	Minimum unique-carrier exposure	mandatory_only	GENERATION_FAILED	16
structured	Minimum unique-carrier exposure	mandatory_only	SOLVED	112
structured	32-start random deletion	full_fixed_point_then_reconstruct	GENERATION_FAILED	16
structured	32-start random deletion	full_fixed_point_then_reconstruct	SOLVED	112
structured	32-start random deletion	mandatory_only	GENERATION_FAILED	16

family	algorithm	preprocessing variant	status	count
structured	32-start random deletion	mandatory_only	MEMORY_LIMIT	14
structured	32-start random deletion	mandatory_only	SOLVED	98
structured	Random-order deletion	full_fixed_point_then_reconstruct	GENERATION_FAILED	16
structured	Random-order deletion	full_fixed_point_then_reconstruct	SOLVED	112
structured	Random-order deletion	mandatory_only	GENERATION_FAILED	16
structured	Random-order deletion	mandatory_only	SOLVED	112
structured	Weighted greedy	full_fixed_point_then_reconstruct	GENERATION_FAILED	16
structured	Weighted greedy	full_fixed_point_then_reconstruct	SOLVED	112
structured	Weighted greedy	mandatory_only	GENERATION_FAILED	16
structured	Weighted greedy	mandatory_only	SOLVED	112
structured	Weighted greedy + reverse deletion	full_fixed_point_then_reconstruct	GENERATION_FAILED	16
structured	Weighted greedy + reverse deletion	full_fixed_point_then_reconstruct	SOLVED	112
structured	Weighted greedy + reverse deletion	mandatory_only	GENERATION_FAILED	16
structured	Weighted greedy + reverse deletion	mandatory_only	SOLVED	112

Size	Preprocess n	Median var. red.	Median row red.	Median log PAR-2 ratio	32-start complete	Improved
N64-R32	28/32	1	1	1.187	28/32	20/28
N64-R64	28/32	1	1	1.29	28/32	19/28
N192-R32	28/32	1	1	2.031	15/32	12/15
N192-R64	28/32	1	1	2.243	27/32	21/27

Table S24 Registered preprocessing and multi-start evidence on structured instances. Reductions are shares removed. The log ratio is $\log(\text{PAR2}_{\text{full}}/\text{PAR2}_{\text{mandatory}})$, so a positive value favors the mandatory-only pipeline in elapsed penalized runtime despite the model-size reduction. Unavailable multi-start endpoints are not coded as zero improvement.

S12.4 Preprocessing, multi-start, and registered models

The complete per-cell preprocessing summaries, per-rule removal counts, and multi-start quantiles remain machine-readable in `journal/aor/tables/preprocessing_summary.csv`, `preprocessing_rule_counts.csv`, and `multistart_summary.csv`. They are not expanded into thousands of PDF rows because the compact table preserves the registered journal comparison while the CSV files preserve every auditable row.

No registered run ended with terminal status `TIME_LIMIT`. The machine-readable timeout-best-bound table is therefore deliberately header-only; it is not omitted and no bound-gap distribution is inferred.

The corresponding committed file is `journal/aor/tables/timeout_best_bound_gaps.csv`.

The registered model suite used the declared structural covariates and retained failed or nonestimable specifications without changing links, penalties, or covariates after outcomes were observed.

Table S25: Registered model fit diagnostics. Failed specifications are reported without post hoc replacement.

model id	status	population n	fitted n	unavailable n	parameter count	rank
family_b_gap_weighted_greedy	FIT	128	111	17	18	18
family_b_gap_cardinality_greedy	FIT	128	111	17	18	18
family_b_gap_weighted_greedy_reverse_delete	FIT	128	111	17	18	18
family_b_gap_heaviest_safe_first	FIT	128	111	17	18	18
family_b_gap_lightest_safe_first	FIT	128	111	17	18	18
family_b_gap_maximum_immediate_burden_release	FIT	128	111	17	18	18
family_b_gap_minimum_unique_carrier_exposure	FIT	128	111	17	18	18
family_b_gap_declared_source_order	FIT	128	111	17	18	18
family_b_gap_random_order_rechecked_deletion	FIT	128	111	17	18	18
family_b_gap_multistart_random_rechecked_deletion_32	FIT	128	97	31	18	18
family_b_gap_joint	FIT	1280	1096	184	45	45
family_a_exact_gap_secondary	FIT	240	230	10	19	19
small_exact_solved_fraction	SEPARATION_DETECTED	600	600	0	23	23
small_exact_log_par2	FIT	600	600	0	23	23
small_exact_log_par2_launch_wave_sensitivity	FIT	600	600	0	24	24
structured_solved_fraction	FIT	2816	2816	0	21	21
structured_log_par2	FIT	2816	2816	0	21	21

model id	status	population n	fitted n	unavailable n	parameter count	rank
structured_log_par2_launch_wave_sensitivity	FIT	2816	2816	0	22	22
adversarial_solved_fraction	SEPARATION_DETected	491	491	0	23	23
adversarial_log_par2	FIT	491	491	0	23	23
adversarial_log_par2_launch_wave_sensitivity	FIT	491	491	0	24	24
family_b_preprocessing_log_par2_difference	RANK_DEFICIENT	128	128	0	24	22
family_b_preprocessing_loglp_node_difference	RANK_DEFICIENT	128	30	98	24	16

These model outputs are retained as the executed registered record, not as valid confirmatory inference. HC3 standard errors do not account for repeated algorithm observations within an instance, and structurally missing generator cells impair estimability. No unregistered clustered refit is substituted.

S12.5 Interpretation limits and future design

The structured generator requested cells that were mathematically incompatible with its incidence constraints. Those failures account for 352 terminal records from 16 source-instance cells and are retained in every denominator. Among generated structured instances, fixed-point preprocessing commonly selected all forced carriers and left an empty residual cover. Family B therefore documents the prevalence and consequences of that preprocessing mechanism in this design; it is weak evidence about residual MIP or dynamic-programming search difficulty.

All recorded times include cold Julia startup and compilation within each subprocess, one observation per run unit, eight-worker contention, and a dirty worktree recorded in the execution metadata. They are execution records, not steady-state speed estimates or a basis for close algorithm rankings. A performance profile is not interpreted because feasibility heuristics, exact methods, and time-limited solvers do not share one registered success target.

A defensible follow-up requires its own registry, seeds, and lock. It should warm algorithms outside timed regions; block repeated clean-tree runs by instance; validate generator feasibility before locking; balance realized nonempty residual structure; define separate feasibility, quality-qualified, and exact-completion targets; and include a bounded certified-exchange method and LP-relaxation diagnostic. These are prospective design requirements, not post hoc additions to v2.

The full coefficients, paired comparisons, heuristic-quality summaries, adversarial tables, plot-data tables, row trace, and SVGs remain generated inputs under `journal/aor/tables/` and `journal/aor/figures/`. The source-to-output map is `journal/aor/tables/ANALYSIS_ARTIFACT_MANIFEST.csv`. The main article uses a compact selection of those inputs; this resource preserves the complete computational records without reproducing the main article's narrative.

The benchmark is synthetic and identity-closure only. It is not a probability sample of financial libraries and supports no population, causal, forecasting, alpha, or deployable-performance claim.

S13 Point-in-Time Financial and Calibrated Mechanism Records

This section supplies the extended records for two distinct evidence layers. The *Point-in-Time Portfolio-Library Retention Study* is retrospective financial evidence from licensed data. The *Calibrated Closure-Option Mechanism Experiment* is design-locked synthetic evidence whose nuisance law uses public-safe aggregate calibration. The first is the article's financial case study; the second isolates the bridge mechanism. Neither is an alpha, forecasting, causal, or deployable-performance experiment.

S13.1 Point-in-time portfolio-library retention study

S13.1.1 Chronology and information firewall

For a compression origin y , formation data from $y - 5$ through $y - 3$ construct the source library and its operating frontier. The compression block from $y - 2$ through y estimates strategy profiles and solves the safe and frontier-only arms. Proposal year $y + 1$ enumerates every grammar-defined innovation option enabled by each retained library and freezes the protected policy choices. Evaluation year $y + 2$ is opened only after those choices are hash-sealed. Proposal values never enter compression; evaluation values never enter compression, option enumeration, uncertainty estimation, or adoption.

The primary universe is point-in-time liquid common equity. Plain ETFs form a separate registered replication and are never pooled with common equities. Eligibility does not require later survival. Observed delisting returns are retained, terminal positions move to cash, and future completeness is not an eligibility screen. One ETF origin fails its predeclared minimum-eligibility gate and remains unavailable in every denominator rather than being silently removed.

The source grammar contains portfolio-level signals, filters, holding horizons, cross-sectional allocation, gross exposure, and exit rules. Capability closure contains executable rule and interface requirements plus point-in-time data, corporate-action and delisting, portfolio-aggregation, and cost/capacity artifacts. The public manifest records 31 complete capabilities in every gate-passed cell. These are declared model artifacts; they do not prove that a real institution's undocumented code or tacit knowledge can be recovered from labels alone.

S13.1.2 Compression evidence and exact rechecks

The innovation-safe arm preserves the entire source frontier and complete declared closure. The comparator preserves only the frontier and is augmented under the safe arm's realized burden cap with a frozen preproposal order. Its action set is nested inside the safe action set before any proposal score is read. Cash is always feasible, and the safe policy inherits the exact comparator choice unless a simultaneous moving-block lower bound for an added option clears the registered hurdle.

3 gate-passed safe cells reduce to an exact preprocessing certificate. The remaining 34 are returned by HiGHS with OPTIMAL termination. Every lifted selection then passes independent original-instance checks for mandatory inactive retention, exact frontier equality, exact identity-closure equality, and exact integer burden. These checks certify the returned library's feasibility and burden. They do not turn the solver's global-optimality status into Lean or exhaustive proof.

The burden reductions in the table are origin-level means of $1 - W(L^{\text{safe}})/W(L)$. Burden is a prespecified validation-computation index, not observed money, staff time, or compute use. The option count is the number of safe-library proposal actions unavailable to the budget-matched frontier-only library. It measures retained opportunity, not realized performance.

Universe	Origins	Mean source burden	Mean safe burden	Mean reduction	Added options	Held-out CE difference
Common equity	19	8169.2	2183.8	73.1%	65.95	0.0000
Plain ETF	18	8909.2	2231.7	74.7%	66.06	-0.0022

Table S26 Point-in-time portfolio-library retention results. Burden is the registered validation-computation index, not currency. Added options are safe-library proposal actions absent from the budget-matched frontier-only library. The held-out column is innovation-safe minus frontier-only certainty-equivalent difference: all 19 common-equity choices coincide, while one of 18 complete ETF choices differs. Safe solutions pass exact original-instance frontier, closure, inactive-retention, and burden rechecks; solver-reported optimality remains solver evidence.

S13.1.3 Origin-level held-out outcomes

The common-equity protected policy never exercises an added innovation option, so all held-out safe-minus-frontier-only contrasts are mechanically zero. In the ETF replication, one protected replacement is exercised and realizes a negative held-out contrast. The failure is retained; no alternative selector is promoted after evaluation.

Table S27: Origin-level point-in-time held-out contrasts. CE differences are innovation-safe minus frontier-only. A dash marks the prespecified failed ETF universe gate; it is retained, not imputed.

Origin	Equity CE diff.	Same choice	ETF CE diff.	Same choice
2005	0.0000	yes	—	—
2006	0.0000	yes	0.0000	yes
2007	0.0000	yes	0.0000	yes
2008	0.0000	yes	0.0000	yes
2009	0.0000	yes	0.0000	yes
2010	0.0000	yes	0.0000	yes
2011	0.0000	yes	0.0000	yes
2012	0.0000	yes	0.0000	yes
2013	0.0000	yes	0.0000	yes
2014	0.0000	yes	0.0000	yes
2015	0.0000	yes	0.0000	yes
2016	0.0000	yes	-0.0390	no
2017	0.0000	yes	0.0000	yes
2018	0.0000	yes	0.0000	yes
2019	0.0000	yes	0.0000	yes
2020	0.0000	yes	0.0000	yes
2021	0.0000	yes	0.0000	yes
2022	0.0000	yes	0.0000	yes
2023	0.0000	yes	0.0000	yes

The origin is the financial analysis unit. Strategy specifications and daily observations are not treated as independent replications. A zero contrast from identical

actions does not establish economic equivalence of the two libraries, and a negative realized contrast at one origin does not identify the population value of closure.

S13.1.4 Registered robustness and diagnostic boundaries

The proposal-stage robustness grid varies transaction costs, risk aversion, and adoption hurdles without reopening compression or using evaluation data. Common-equity safe and comparator choices coincide throughout. ETF differences remain confined to the same origin and vanish at the higher cost settings. Leave-one-origin-out summaries perform aggregation only, with no reselection. A larger universe cap and a permuted-closure sham that were not frozen before proposal outcomes became known remain explicitly unavailable.

The registered forced-maximum policy omits the protected cash/adoption rule and is retained only as a negative control. Full-family best-candidate, White reality-check, SPA, Deflated-Sharpe, and PBO summaries are search diagnostics; they never feed back into a policy choice. Capacity records likewise score frozen policies without reselection over the declared spread, impact, AUM, and participation-cap grid. Missing capacity inputs, incomplete return paths, and hard participation-cap breaches remain failures rather than dropped rows.

S13.2 Calibrated closure-option mechanism experiment

S13.2.1 Calibration and registered regimes

The mechanism experiment begins from two equal-burden libraries with equal current operating frontiers. Only the innovation-safe library retains the bridge capability for a declared descendant project. Its net bridge margin is

$$M = \beta^d \rho^d \pi G - \kappa,$$

and the oracle adopts exactly when $M > 0$. The frontier-only project is closure-infeasible.

The noise calibration uses every complete proposal candidate path in the gate-passed point-in-time cells: 2479 paths and 623647 residual observations. Cross-strategy date means are removed within cells before estimating scale, lag-one dependence, and tail thickness. No sign or treatment margin is selected from financial outcomes. An independent null simulation fixes the learner's one-sided threshold; disjoint seeds generate test worlds; and the powered margin is fixed prospectively at twice that threshold.

The low-dose oracle has positive value, but the learned-value interval crosses zero and adoption remains low. At the powered dose, both oracle and learned mean intervals are positive and learned adoption exceeds the registered gate. The null point adoption rate passes its prespecified finite-test tolerance, but its Wilson upper endpoint slightly exceeds that tolerance; both facts are reported. Adverse adoption is near zero, and every world passes the structural equal-frontier, equal-burden, and bridge-availability assertions.

Regime	True margin	Oracle mean [95% CI]	Learned mean [95% CI]	Adoption
Adverse margin	-0.0270	0.0000 [0.0000, 0.0000]	-0.0000 [-0.0000, 0.0000]	0.05%
Null margin	0.0000	0.0000 [0.0000, 0.0000]	-0.0002 [-0.0003, 0.0000]	5.76%
Low positive dose	0.0025	0.0026 [0.0019, 0.0032]	0.0001 [-0.0001, 0.0002]	7.35%
Powered positive dose	0.0540	0.0541 [0.0529, 0.0554]	0.0512 [0.0500, 0.0525]	94.17%

Table S28 Calibrated closure-option mechanism experiment, with 4096 synthetic worlds per prespecified regime. Current frontier and burden are equal by construction; only the innovation-safe library can admit the designated bridge descendant. Intervals are registered finite-simulation uncertainty summaries. This is synthetic mechanism evidence, not a market-return, alpha, forecasting, or causal estimate.

S13.2.2 Evidence status and interpretation

The mechanism result is exact with respect to its structural fixture and floating-point synthetic evidence with respect to its simulated value and uncertainty summaries. The complete world ledger is independently regenerated, its record hashes are checked, and the summary is reproduced. These checks are not formal proof of a market effect. The powered margin is a prospective power dose, not an estimated return premium, and the calibrated noise distribution does not transform the simulation into securities-panel evidence.

Taken together, the studies support a bounded managerial conclusion. Safe compression can preserve a materially larger research-option menu at much lower declared maintenance burden, but the decision to exercise any option requires a separate evidence threshold. The retrospective panel shows disciplined abstention and one adverse exercise; the synthetic experiment shows that this outcome can coexist with a valid bridge mechanism when small effects are difficult to distinguish from proposal noise.

S13.3 Licensed-data and provenance boundary

All public financial tables in this section are generated from sealed aggregate manifests. Raw CRSP/WRDS rows, security identifiers, and return paths are excluded. The exact internal experiment identifiers and hash-bound paths remain in the repository provenance map so that design and result seals can be verified; publication-facing scientific names are used here. Reproduction of the source-data stages requires independent licensed access. Regeneration and audit of every published aggregate table requires only the committed public-safe artifacts.

S14 Solver Diagnostics and Independent Exactness Records

The baseline binary covering model fixes mandatory columns, places one at-least-one constraint on every residual tagged row, and minimizes the exact weighted burden after a verified integer scaling. HiGHS runs record solver and wrapper versions, deterministic controls, time limit, gap tolerances, primal, dual, and termination status, objective,

best bound, reported gap, solve time, node count where exposed, preprocessing audit, warm-start source, selected original identifiers, and log location.

Every returned vector must already be binary under the declared exact acceptance rule; no fractional candidate is rounded. Reconstruction is followed by independent exact checks of mandatory retention, tagged coverage, burden, the original frontier, and the original identity closure. A returned `OPTIMAL` status is solver evidence plus an exact feasibility recheck, not exhaustive enumeration, a Lean theorem, or a formal optimality proof. When exact preprocessing empties the residual model, the record states that no solver was invoked.

The authoritative benchmark audit is `experiments/algorithmic_compression_v2/RESULT_AUDIT.md`; machine certificates and hashes are below `experiments/algorithmic_compression_v2/results/audit/certificates/` and `experiments/algorithmic_compression_v2/results/ARTIFACT_MANIFEST.csv`. The complete MIP logs, worker handshakes, stdout, stderr, instances, and raw TOML records are retained under the committed v2 result tree. No second open-source MIP solver was pinned, so medium and large cases receive exact feasibility and burden rechecks plus HiGHS bound/status/log diagnostics rather than a second-solver optimality claim.

For the two financial source instances and each of the three prospectively locked weight schedules, fixed-point exact preprocessing leaves zero residual strategies and zero residual requirements. The resulting six certificates prove the optimum value by the human preprocessing propositions together with exact finite computation and original-instance rechecks. They are not Lean verified and do not use a solver status as proof. The authoritative rows are `experiments/results/summaries/financial_resource_optimization_weight_robustness_empty_residual_certificates.csv`.

The audit command is nonmutating with respect to saved result records:

```
make aor-benchmark-v2-audit
```

It reads the committed raw records and certificates; it does not rerun any solver or heuristic.

S15 Theorem and Evidence Correspondence

The authoritative ledger is `THEOREM_LEDGER.md`; the table below is its journal-facing correspondence. A supporting finite fixture never upgrades the status of a universal result, and solver evidence establishes none of the mathematical claims in this table.

Table S29: Evidence status of the journal theory and algorithm results.

Result	Ledger	Complete-result status	Supporting evidence and boundary
Tagged-cover equivalence and weighted objective identity	SC-TAG	Lean verified	Human proof; exact Julia boundary fixtures. The Lean result covers the encoded universal biconditional and burden identity, not complexity or approximation.
Identity-closure completeness	NP- SC-COMP	Human proof only	Exact reduction fixtures and Julia tests; no Lean complexity framework.
Unbounded heaviest-safe-first gap	SC-GAP	Human proof only	Exact rational family fixtures and Julia tests; Lean checks one finite arithmetic case only.
Preprocessing propositions	SC-PRE	Human proof only	Lean verifies selected finite feasibility cores; objective preservation, reconstruction, propagation, and fixed-point claims are exact fixtures and Julia tested, not formally verified.
Enumeration and requirement-mask DP correctness, complexity, and FPT status	SC-DP	Human proof only	Exact cross-method fixtures and Julia tests; Lean mask identities do not formalize the recurrence invariant or complexity bound.
Weighted-greedy transfer and reverse deletion	SC-GREEDY	Human proof only	The external Chvátal theorem is not re-formalized; finite ratios are exact fixtures and Julia tested.
Rechecked-deletion feasibility	SC-DEL-SUITE / OP T-T2T4	Lean verified for an abstract supplied safe trace; human proof only for concrete-suite termination and final-scan irreducibility	Julia tests cover scoring, stale-certificate invalidation, reproducible randomness, counters, feasibility, and irreducibility. No approximation quality follows.

The Lean axiom audits recorded for the tagged-cover and preprocessing declarations print only `propext`, `Classical.choice`, and `Quot.sound`. The shared Julia–Lean fixture prevents manual transcription of selected finite rational values. The detailed source hashes and roles are in `ARTIFACT_MANIFEST.md`; formal sources are under `formal/StrategyInnovation/Optimization/`, `formal/StrategyInnovation/Fixtures/`, and `formal/StrategyInnovation/Audit/`.

S16 Reproducibility Commands and Artifact Records

This section records the execution boundaries, configuration histories, hashes, and complete artifact locations for Sections S10–S14. The commands are reproduction instructions only; compiling this resource does not run Julia or regenerate any committed result. The public repository is available at

<https://github.com/dramirez37/financial-strategy-library-compression>.

S16.1 Canonical and presentation records

The canonical solve is deterministic and uses no random seed:

```
julia --project=julia \  
  julia/scripts/solve_unified_canonical_benchmark.jl
```

The manuscript presentation layer is Julia-generated:

```
julia --project=julia \  
  julia/scripts/generate_manuscript_numerical_artifacts.jl  
julia --project=julia \  
  julia/scripts/generate_manuscript_numerical_artifacts.jl --check
```

The first command validates the committed convergence, compression, decomposition, geometry, grammar, mechanism, and uncertainty tables before writing manuscript artifacts. The second is nonmutating and rejects byte drift. Complete canonical numerical outputs are listed in Section S10.

S16.2 Randomized-study configuration history

The registered v2 experiment uses the fixed maximum $N = 1024$, master seed 6075990691714899803, and the complete trial and derived-seed registry at

`experiments/randomized_library_v2/TRIAL_REGISTRY.csv`.

The initial design, sequential-precision amendment, and executable-protocol amendment are retained at

`experiments/randomized_library_v2/DESIGN_LOCK.json`,

```
experiments/randomized_library_v2/DESIGN_LOCK_AMENDMENT_1.json,  
experiments/randomized_library_v2/DESIGN_LOCK_AMENDMENT_2.json.
```

Their aggregate hashes are, respectively,

```
0b012dfc14f4ea57b0d34877a68d9a546cd499d2270903e772d78b21425d14db  
d9122d258b8fe62d5872c41b0a418b1351a5798b25ddb99ec73b726e324d244f  
8c278c07d998ba118d98c78cc1373a47ab63127f00d606c0042c006dac11e7be
```

The registered run, byte-drift replay, and independent result-reader audit are:

```
julia --project=julia \  
  julia/scripts/run_randomized_library_stress_v2.jl  
julia --project=julia \  
  julia/scripts/run_randomized_library_stress_v2.jl --check  
julia --project=julia \  
  julia/scripts/audit_randomized_library_v2_results.jl
```

All theorem-facing calculations and point estimates use `Rational{BigInt}`. Decimal Wilson intervals, Monte Carlo standard errors, and SVG coordinates are presentation diagnostics only. The complete cumulative source is

```
experiments/results/summaries/randomized_library_v2_stability_summary.  
                                csv,
```

and the 896-row two-level slice table is

```
experiments/results/summaries/randomized_library_v2_stability_factor_s  
                                ummary.csv.
```

The data-linked stabilization plot is

```
manuscript/figures/randomized_library_v2_stability.svg.
```

The exact final counts are 398 positive frontier-only losses, zero positive innovation-safe losses, 388 silent generative assets among 5,120 source assets, 256 substitutions, 141 complementarities, and 627 zero interactions. All interaction counts use four raw witnesses. These diagnostics describe Monte Carlo precision within the registered finite generator, not uncertainty about a real population.

The separately locked resource extension reuses only the frozen v2 registry. Its exact run and byte-drift check are

```
julia --project=julia \  
  julia/scripts/run_randomized_library_optimization_extension.jl  
julia --project=julia \  
  julia/scripts/audit_randomized_library_v2_results.jl
```

```
julia/scripts/run_randomized_library_optimization_extension.jl \  
--check
```

Every trial enumerates all 32 inactive-containing sublibraries. The readable exact summary is

RANDOMIZED_LIBRARY_OPTIMIZATION_EXTENSION_V1.md.

The parent report and frozen pilot remain separately indexed by RANDOMIZED_LIBRARY_REPORT_V2.md and RANDOMIZED_LIBRARY_REPORT.md; they are not pooled.

S16.3 Financial configuration and amendment history

Both financial audits use the licensed CRSP/WRDS snapshot associated with source-repository commit 3146029aa71dfe7639ede9ca79f81602165125e8. Preparation scripts perform no download. The terminal audit uses seed 6075990691714899801; the annual walk-forward audit uses seed 6075990691714899802. Their ignored panels contain 106,975 and 427,900 rows, each with 4,279 complete common dates. Raw and row-level data are nonredistributable.

The terminal configuration, audit, results, and limitations records are

```
experiments/configs/financial_terminal_audit.toml,  
experiments/financial_terminal_audit/DATA_AUDIT.md,  
experiments/financial_terminal_audit/DRAFT_RESULTS.md,  
experiments/financial_terminal_audit/EMPIRICAL_LIMITATIONS.md.
```

The annual walk-forward audit's initial lock, six retained amendments, amendment history, final lock, frozen universe, audit, results, and limitations are

```
experiments/financial_annual_walkforward_audit/DESIGN_LOCK_INITIAL.json,  
experiments/financial_annual_walkforward_audit/DESIGN_LOCK_AMENDED_1.json through  
experiments/financial_annual_walkforward_audit/DESIGN_LOCK_AMENDED_6.json,  
experiments/financial_annual_walkforward_audit/DESIGN_AMENDMENTS.md,  
experiments/financial_annual_walkforward_audit/DESIGN_LOCK.json,  
experiments/financial_annual_walkforward_audit/universe_selection.toml,  
experiments/financial_annual_walkforward_audit/DATA_AUDIT.md,  
experiments/financial_annual_walkforward_audit/DRAFT_RESULTS.md,  
experiments/financial_annual_walkforward_audit/EMPIRICAL_LIMITATIONS.md.
```

The initial protocol, sparse-support amendment, and report-only corrections remain in that versioned history. Reruns must preserve negative and mixed outcomes unless a new experiment is registered.

The cross-audit financial resource optimization is a separate exact post-solve analysis of the two committed aggregate audit records; it is not a third licensed-data audit and does not read row-level inputs. Its nonmutating lock and certificate checks are

```
julia --project=julia \  
  julia/scripts/lock_financial_resource_optimization.jl --check  
julia --project=julia \  
  julia/scripts/verify_financial_resource_optimization_outputs.jl
```

The registered design and readable exact report are

```
experiments/financial_resource_optimization/DESIGN_LOCK.json,  
  FINANCIAL_RESOURCE_OPTIMIZATION.md.
```

S16.4 Complete artifact index

ARTIFACT_MANIFEST.md is the authoritative configuration, data, decision, RNG, output-hash, and correction-history index. REPRODUCIBILITY.md records the complete command surface and component-specific gates. EMPIRICAL_INFORMATION_SET_AUDIT.md records the code-level financial execution order and quantity-by-quantity information classification.

Within `experiments/results/summaries/`, the following prefix families exhaust the historical study records used in the main paper and Sections S10–S13 of this resource:

Registered parent study. The parent-study prefix is

```
randomized_library_v2_*
```

It excludes the separately registered optimization-extension prefix below. This family contains trials, assets, profiles, modules, closures, kernels, projects, pruning, actions, raw-witness manifests, rectangle corners and transitions, interaction signs, method, factor, relationship, stability, and summary records.

Randomized resource extension. `randomized_library_v2_optimization_v1_*`. This family contains trial, sublibrary, compression, capacity, breakpoint, penalized, elasticity, factor-summary, and aggregate-summary records.

Exact response experiments. `unified_comparative_statics_*` and `theorem_mechanism_*`. These families contain exact fixtures, the 2,430-row interaction response, sign checks, policy-map rows, decomposition, pruning-loss, coverage, selection, metadata, and aggregate summaries.

Locked terminal financial audit. `financial_terminal_audit_*`. This namespace is disjoint from the annual walk-forward prefix below. The family contains grammar, candidate, pruning, ranking, decomposition, uncertainty, cost-sensitivity, mechanism, figure-source, and status records.

Annual walk-forward financial audit. `financial_annual_walkforward_audit_*`. This family contains grammar, candidate, pruning, selection, episode-ranking, ranking,

decomposition, uncertainty, cost-sensitivity, mechanism, universe-figure, and status records.

Cross-audit financial resource optimization. `financial_resource_optimization_*`. This family contains registered weights, selected-library memberships, exact frontier certificates, solution summaries, and status metadata derived from the two aggregate audit records.

Registered Algorithmic Compression Benchmark v2. The design, registries, amendments, locks, execution record, raw terminal matrix, audit certificates, logs, and artifact manifest are under `experiments/algorithmic_compression_v2/`. This namespace is separate from the frozen `randomized_library_v2_*` historical study and from the unincorporated v1 development output.

Locked financial weight robustness. `financial_resource_optimization_weight_robustness_*`. This family contains the prospective schedule protocol and lock outputs, public-safe selections, burdens, schedule overlaps, rank stability, unique-carrier stability, and six empty-residual optimum certificates. It contains no raw licensed market rows.

The corresponding supplement figures are

```
manuscript/figures/dynamic_research_policy_regions.tex,  
manuscript/figures/randomized_library_v2_stability.svg,  
manuscript/figures/financial_coverage_comparison.tex.
```

The theorem ledger, formal correspondence, and validation matrix retain mathematical validity, Lean kernel verification, Julia validation, and empirical relevance as separate evidence dimensions.

S16.5 Journal-program commands

The registered v2 final benchmark is not rerun to compile this resource. Its nonmutating artifact audit is

```
make aor-benchmark-v2-audit
```

The focused journal theory, algorithm, and evidence-status gates are

```
make aor-algorithm-tests  
make aor-evidence-status-tests
```

The two licensed financial workflows remain subject to the documented local input contracts and redistribution boundary. The public aggregate financial algorithm outputs can be verified without opening raw licensed rows through the commands recorded in `REPRODUCIBILITY.md` and the resource-optimization verification script shown above.

This Online Resource compiles independently from its own directory:

```
./build.sh
```

The build performs LaTeX compilation followed by missing-input, undefined citation/reference, duplicate-label, and S-prefix audits. It creates no mathematical, experimental, solver, or financial result.

Generative-AI disclosure

OpenAI ChatGPT and Codex were used during preparation of this Online Resource to assist with LaTeX drafting and editing, software development and review, literature-metadata checking, and repository-based analysis and reproducibility tasks. The author reviewed, revised, and verified all AI-assisted material and remains solely responsible for the resource. No AI system is listed as an author, and no generative-AI image is used.

References

- Chvátal, V.: A greedy heuristic for the set-covering problem. *Mathematics of Operations Research* 4(3), 233–235 (1979) <https://doi.org/10.1287/moor.4.3.233>
- Karp, R.M.: Reducibility among combinatorial problems. In: Miller, R.E., Thatcher, J.W., Bohlinger, J.D. (eds.) *Complexity of Computer Computations*, pp. 85–103. Springer, Boston, MA (1972). https://doi.org/10.1007/978-1-4684-2001-2_9 . https://link.springer.com/chapter/10.1007/978-1-4684-2001-2_9